Reviewer: Matthew P. Koenig
Date: 2/27/2026

CONTRACT No. B - 13517

INDEX

PROJECT	STRUCTURE	TYPE	SPAN	OVER	STATION
F-068-6 (1)	44-70-6634	CONT. COMPOSITE PRESTR. CONC. I - BEAM	3 SPANS 49'-3, 50'-0, 49'-3 SKEW 15° RT	MUD CREEK	617+95 "A"

SHEET NO	SHEET DESIGNATION	SUBJECT	F.H.W.A. APPROVAL
1	ONE SHEET	INDEX AND TITLE	
2	ONE SHEET	TYPICAL CROSS SECTIONS	
3	ONE SHEET	PLAN & PROFILE LINE "A"	
4	ONE SHEET	PLAN & PROFILE LINE "S1A" & "S2A"	
5	ONE SHEET	DETAILS	
6	ONE SHEET	SOIL BORINGS	
7	C1	LAYOUT	
8	C2	GENERAL PLAN	
9	C3	BENT N°1 & BENT N°4 DETAILS	
10	C4	BENT N°1 & BENT N°4 DETAILS & BILL OF MATERIALS	
11	C5	PIERS N°2 & 3 DETAILS & BILL OF MATERIALS	
12	C6	SUPERSTRUCTURE DETAILS	
13	C7	SUPERSTRUCTURE DETAILS & BILL OF MATERIALS	
14	ONE SHEET	MISCELLANEOUS TABLES	
15	ONE SHEET	BRIDGE SUMMARY	
16	ONE SHEET	BRIDGE ESTIMATE OF QUANTITIES	
17-35	NINETEEN SHEETS	CROSS SECTIONS	

STATE OF INDIANA
INDIANA STATE HIGHWAY COMMISSION

BRIDGE PLANS

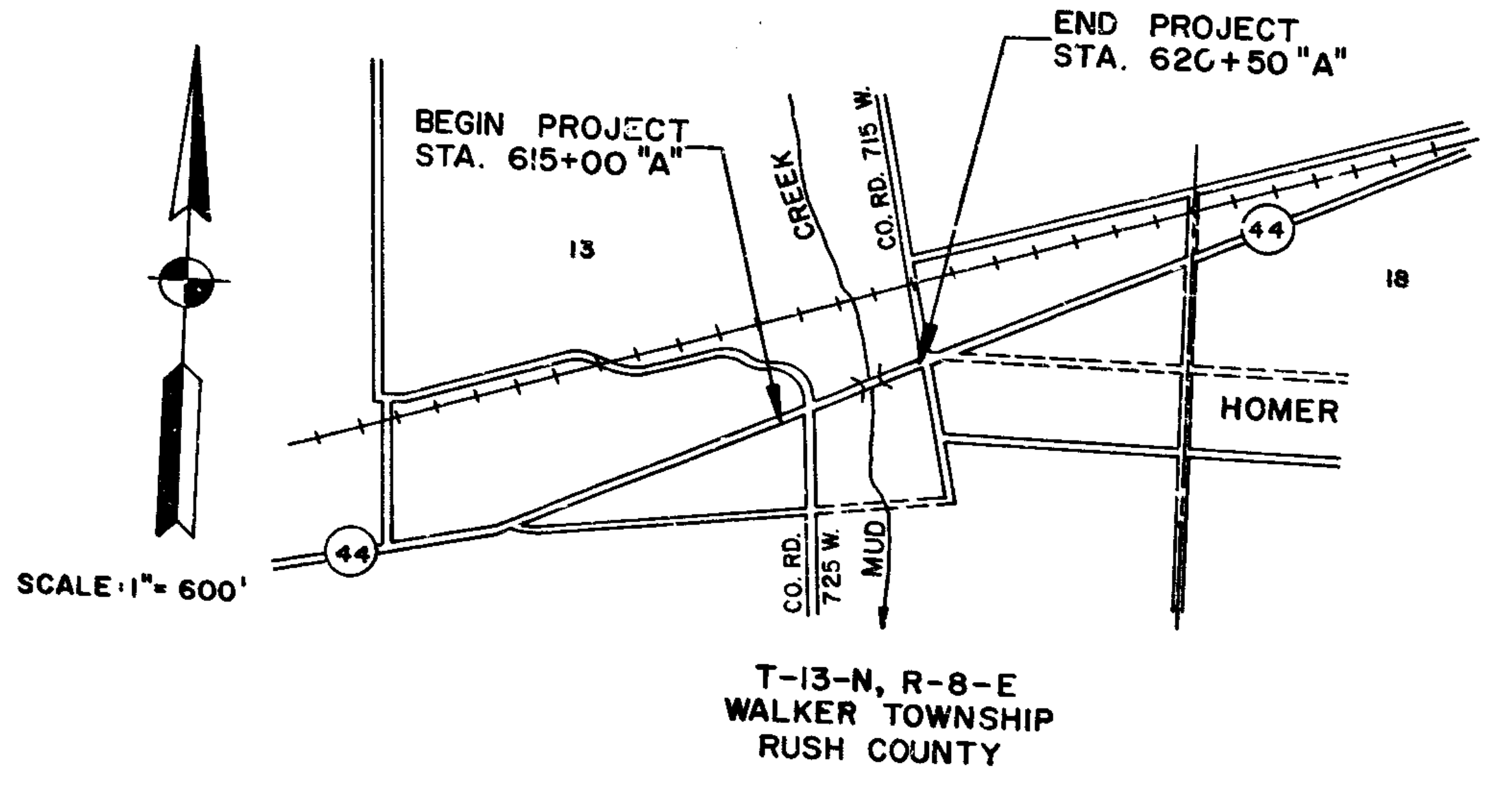
FOR SPANS OVER 20 FEET
ON
STATE ROAD NO. 44

PROJECT NO. F - 068-6

(1) P.E.
(1) R/W
(1) CONST.

BEGINNING AT A POINT ON THE CENTERLINE OF S.R. 44 APPROXIMATELY 105.3 FEET WESTERLY FROM THE INTERSECTION OF S.R. 44 AND COUNTY ROAD 725 W. THENCE EXTENDING EASTERLY 530.0 FEET TO A POINT ON THE CENTERLINE OF S.R. 44 APPROXIMATELY 18.4 FEET WESTERLY FROM THE INTERSECTION OF S.R. 44 AND COUNTY ROAD 715 W ALL IN SECTION 13, T-13-N, R-8-E, WALKER TOWNSHIP, RUSH COUNTY

BRIDGE	LENGTH:	0.029	MI.
ROADWAY	LENGTH:	0.075	MI.
TOTAL	LENGTH:	0.104	MI.
MAX.	GRADE :	1.35	%



T-13-N, R-8-E
WALKER TOWNSHIP
RUSH COUNTY

BRIDGES OVER 20' SPAN					
FEDERAL REGION NO.	STATE	PROJECT NO.	FISCAL YEAR	SHEET NO.	TOTAL SHEETS
5	IND.	F-068-6(1)	1979	1	73

INDEX CONTINUED
STANDARD DRAWINGS

SHEET NO.	SHEET DESIGNATION	SUBJECT	F.H.W.A. APPROVAL	ADOPTED "A" REVISION "R"
36	BRIDGE STD. BR1	ALUMINUM BRIDGE RAILING		
37	BRIDGE STD. BR2	ALUMINUM BRIDGE RAILING DETAILS	12-16-80	R-11-3-80
38	BRIDGE STD. BR3	STEEL BRIDGE RAILING	5-10-79	R-12-1-78
39	BRIDGE STD. BR4	STEEL BRIDGE RAILING DETAILS	12-16-80	R-11-3-80
	BRIDGE STD. BR5	RAILING CONNECTION DETAILS	5-10-79	R-12-1-78
	BRIDGE STD. BR6	RAILING CONNECTION DETAILS		
40	BRIDGE STD. C1	MISCELLANEOUS DETAILS		
	BRIDGE STD. C2	MISCELLANEOUS DETAILS	12-21-81	R-12-7-81
41	BRIDGE STD. C3	MISCELLANEOUS DETAILS		
42	BRIDGE STD. C4	MISCELLANEOUS DETAILS	12-21-81	R-12-7-81
	BRIDGE STD. D	CASTING DETAILS ROADWAY DRAINS	12-4-74	R-10-29-74
[illegible]	BRIDGE STD. M2	TEMPORARY RUNAROUND (Obsolete)		
	BRIDGE STD.			[illegible]
44	BRIDGE STD. PB1	PRESTRESSED CONCRETE TYPE I I-BEAMS		
	BRIDGE STD. PB	PRESTRESSED CONCRETE TYPE I-BEAMS	4-16-62	A-1-16-62
	BRIDGE STD. PB6	PRESTRESSED BOX BEAMS		
	BRIDGE STD. PB	PRESTRESSED COMPOSITE BOX BEAMS WIDE		
	BRIDGE STD. PB	PRESTRESSED COMPOSITE BOX BEAMS WIDE		
45	BRIDGE STD. PB10	TOLERANCES FOR FABRICATION OF PRESTRESSED BEAMS		
46	BRIDGE STD. PB11	ELASTOMERIC BEARING PAD DETAILS	8-14-63	A-11-9-62
	BRIDGE STD.		9-21-79	R-8-1-72
	BRIDGE STD.			
	BRIDGE STD. R2A	BRIDGE LIGHTING DETAILS		
	BRIDGE STD. R2B			
47	BRIDGE STD. S1	MISCELLANEOUS DETAILS		
	BRIDGE STD. SH1	STEEL SHOE DETAILS	1-17-72	R-8-2-71
48	BRIDGE STD. T SHEET A	STANDARD TEMPORARY BRIDGE		
49	BRIDGE STD. T SHEET B	STANDARD TEMPORARY BRIDGE		R-12-15-78
	BRIDGE STD.		1-12-60	R-11-13-39
	BRIDGE STD.			
	BRIDGE STD.			
	BRIDGE STD.			
	BRIDGE STD.			
50	ROAD STD. SHEET A	STANDARD PAVEMENT JOINTS		
	ROAD STD. SHEET B	STANDARD PAVEMENT JOINTS	6-3-81	R-4-1-81
51	ROAD STD. SHEET MA	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MA	MISCELLANEOUS STANDARDS	9-29-71	R-1-4-71
	ROAD STD. SHEET MB	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MB2	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS		
52	ROAD STD. SHEET MC	MISCELLANEOUS STANDARDS		
53	ROAD STD. SHEET MC1	MISCELLANEOUS STANDARDS	9-14-78	R-8-1-78
54	ROAD STD. SHEET MD	MISCELLANEOUS STANDARDS	10-21-76	R-7-1-76
55	ROAD STD. SHEET MD1	MISCELLANEOUS STANDARDS	6-14-74	R-1-2-74
	ROAD STD. SHEET MD	MISCELLANEOUS STANDARDS	6-14-74	R-1-2-74
56	ROAD STD. SHEET ME	MISCELLANEOUS STANDARDS		
57	ROAD STD. SHEET ME2	MISCELLANEOUS STANDARDS	12-22-80	R-10-1-80
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS	11-2-78	R-10-2-78
	ROAD STD. SHEET MH	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MH	MISCELLANEOUS STANDARDS		
58	ROAD STD. SHEET MI-1	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS	11-15-79	R-8-1-79
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS		
59	ROAD STD. SHEET MN	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS	12-22-80	A-10-1-80
60	ROAD STD. SHEET MP	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS	6-6-80	R-4-1-80
	ROAD STD. SHEET MQ	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MR	MISCELLANEOUS STANDARDS		
61	ROAD STD. SHEET CB2	TEMPORARY CONCRETE BARRIER		
	ROAD STD.	STANDARD REINF. CONCRETE BOX CULVERTS	6-3-81	R-4-1-81
	ROAD STD.	STANDARD REINF. CONCRETE CULVERTS		
62	ROAD STD. SHEET GR1	GUARD RAIL CLASS "A"		
	ROAD STD. SHEET GR	GUARD RAIL CLASS	10-24-80	R-8-1-80
63	ROAD STD. SHEET GR4	GUARD RAIL CLASS "G"		
64	ROAD STD. SHEET GR5	ALUMINUM GUARD RAIL DETAILS	5-17-79	R-2-1-79
65	ROAD STD. SHEET GR6	STEEL TUBE GUARD RAIL DETAILS	12-6-76	R-9-1-76
	ROAD STD. SHEET GR		11-2-78	R-8-1-78
66	ROAD STD. SHEET GR10	GUARD RAIL BURIED ENDS		
	ROAD STD.		6-6-80	R-4-1-80
67	ROAD STD. SHEET 9	TRAFFIC SIGN DETAILS		
	ROAD STD. SHEET 1 DETOURS	STANDARD DETOUR SIGNS		R-4-1-79
	ROAD STD. SHEET 1A DETOURS	STANDARD DETOUR SIGNS		
	ROAD STD. SHEET 1B DETOURS	STANDARD DETOUR SIGNS		
68	ROAD STD. SHEET 2 DETOURS	STANDARD DETOUR SIGNS		
69	ROAD STD. SHEET 2A DETOURS	STANDARD DETOUR SIGNS	5-10-80	R-5-1-80
70	ROAD STD. SHEET 3 DETOURS	STANDARD DETOUR SIGNS	8-25-81	R-8-3-81
71	ROAD STD. SHEET 3A DETOURS	STANDARD DETOUR SIGNS	10-20-80	R-9-1-80
72	ROAD STD. SHEET 4 DETOURS	STANDARD DETOUR SIGNS	5-18-77	R-4-1-77
73	ROAD STD. SHEET 5 DETOURS	STANDARD DETOUR SIGNS	5-18-77	R-4-1-77
	ROAD STD. SHEET 5A DETOURS	STANDARD DETOUR SIGNS	10-20-80	R-9-1-80

* F.H.W.A. APPROVAL PENDING

TRAFFIC DATA		S.R. 44
A.D.T. (1978)		3320 V.P.D.
A.D.T. (1998 PROJECTED)		5290 V.P.D.
D.H.V. (1998 PROJECTED)		530 V.P.H.
TRUCKS		D.H.V. 6 % A.D.T. 10 %
DESIGN SPEED		60 M.P.H.
ACCESS CONTROL		NONE

PREPARED & RECOMMENDED BY
REID, QUEBE, ALLISON, WILCOX & ASSOCIATES INC.
INDIANAPOLIS, INDIANA
Raman Patel 5/16/80

RAMAN B. PATEL
No. 13050

NOTE:—
WHEREVER "INDIANA STATE HIGHWAY COMMISSION" APPEARS IN THESE PLANS, IT SHALL BE INTERPRETED AS "INDIANA DEPARTMENT OF HIGHWAYS" EXCEPT THE 1978 INDIANA STATE HIGHWAY COMMISSION SPECIFICATIONS SHALL BE USED.

APPROVED 8-7-80
CHIEF HIGHWAY ENGINEER—INDIANA STATE HIGHWAY COMMISSION

INDIANA STATE HIGHWAY COMMISSION STANDARD SPECIFICATIONS DATED 1978 TO BE USED WITH THESE PLANS.

DATE	REVISIONS SHEET NO.
05-06-81	3, 4, 7
8-18-81	3, 4, 5, 7, 14, 16 & 23
01-26-82	2, 8, 12, 15 & 16, Deleted Sh. 43
02-22-82	2, 5, 8, 16

DATE	REVISIONS SHEET NO.

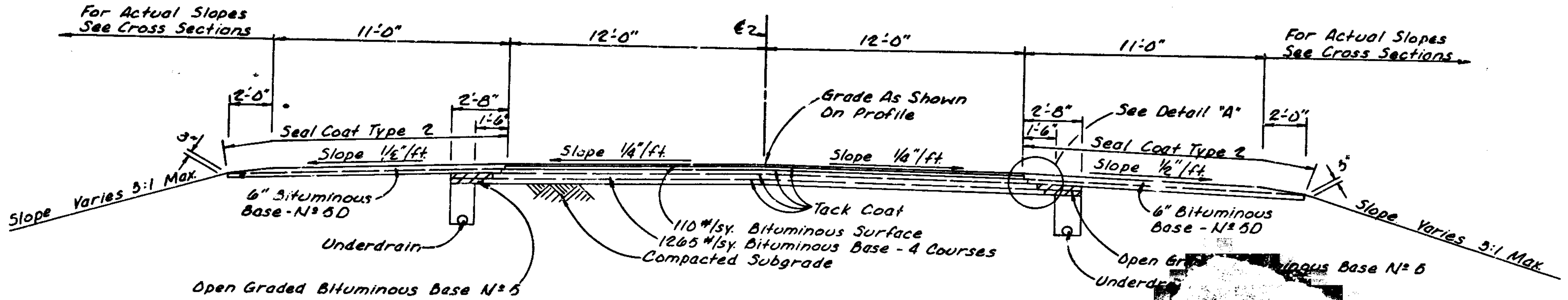
RECOMMENDED FOR APPROVAL 8-6-80
E. W. Walters
ENGINEER OF BRIDGE DESIGN, INDIANA STATE HIGHWAY COMMISSION

No. 7059

FEDERAL HIGHWAY ADMINISTRATION
DEPARTMENT OF TRANSPORTATION
APPROVED:-
DIVISION ADMINISTRATOR DATE

BRIDGE FILE:- 44-70-6634

FEDERAL ROAD REGION NO.	STATE	PROJECT NO.	FISCAL YEAR	SHEET NO.	TOTAL SHEETS
	IND.	F-068-6(1)	1979	2	73

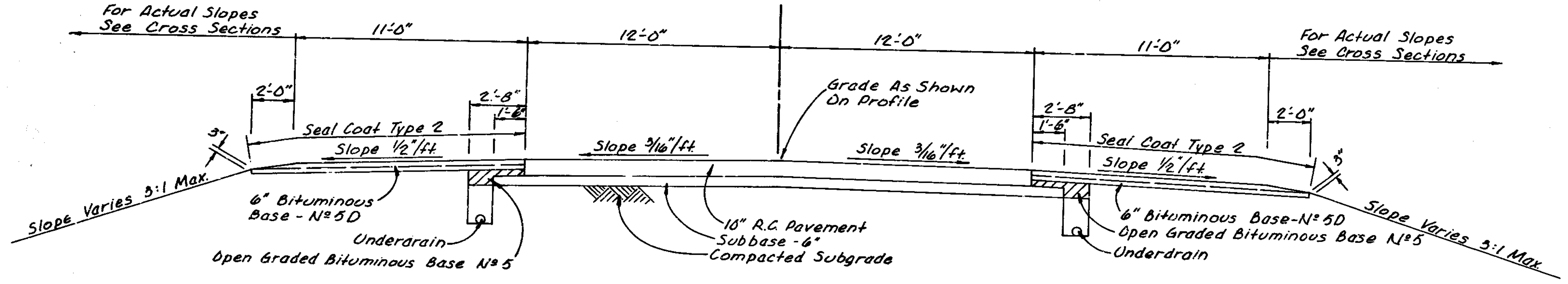

For Actual Slopes
See Cross Sections
11'-0"
12'-0"
℄
12'-0"
11'-0"
For Actual Slopes
See Cross Sections
2'-0"
2'-8"
1'-6"
Seal Coat Type 2
Slope 1/2"/ft
Slope 1/4"/ft
Grade As Shown
On Profile
Slope 1/4"/ft
2'-8"
1'-6"
See Detail "A"
2'-0"
Seal Coat Type 2
Slope 1/2"/ft
3"
3"
Slope Varies 3:1 Max.
6" Bituminous
Base - N° 5D
Underdrain
Open Graded Bituminous Base N° 5
Tack Coat
110 #/sy. Bituminous Surface
1265 #/sy. Bituminous Base - 4 Courses
Compacted Subgrade
6" Bituminous
Base - N° 5D
Open Gr[illegible]inous Base N° 5
Underd[illegible]
Slope Varies 3:1 Max.

TANGENT SECTION

STA. 615+00.00 TO STA. 616+94.99 "A"
STA. 618+95.01 TO STA. 620+50.00 "A"

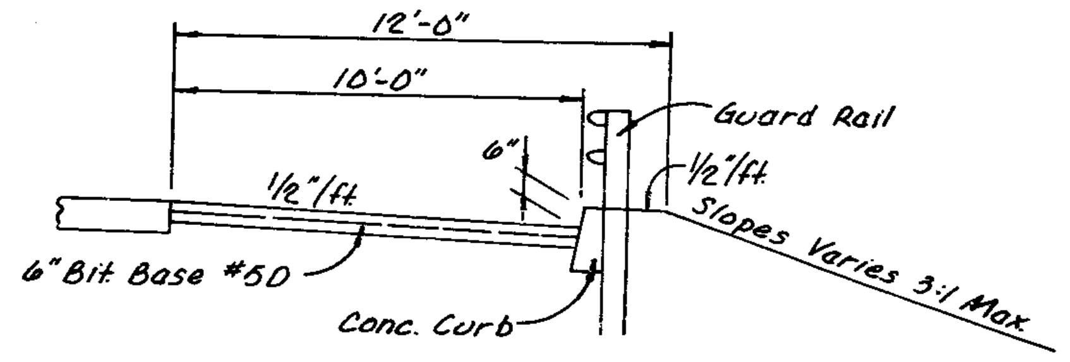
NOTES

1- BITUMINOUS SURFACE SHALL BE HOT ASPHALT CONCRETE SURFACE TYPE "11B"
2- FOR UNDERDRAIN DETAILS SEE MISC. STANDARD SHEET "MN"
3- FOR REINFORCED CONCRETE PAVEMENT DETAILS - SEE STANDARD PAVEMENT JOINTS SHEET A
4- THE QUANTITY CROWN-VETCH SEEDING, SHOWN ON THE ESTIMATE OF QUANTITIES SHEET IS TO BE USED AT THOSE LOCATIONS WHERE THE SLOPES ARE 3:1 OR STEEPER OR IN AN AREA REQUIRING SAND CUT OR SAND FILLS OR AS DIRECTED BY THE ENGINEER.

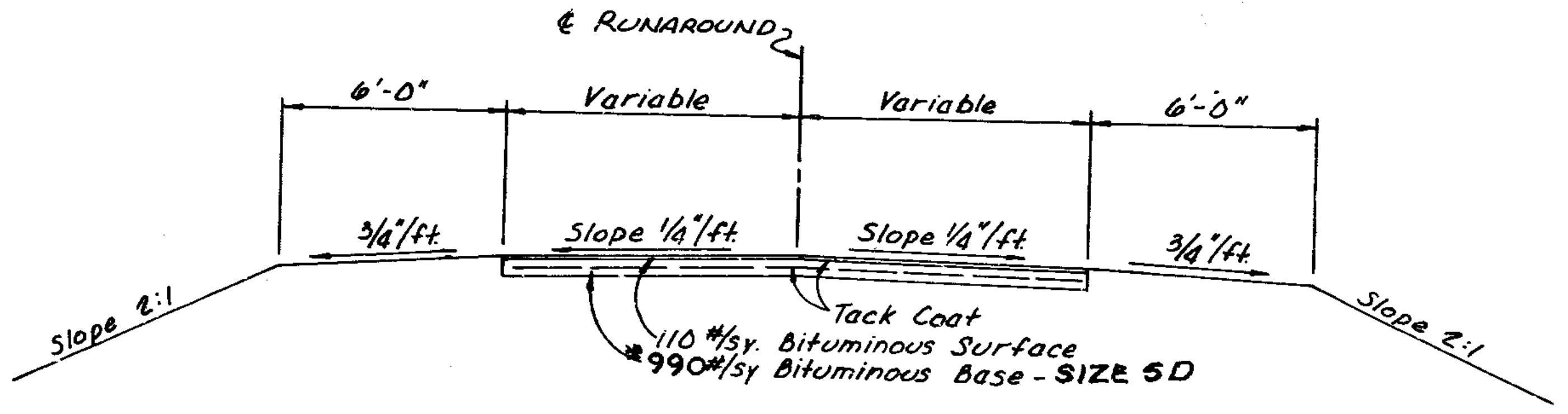

For Actual Slopes
See Cross Sections
11'-0"
12'-0"
12'-0"
11'-0"
For Actual Slopes
See Cross Sections
2'-0"
2'-8"
1'-6"
Seal Coat Type 2
Slope 1/2"/ft
Slope 3/16"/ft
Grade As Shown
On Profile
Slope 3/16"/ft
2'-8"
1'-6"
Seal Coat Type 2
Slope 1/2"/ft
2'-0"
3"
3"
Slope Varies 3:1 Max.
6" Bituminous
Base - N° 5D
Underdrain
Open Graded Bituminous Base N° 5
10" R.C. Pavement
Subbase - 6"
Compacted Subgrade
6" Bituminous Base - N° 5D
Open Graded Bituminous Base N° 5
Underdrain
Slope Varies 3:1 Max.

TANGENT SECTION

STA. 616+94.99 TO STA. 617+18.71 "A"
STA. 618+71.29 TO STA. 618+95.01 "A"


12'-0"
10'-0"
Guard Rail
6"
1/2"/ft
1/2"/ft
Slopes Varies 3:1 Max.
6" Bit Base #5D
Conc. Curb

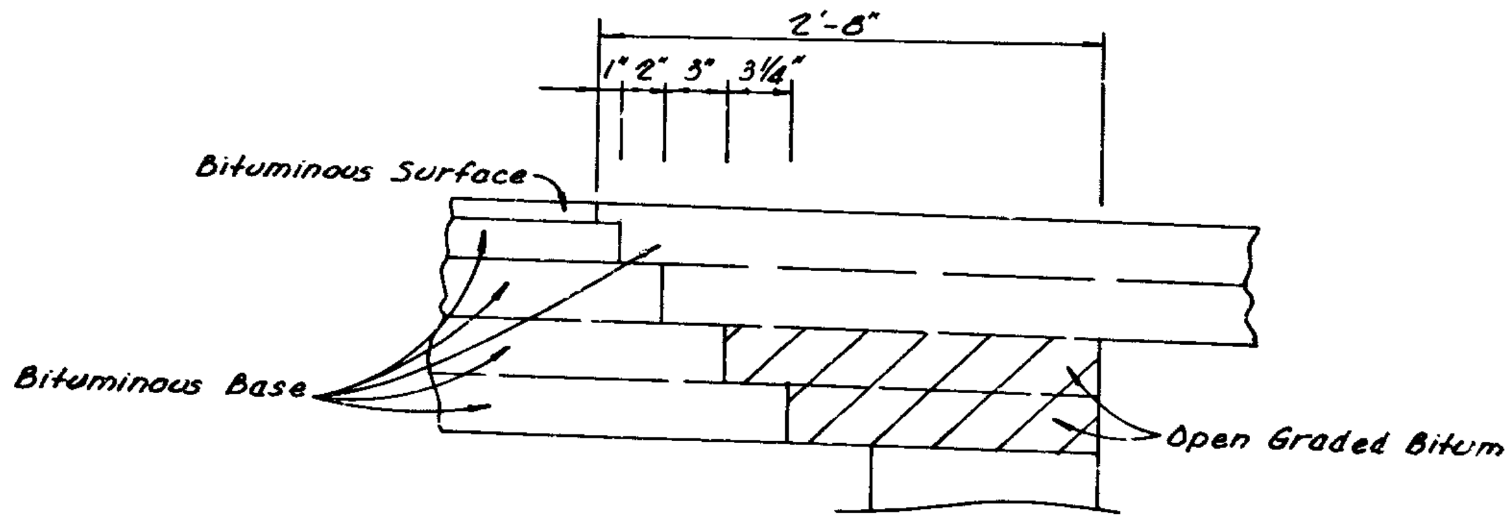
CURBED SHOULDER


℄ RUNAROUND
6'-0"
Variable
Variable
6'-0"
3/4"/ft
Slope 1/4"/ft
Slope 1/4"/ft
3/4"/ft
Slope 2:1
Slope 2:1
Tack Coat
110 #/sy. Bituminous Surface
*990#/sy Bituminous Base - SIZE 5D

PAVED RUNAROUND

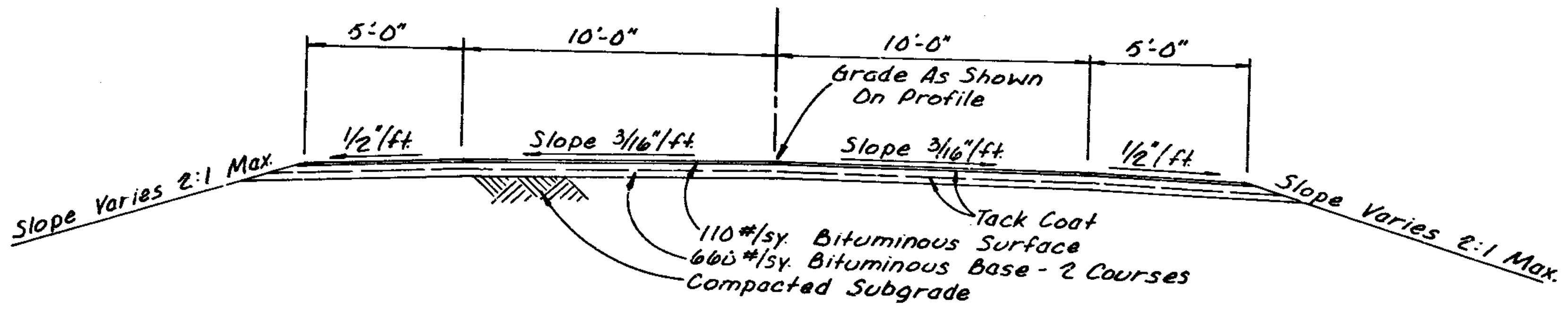
NOT TO SCALE

* This item to be placed if the use of the runaround is to be extended over the winter.


2'-8"
1" 2" 3" 3 1/4"
Bituminous Surface
Bituminous Base
Open Graded Bituminous Base N° 5

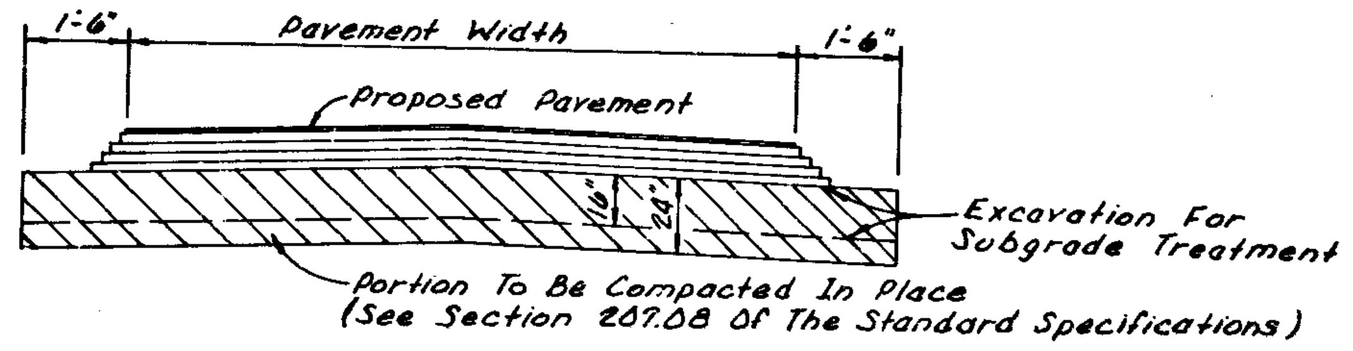
DETAIL "A"

NOT TO SCALE


5'-0"
10'-0"
10'-0"
5'-0"
Grade As Shown
On Profile
1/2"/ft
Slope 3/16"/ft
Slope 3/16"/ft
1/2"/ft
Slope Varies 2:1 Max.
Slope Varies 2:1 Max.
Tack Coat
110 #/sy. Bituminous Surface
660 #/sy. Bituminous Base - 2 Courses
Compacted Subgrade

TANGENT SECTION

STA. 50+12.87 TO STA. 51+00.00 "S-1-A"


1'-6"
Pavement Width
1'-6"
Proposed Pavement
16"
24"
Excavation For
Subgrade Treatment
Portion To Be Compacted In Place
(See Section 207.08 Of The Standard Specifications)

TYPICAL SECTION FOR SUBGRADE TREATMENT

NOT TO SCALE

TYPICAL CROSS SECTIONS

SCALE:- 1/4" = 1'-0"

RECOMMENDED FOR APPROVAL
5/16/80

No. 13050

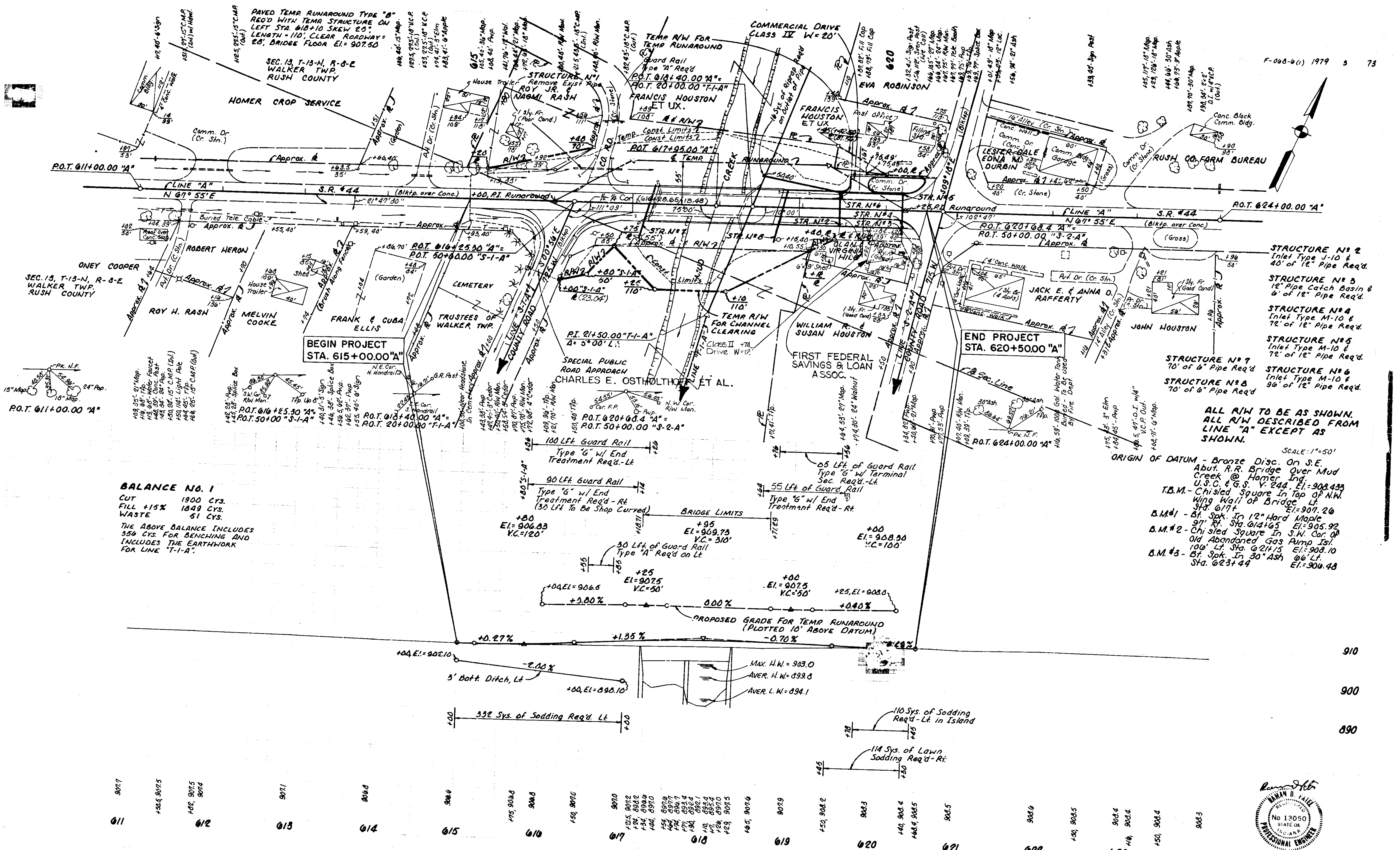
Rev.: 02-22-82 Paved Runaround
Rev.: 01-26-82 Do Not Spay Sign Detail and Notes Deleted.

Contr. No. B-13517

TP110

1975

PROJECT NO.	LINE	SHEET NO.	TOTAL SHEETS	FILE
F-068-6(1)		2	73	44-70-6634


PAVED TEMP. RUNAROUND TYPE "B" REQ'D WITH TEMP. STRUCTURE ON LEFT STA. 618+10 SKEW 25°. LENGTH = 110', CLEAR ROADWAY = 28', BRIDGE FLOOR El. = 907.50
SEC. 13, T-13-N, R-8-E
WALKER TWP.
RUSH COUNTY
HOMER CROP SERVICE
STRUCTURE Nº1
Remove Exist Pipe
ROY JR. & NAOMI RASH
TEMP R/W FOR TEMP RUNAROUND
Guard Rail Type "A" Req'd
P.O.T. 618+40.00 "A" = P.O.T. 20+00.00 "T-1-A"
FRANCIS HOUSTON ET UX.
COMMERCIAL DRIVE CLASS IV W=20'
CREEK
RUNAROUND
EVA ROBINSON
Post Office
LESTER DALE & EDNA M. DURBIN
Comm. Garage
RUSH CO. FARM BUREAU
Conc. Block Comm. Bldg.
P.O.T. 611+00.00 "A"
LINE "A"
N 67° 55' E
S.R. #44
(Blktp. over Conc.)
+00, P.I. Runaround
P.O.T. 617+95.00 "A"
Temp. Const. Limits
STR. Nº6
STR. Nº5
STR. Nº4
STR. Nº2
STR. Nº3
STR. Nº8
STR. Nº7
+25, P.I. Runaround
P.O.T. 620+68.4 "A" = P.O.T. 50+00.00 "S-2-A"
P.O.T. 624+00.00 "A"
(Grass)
Buried Tele. Cable
ROBERT HERON
ONEY COOPER
SEC. 13, T-13-N, R-8-E
WALKER TWP.
RUSH COUNTY
ROY H. RASH
MELVIN COOKE
P.O.T. 616+25.30 "A" = P.O.T. 50+00.00 "S-1-A"
FRANK & CUBA ELLIS
CEMETERY
TRUSTEES OF WALKER TWP.
LINE "S-1-A"
COUNTY ROAD
+00 "S-1-A"
+80 "S-1-A"
Const. Limits
TEMP R/W FOR CHANNEL CLEARING
Class II Drive W=12'
WILLIAM R. & SUSAN HOUSTON
ALAN A. & VIRGINIA HILL
FIRST FEDERAL SAVINGS & LOAN ASSOC.
LINE "S-2-A"
COUNTY ROAD
JACK E. & ANNA D. RAFFERTY
JOHN HOUSTON
BEGIN PROJECT STA. 615+00.00 "A"
END PROJECT STA. 620+50.00 "A"
P.I. 21+50.00 "T-1-A"
Δ = 5° 00' Lt.
SPECIAL PUBLIC ROAD APPROACH
CHARLES E. OSTHOLTHOFF ET AL.
P.O.T. 618+40.00 "A" = P.O.T. 20+00.00 "T-1-A"
P.O.T. 616+25.30 "A" = P.O.T. 50+00 "S-1-A"
P.O.T. 620+68.4 "A" = P.O.T. 50+00.00 "S-2-A"
P.O.T. 624+00.00 "A"
P.O.T. 611+00.00 "A"
Sec. Line
100 Lft. Guard Rail Type "G" w/ End Treatment Req'd - Lt.
90 Lft. Guard Rail Type "G" w/ End Treatment Req'd - Rt. (30 Lft. To Be Shop Curved)
65 Lft. of Guard Rail Type "G" w/ Terminal Sec. Req'd - Lt.
55 Lft. of Guard Rail Type "G" w/ End Treatment Req'd - Rt.
BRIDGE LIMITS
+80 El. = 906.83 V.C. = 120'
+95 El. = 909.73 V.C. = 310'
+00 El. = 908.90 V.C. = 100'
30 Lft. of Guard Rail Type "A" Req'd on Lt.
+25 El. = 907.5 V.C. = 50'
+00 El. = 907.5 V.C. = 50'
+04, El. = 906.6
+25, El. = 908.0
+3.80%
0.00%
+0.40%
PROPOSED GRADE FOR TEMP. RUNAROUND (PLOTTED 10' ABOVE DATUM)
+0.27%
+1.35%
-0.70%
+04, El. = 902.10
-2.00%
5' Bott. Ditch, Lt.
+60, El. = 898.10
MAX. H.W. = 903.0
AVER. H.W. = 899.8
AVER. L.W. = 894.1
332 Sys. of Sodding Req'd Lt.
110 Sys. of Sodding Req'd - Lt. in Island
114 Sys. of Lawn Sodding Req'd - Rt.
910
900
890
611
612
613
614
615
616
617
618
619
620
621
622
623
624

STRUCTURE Nº 2
Inlet Type J-10 & 40' of 12" Pipe Req'd.

STRUCTURE Nº 3
12" Pipe Catch Basin & 6' of 12" Pipe Req'd.

STRUCTURE Nº 4
Inlet Type M-10 & 72' of 12" Pipe Req'd.

STRUCTURE Nº 5
Inlet Type M-10 & 72' of 12" Pipe Req'd.

STRUCTURE Nº 6
Inlet Type M-10 & 96' of 12" Pipe Req'd.

STRUCTURE Nº 7
70' of 6" Pipe Req'd

STRUCTURE Nº 8
70' of 6" Pipe Req'd

ALL R/W TO BE AS SHOWN. ALL R/W DESCRIBED FROM LINE "A" EXCEPT AS SHOWN.

SCALE: 1" = 50'

ORIGIN OF DATUM - Bronze Disc. On S.E. Abut. R.R. Bridge Over Mud Creek @ Homer Ind. U.S.C. & G.S. Y-244, El. = 908.433
T.B.M. - Chisled Square In Top Of N.W. Wing Wall of Bridge Lt. Sta. 617+ El. = 907.26
B.M.#1 - Bt. Spk. In 12" Hard Maple 97' Rt. Sta. 614+65 El. = 905.92
B.M.#2 - Chisled Square In S.W. Cor. Of Old Abandoned Gas Pump Isl. 106' Lt. Sta. 621+15 El. = 908.10
B.M.#3 - Bt. Spk. In 30" Ash 66' Lt. Sta. 623+44 El. = 906.48

BALANCE NO. 1

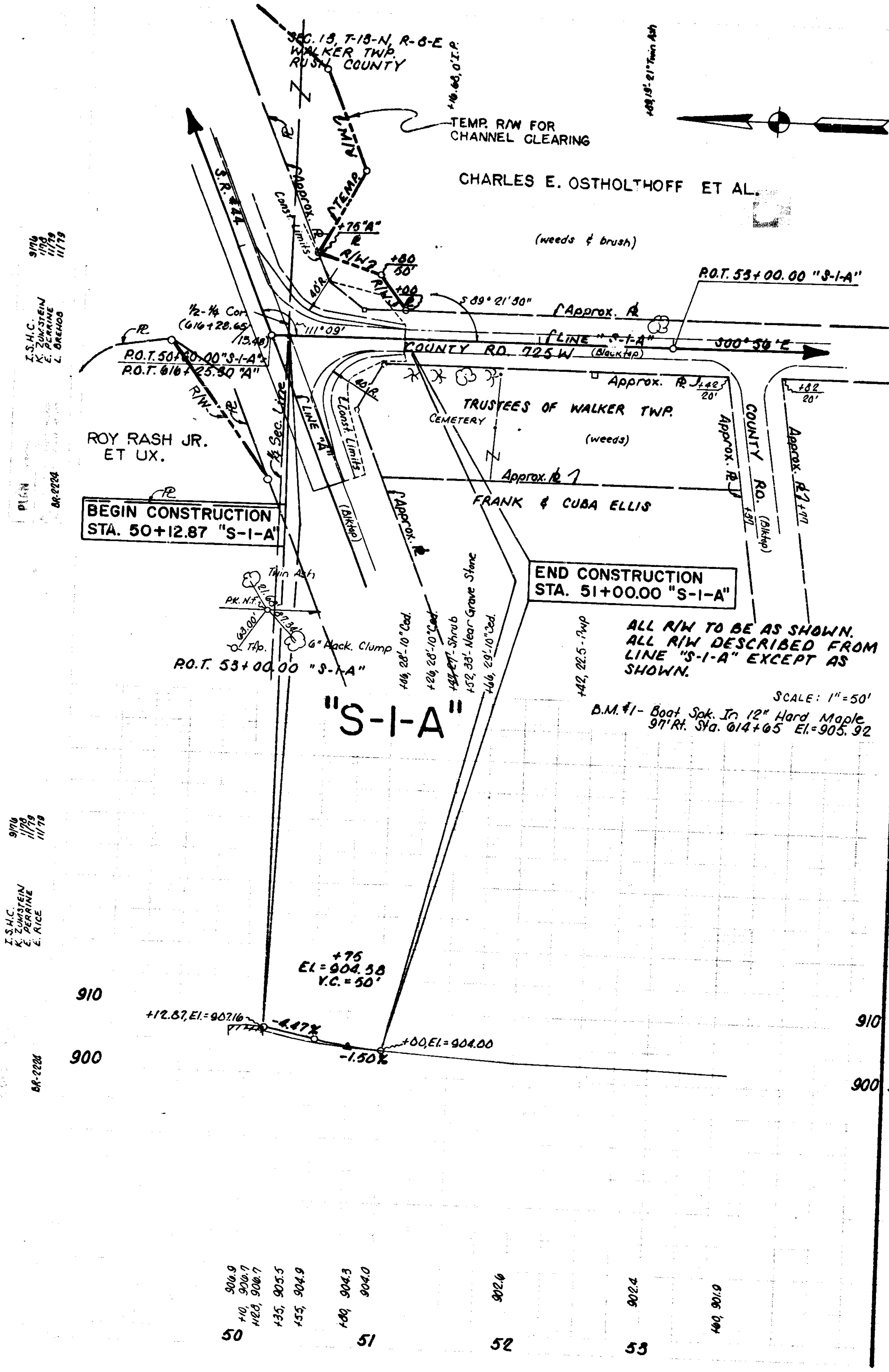
CUT	1900 CYS.
FILL +15%	1849 CYS.
WASTE	51 CYS.

THE ABOVE BALANCE INCLUDES 356 CYS. FOR BENCHING AND INCLUDES THE EARTHWORK FOR LINE "T-1-A".

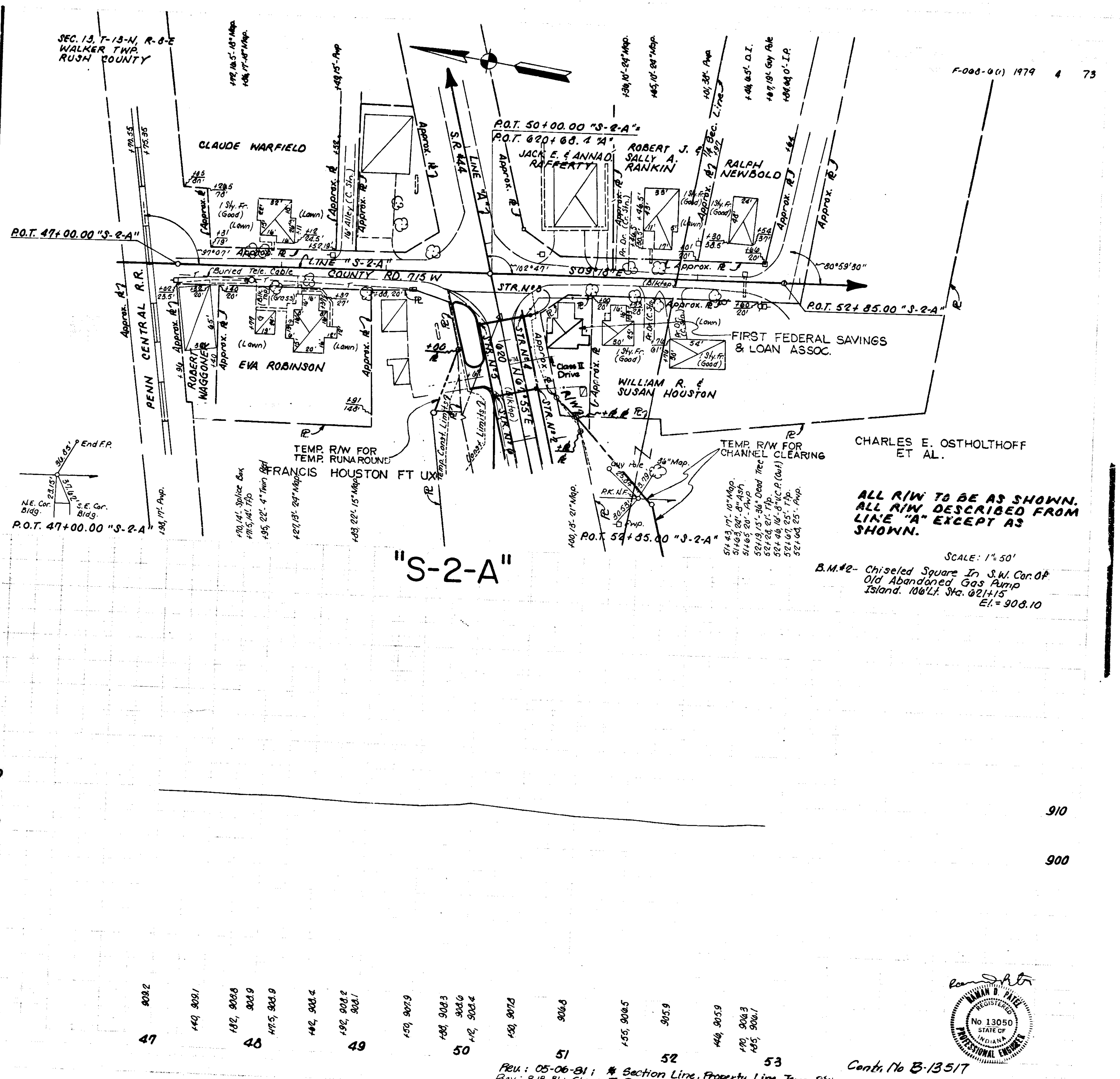
RAMAN D. PATEL
No. 13050

Rev: 05-06-81; ℄ Section Line, Property Line, Temp. R/W
Rev: 8-13-81; Class II Dr., 619+74 Rt., Type 'G' Guard Rail

Contr. No. B-13517


SEC. 13, T-13-N, R-8-E
WALKER TWP.
RUSH COUNTY
TEMP. R/W FOR CHANNEL CLEARING
CHARLES E. OSTHOLTHOFF ET AL.
(weeds & brush)
S.R. 244
TEMP. R/W
P.O.T. 53+00.00 "S-1-A"
LINE "S-1-A"
COUNTY RD. 725 W (Bleu.Hp)
P.O.T. 50+00.00 "S-1-A" =
P.O.T. 616+25.50 "A"
S 89° 21' 30" E
ROY RASH JR. ET UX.
TRUSTEES OF WALKER TWP.
CEMETERY
(weeds)
FRANK & CUBA ELLIS
COUNTY RD.
BEGIN CONSTRUCTION
STA. 50+12.87 "S-1-A"
END CONSTRUCTION
STA. 51+00.00 "S-1-A"
P.O.T. 53+00.00 "S-1-A"
"S-1-A"
SCALE: 1" = 50'
B.M. #1 - Boat Spk. In 12" Hard Maple 97' Rt. Sta. 614+65 El. = 905.92
+75
El. = 904.38
V.C. = 50'
+12.87, El. = 907.16
-4.47%
-1.50%
+00, El. = 904.00
910
900
50
51
52
53

ALL R/W TO BE AS SHOWN. ALL R/W DESCRIBED FROM LINE "S-1-A" EXCEPT AS SHOWN.


SEC. 13, T-13-N, R-8-E
WALKER TWP.
RUSH COUNTY
CLAUDE WARFIELD
P.O.T. 47+00.00 "S-2-A"
PENN CENTRAL R.R.
ROBERT WAGGONER
EVA ROBINSON
LINE "S-2-A"
COUNTY RD. 715 W
Buried Tele. Cable
S.R. 244
LINE "A"
P.O.T. 50+00.00 "S-2-A" =
P.O.T. 620+68.4 "A"
JACK E. & ANNA D. RAFFERTY
ROBERT J. & SALLY A. RANKIN
RALPH NEWBOLD
S 09° 13' E
P.O.T. 52+85.00 "S-2-A"
FIRST FEDERAL SAVINGS & LOAN ASSOC.
Class II Drive
WILLIAM R. & SUSAN HOUSTON
TEMP. R/W FOR TEMP. RUNAROUND
FRANCIS HOUSTON ET UX
TEMP. R/W FOR CHANNEL CLEARING
CHARLES E. OSTHOLTHOFF ET AL.
P.O.T. 47+00.00 "S-2-A"
P.O.T. 52+85.00 "S-2-A"
"S-2-A"
910
900
47
48
49
50
51
52
53

ALL R/W TO BE AS SHOWN. ALL R/W DESCRIBED FROM LINE "A" EXCEPT AS SHOWN.

SCALE: 1" = 50'

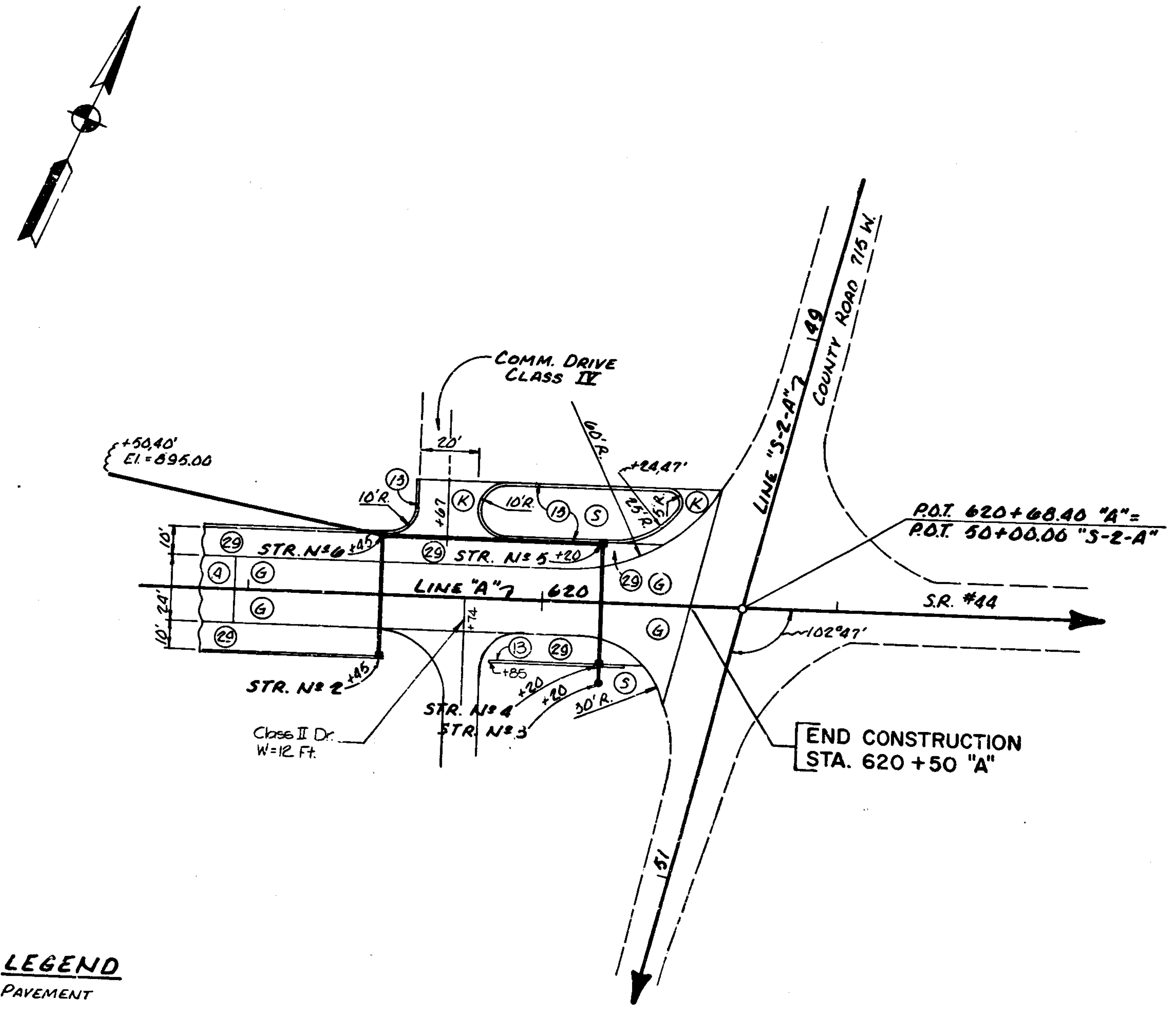
B.M. #2 - Chiseled Square In S.W. Cor. Of Old Abandoned Gas Pump Island. 106' Lt. Sta. 621+15 El. = 908.10

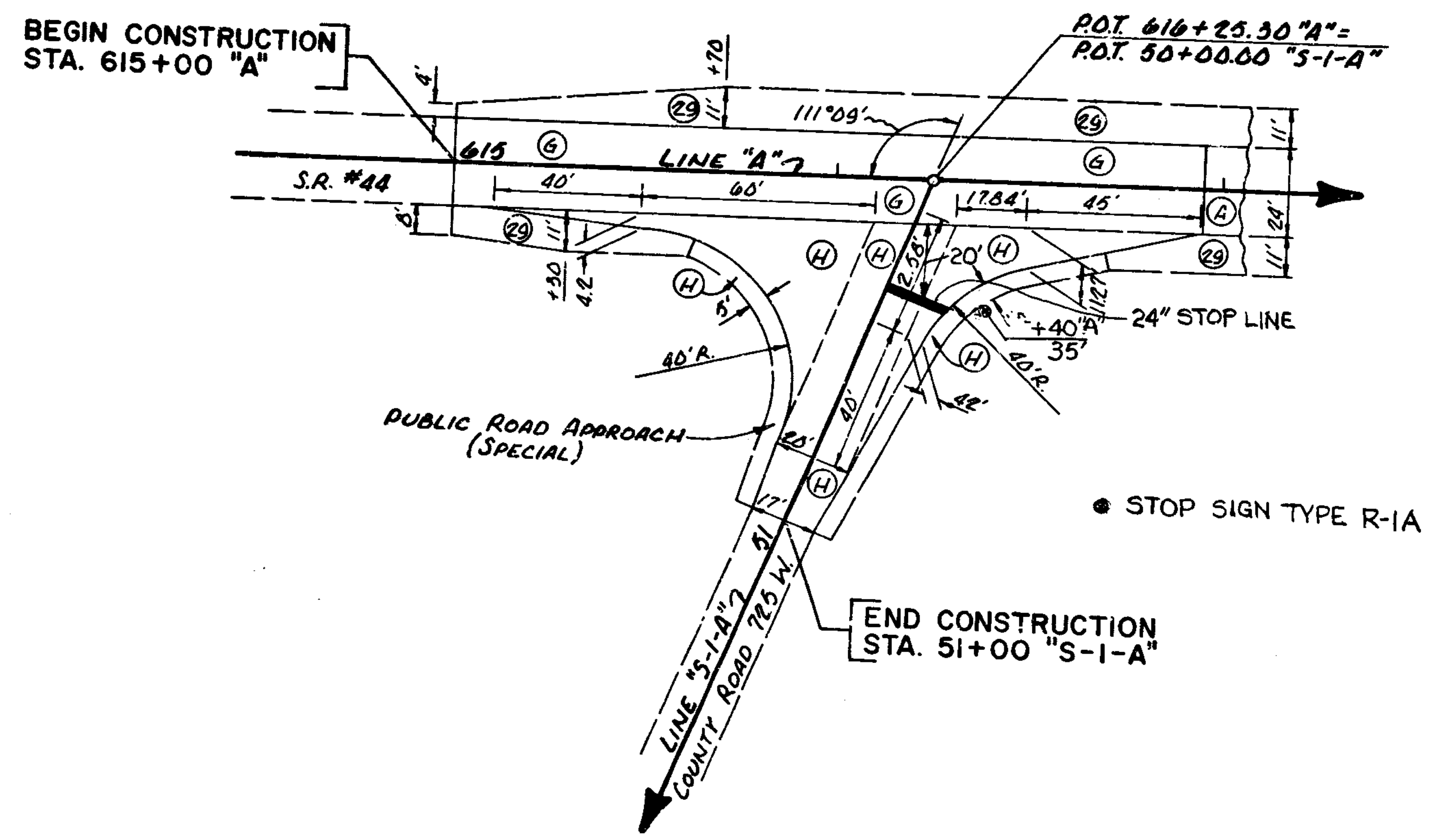
Raman D. Patel
No 13050

Rev: 05-06-81; ℄ Section Line, Property Line, Temp. R/W
Rev: 8-18-81, Class II Dr. 619+74 Rt.

Contr. No B-13517

FEDERAL ROAD REGION NO.	STATE	PROJECT NO.	FISCAL YEAR	SHEET NO.	TOTAL SHEETS
5	IND.	F-068-6(1)	1979	5	73


COMM. DRIVE CLASS IV
+50.40'
El. = 895.00
20'
10'R.
13
K
10'R.
13
S
60'R.
+24.47'
25'R.
5'R.
K
+67
29
STR. Nº 6 +45
29
STR. Nº 5 +20
A
G
G
29
LINE "A"
620
29
G
G
+74
13
29
+85
S
30'R.
STR. Nº 2 +45
STR. Nº 4 +20
STR. Nº 3 +20
Class II Dr.
W=12 Ft.
10'
24'
10'
LINE "S-2-A"
49
COUNTY ROAD 715 W.
P.O.T. 620+68.40 "A" =
P.O.T. 50+00.00 "S-2-A"
S.R. #44
102°47'
51
END CONSTRUCTION
STA. 620+50 "A"


BEGIN CONSTRUCTION
STA. 615+00 "A"
4'
+70
29
11'
111°09'
P.O.T. 616+25.30 "A" =
P.O.T. 50+00.00 "S-1-A"
29
615
G
LINE "A"
G
S.R. #44
40'
60'
G
17.84'
45'
A
11'
24'
11'
8'
29
11'
+30
4.2
H
H
H
2.58'
20'
H
29
12.7'
+40 "A"
35'
24" STOP LINE
5'
40'R.
H
40'R.
42'
PUBLIC ROAD APPROACH (SPECIAL)
20'
40'
17'
H
51
LINE "S-1-A"
COUNTY ROAD 725 W.
END CONSTRUCTION
STA. 51+00 "S-1-A"
● STOP SIGN TYPE R-1A

LEGEND

- Ⓐ 10" R.C. PAVEMENT
- Ⓖ 12½" FULL DEPTH FLEXIBLE PAVEMENT
- Ⓗ BITUMINOUS MIXTURE FOR APPROACHES - 7"
- Ⓚ BITUMINOUS MIXTURE FOR APPROACHES - 6"
- Ⓢ SODDING
- ⑬ CONCRETE CURB
- ㉙ BITUMINOUS SHOULDER - 6"

DETAILS

SCALE: 1" = 30'

Rev.: 2-22-82 STOP LINE, STOP SIGN TYPE R-1A

Rev: 8-18-81 ; Class II Dr. 619+74 Rt. ; Conc. Crub Limits

44-70-6634
B-13517

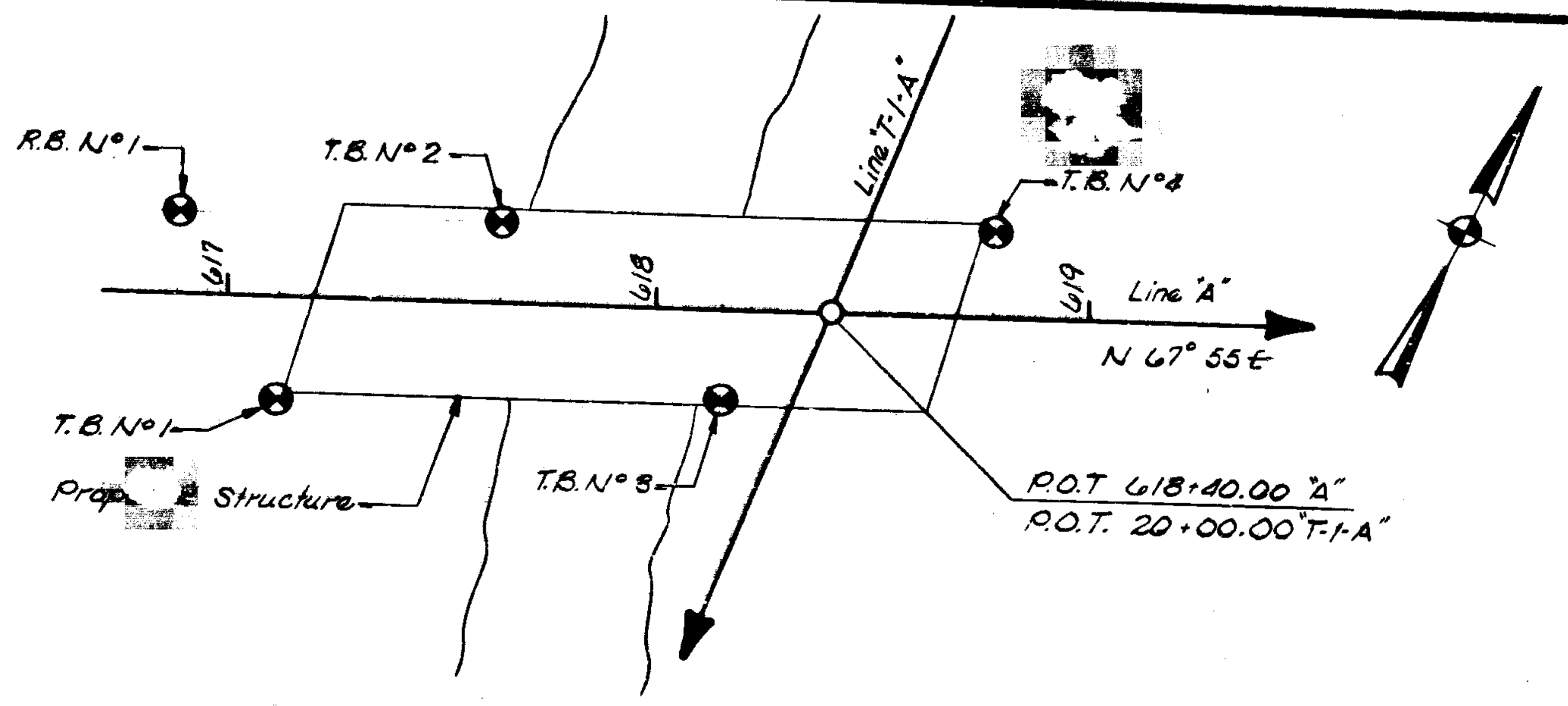
RAMAN D. PATEL
REGISTERED
No. 13050
STATE OF INDIANA
PROFESSIONAL ENGINEER

TP107

1976

F-068-6(1)

10-25-79 JCF
1:30 - 4:30 - 5
10-26-79 JCF
9:30 - 4:30 - 8
10-29-79 JCF
7.30 - 10.30 - 3



PLAN
Scale: 1" = 30'-0

TEST BORING N°	T.B. N° 1	T.B. N° 2	T.B. N° 3	T.B. N° 4	R.B. N° 1
STATION (A)	617+12	617+63	618+15	618+78	616+88
OFFSET	25' Rt.	20' Lt.	22' Rt.	21' Lt.	20' Lt.
SURFACE ELEV.	898.4	898.4	899.1	906.0	906.1

T.B. N° 1

Sample N°	Sample Elev.	N	Description
	898.4		Ground Level
	897.4		Dark brown moist soft silty clay loam (Top Soil)
1	895.9	2	
2	893.4	2	895.4 Brownish gray very moist very soft silty clay w/trace gravel (backfill for wingwall)
3_2	890.9_2	3_2	891.4 Gray moist very soft clay loam w/ sand seams (backfill for wingwall)
4	888.4	14	890.4 Gray wet very soft silty clay w/ little gravel (backfill for wingwall)
5	883.4	25	
6	878.4	22	Gray moist to slightly moist very stiff loam w/ trace to little gravel
7	873.4	20	Gray slightly moist very stiff silty loam w/ little gravel
	872.4		
8	868.4	41	Gray wet dense coarse sand w/ little gravel
9	863.4	46	
10	858.4	13	
	854.4		
11	853.4	93	Golden brown slightly moist hard sandy loam
12	849.7	[51 for .2']	
	847.0		Tan hard fine-grained slightly vuggy dolomatic limestone
	844.7		Light gray hard medium-grained limestone w/ styolite

End of Boring
Depth of Boring ~ 53.7'

T.B. N° 2

Sample N°	Sample Elev.	N	Description
	898.4		Ground Level
1	894.9	3	Brown very moist very soft silty clay w/ trace gravel
	893.4		
2	892.4	3	891.9 Brown wet very loose sand
3	889.9	1	889.4 Gray very moist very soft silty clay w/ leaves & wood
4	887.4	4	
	884.6		Gray very moist soft silty clay loam w/ little gravel & sticks of wood
5	882.4	17	
6	877.4	29	Gray moist very stiff loam w/ trace to little gravel (Till)
	875.9		
7	872.4	32	Multi-colored (clean) to gray wet dense to very dense sandy gravel
8	867.4	[84 for .9']	
	864.9		Brown moist hard silty clay loam w/ trace gravel (older weathered Till)
	863.7		
9	862.4	50	
	860.4		Golden brown wet dense fine sand
10	857.4	89	Golden brown slightly moist hard sand loam w/ trace gravel (older weathered Till)
11	852.4	32	El. 851.4
	850.4		Apprently hard limestone or dolomite bedrock

End of Boring
Depth of Boring ~ 48'

T.B. N° 3

Sample N°	Sample Elev.	N	Description
	899.1		Ground Level
1	896.1	5	Dark brown & gray very moist soft silty clay loam El. 895.6_2
2	893.6	3	Gray very moist very soft silty clay El. 893.1_2
3	891.1	13	Gray moist stiff silty clay loam w/ trace gravel El. 890.6_2
4	888.6	27	Gray slightly moist very stiff to hard loam w/ trace gravel (Till)
5	883.6	18	
6	878.6	41	
	875.5		
7	873.6	9	Gray wet loose fine sand w/ trace gravel (source of Artesian water)
	872.1		
8	868.6	34	Gray wet dense fine to coarse gravelly sand w/ thin layers of coarse gravel
	865.1		
9	863.6	35	Multi-colored clean wet dense coarse sand w/ trace gravel
	862.1		
10	858.6	53	Brown moist hard silty clay loam w/ trace to little gravel (older weathered Till)
	857.1		
11	853.6	76	Brown moist hard clay loam w/ little to some gravel
12	848.6	43	El. 848.1_2

End of Boring
Depth of Boring ~ 51'

T.B. N° 4

Sample N°	Sample Elev.	N	Description
	906.0		Ground Level
1	903.5	4	Brown moist soft to medium stiff silty clay loam w/ trace gravel (Fill)
2	901.0	8	
3	898.5	7	El. 898.0
4	896.0	7	Gray very moist medium stiff silty clay loam w/ thin layers of silty loam (natural ground)
	894.0		
5	891.0	10	Gray very moist to moist medium stiff to very stiff loam w/ trace gravel (Till)
6	886.0	10	
7	881.0	13	
8	876.0	25	
	874.0		
9	871.0	26	Gray wet medium dense coarse sand w/ little gravel
	866.8		Brown moist hard silty clay loam w/ trace gravel (old weathered fill) El. 865.0
10	866.0	82	
	862.3		Gray wet dense coarse sand w/ trace gravel
11	861.0	46	Light & dark brown mottled moist hard silty clay loam w/ trace gravel (older weathered fill)

End of Boring
Depth of Boring ~ 45'

R.B. N° 1

Sample N°	Sample Elev.	N	Description
	906.1		Ground Level
1_2	903.6_2	5_2	Brown moist soft silty clay loam (Fill) El. 904.1
2	901.1	4	Brown moist soft clay loam (Fill)
	899.6		
3	898.6	4	Gray moist to slightly moist soft to medium stiff silty clay (natural ground)
4	896.1	9	
5_2	893.6_2	4_2	El. 894.1_2
6	891.1	7	Brown & gray mottled moist soft to medium stiff silty clay loam w/ trace gravel El. 890.6
7	888.6	17	Gray moist very stiff loam w/ trace to little gravel (Till)
8	886.1	15	

End of Boring
Depth of Boring ~ 20'

"N" - Indicates the number of blows required to drive a 1⅜" I.D., 2" O.D. Split Spoon Sampler 12" by means of a 140 lb weight falling 30".

DESIGNED JLF C'K'D
DRAWN JLF 10-79 C'K'D DEA 2/80
TRACED C'K'D

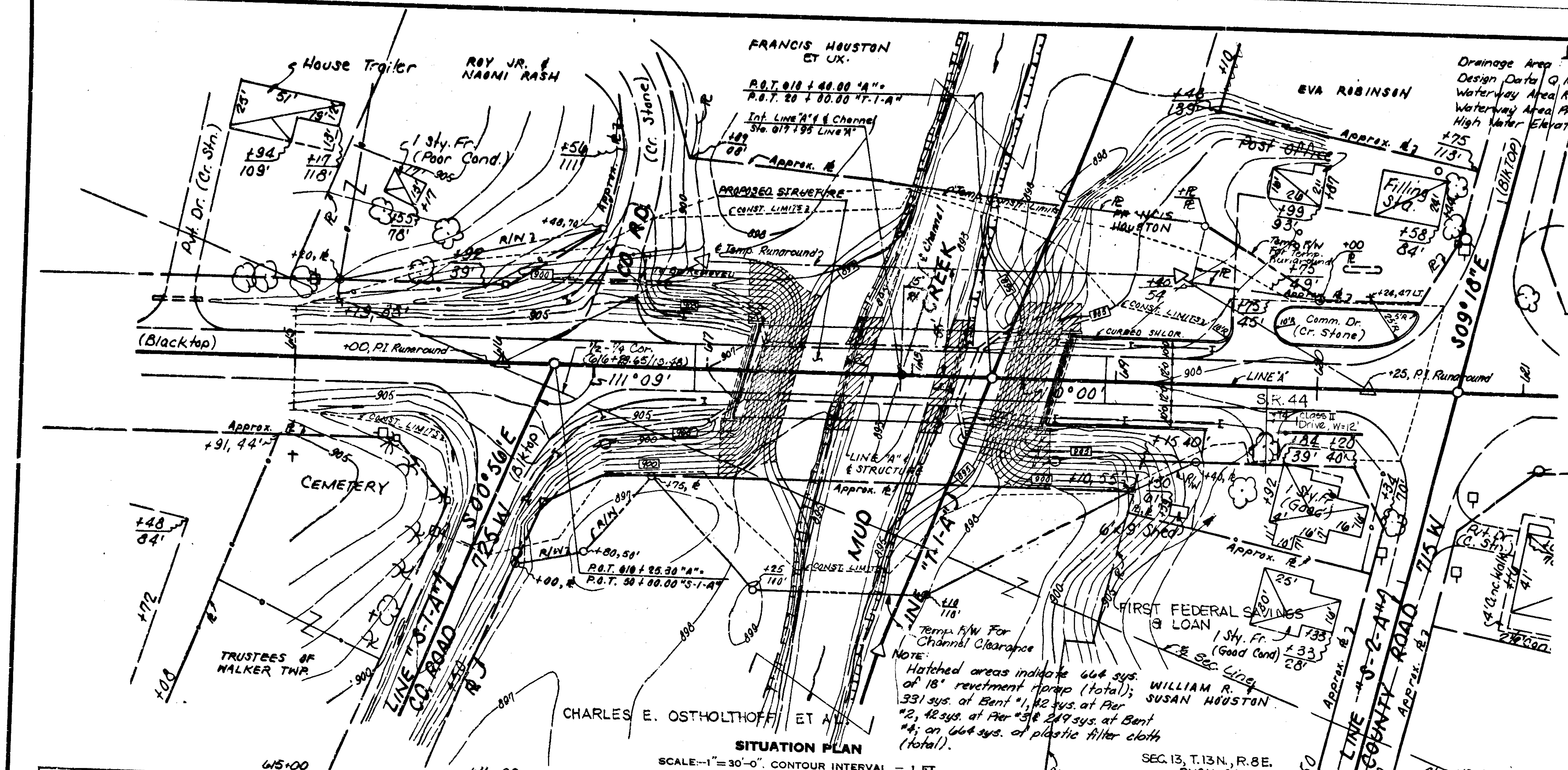
TP118

SOIL BORINGS

INDIANA STATE HIGHWAY COMMISSION

SCALE:- As Noted
DATE:- 5-16-80
Submitted for Approval
DRAWING:- OF
SHEET:- 6 OF 78
PROJECT:- F-068-6(1)
CONTRACT NO. B-13517
BRIDGE FILE:- 44-70-6634

RAMAN D. PATEL
No. 13050

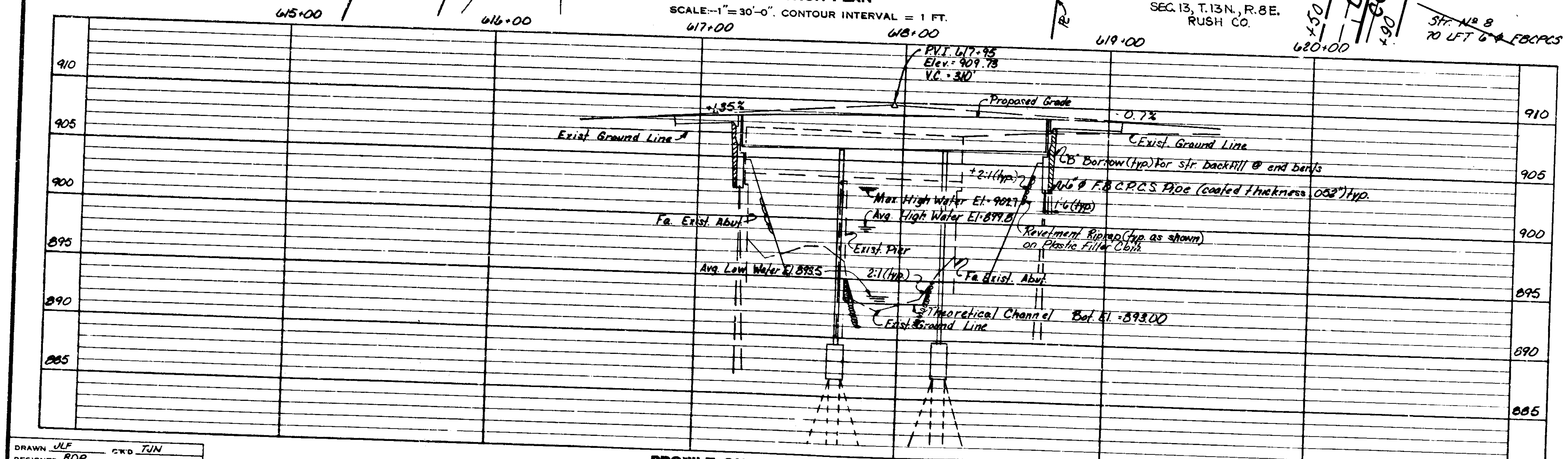

House Trailer
ROY JR. & NAOMI RASH
FRANCIS HOUSTON ET UX.
EVA ROBINSON
1 Sty. Fr. (Poor Cond.)
Pvt. Dr. (Cr. Stn.)
CO. RD. (Cr. Stone)
P.O.T. 618 + 40.00 "A" = P.O.T. 20 + 00.00 "T-1-A"
Int. Line "A" & ℄ Channel Sta. 617+95 Line "A"
PROPOSED STRUCTURE
℄ Temp. Runaround
Post Office
Filling Sta.
(Blacktop)
+00, P.I. Runaround
+25, P.I. Runaround
LINE "A"
S.R.44
MUD CREEK
CEMETERY
TRUSTEES OF WALKER TWP.
LINE "S-1-A" CO. ROAD 725 W (Blktop)
P.O.T. 618 + 95.30 "A" = P.O.T. 50 + 00.00 "S-1-A"
LINE "T-1-A"
Temp. R/W For Channel Clearance
FIRST FEDERAL SAVINGS & LOAN
1 Sty. Fr. (Good Cond)
WILLIAM R. & SUSAN HOUSTON
℄ Sec. Line
LINE "S-2-A" COUNTY ROAD 715 W
S09°18"E
(Blktop)
CHARLES E. OSTHOLTHOFF ET AL
Str. Nº 8 70 LFT 6' & F.B.C.P.S PIPE

NOTE:
Hatched areas indicate 664 sys. of 18" revetment riprap (total); 331 sys. at Bent #1, 42 sys. at Pier #2, 42 sys. at Pier #3 & 249 sys. at Bent #4; on 664 sys. of plastic filter cloth (total).

SITUATION PLAN

SCALE:-1"=30'-0". CONTOUR INTERVAL = 1 FT.

SEC. 13, T.13N., R.8E.
RUSH CO.


615+00
616+00
617+00
618+00
619+00
620+00
P.V.I. 617+95
Elev.=909.73
V.C.=310'
+1.35%
Proposed Grade
-0.7%
Exist. Ground Line
"B" Borrow (typ) for str. backfill @ end bents
12" & F.B.C.P.C.S. Pipe (coated thickness .003") typ.
Max. High Water El.=902.7
Avg. High Water El.=899.8
Revetment Riprap (typ. as shown) on Plastic Filter Cloth
Fa. Exist. Abut.
Exist. Pier
Avg. Low Water El. 893.5
2:1 (typ.)
Theoretical Channel Bot. El.=893.00
910
905
900
895
890
885

PROFILE ON PROPOSED ℄ ROADWAY

SCALES: HORIZ. 1"=30'-0" VERT. 1"=5'-0"

DRAWN JLF	C'K'D TJN		
DESIGNED RDP	C'K'D		
TRACED JLF	C'K'D DEA		

WATERWAY DATA

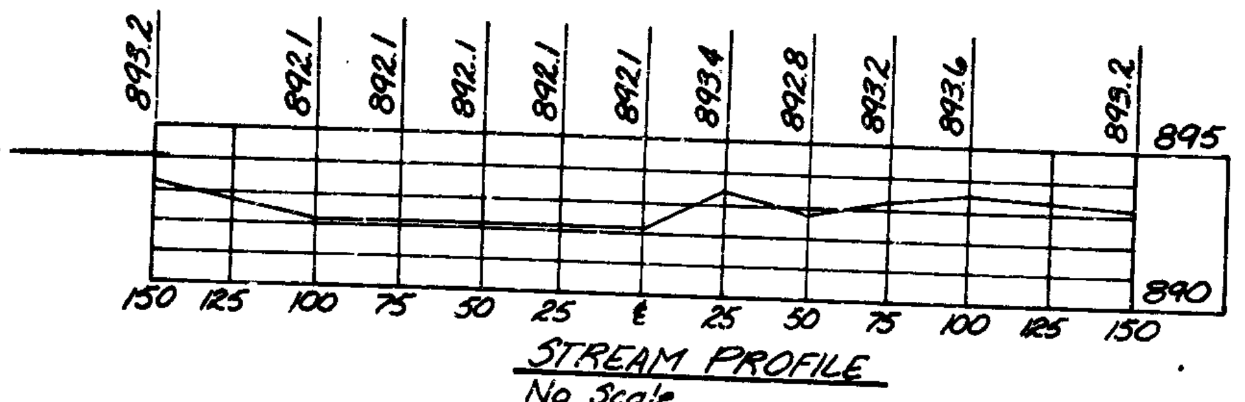
Drainage Area = 32.0 Sq. Miles
Design Data Q 100: = 2,600 C.F.S.
Waterway Area Req'd.: = 749 Sq. Ft.
Waterway Area Provided: = 806 Sq. Ft.
High Water Elevation Q 100: = 903.00

NOTE:
For Bench Mark data & reference points, see "Plan & Profile Sheet", Sheet Nº 3

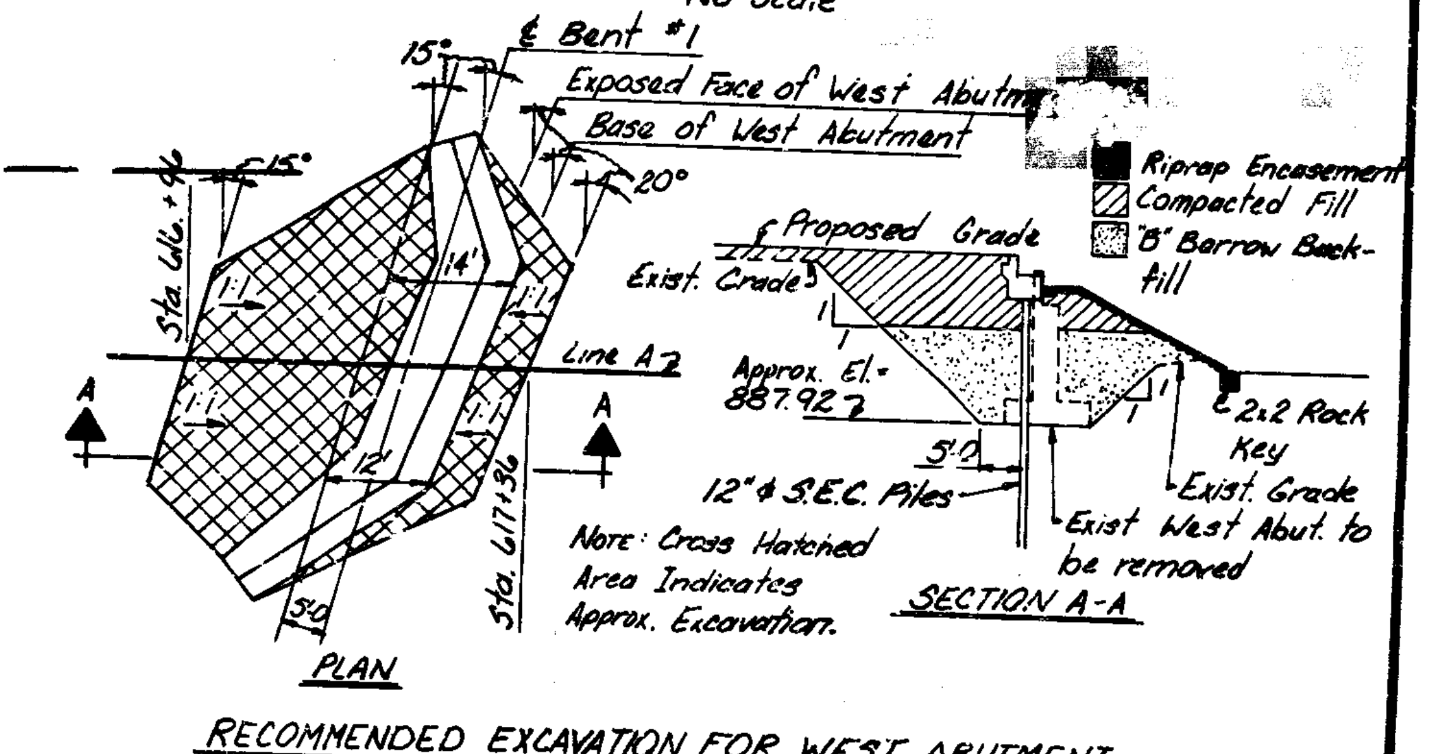
BRIDGES OVER 20' SPAN

PUB. ROAD REG. NO.	STATE	PROJECT NO.	FISCAL YEAR	SHEET NO.	TOTAL SHEETS
5	IND.	F-068-6(1)	1979	7	73

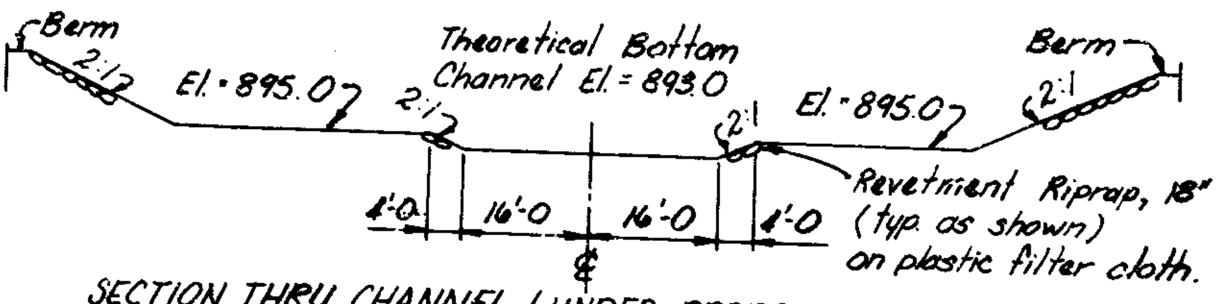
UTILITY OWNERS
ELECTRIC:
Indiana Public Service Co.
248 N. Main. Rushville, Ind.
TELEPHONE:
General Telephone Co.
Rushville, Ind.



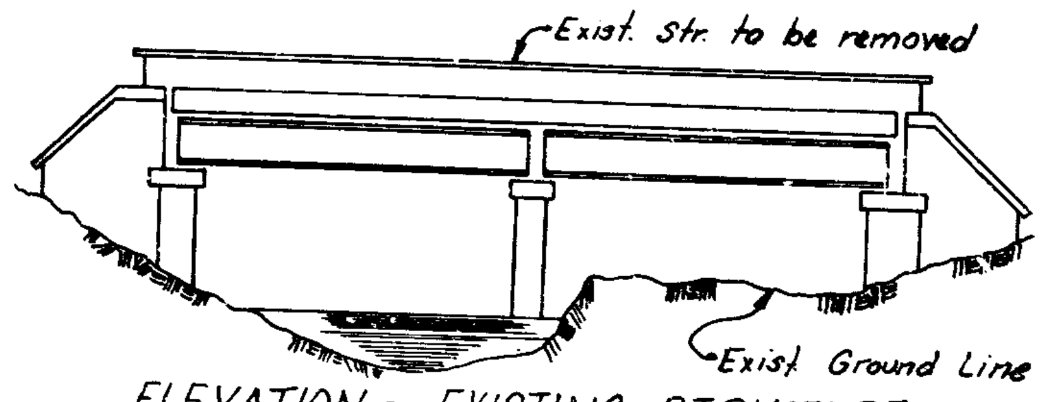
STREAM PROFILE
No Scale


℄ Bent #1
Exposed Face of West Abutment
Base of West Abutment
Riprap Encasement
Compacted Fill
"B" Borrow Backfill
Proposed Grade
Exist. Grade
Line A
Approx. El.=887.92
12" & S.E.C. Piles
2:2 Rock Key
Exist. Grade
Exist. West Abut. to be removed
PLAN
SECTION A-A
Note: Cross Hatched Area Indicates Approx. Excavation.

RECOMMENDED EXCAVATION FOR WEST ABUTMENT
No Scale


Berm
El.=895.0
Theoretical Bottom Channel El.=893.0
El.=895.0
Berm
2:1
4'-0
16'-0
16'-0
4'-0
℄
Revetment Riprap, 18" (typ. as shown) on plastic filter cloth.

SECTION THRU CHANNEL (UNDER PROPOSED STRUCTURE)
No Scale


Exist. Str. to be removed
Exist. Ground Line

ELEVATION ~ EXISTING STRUCTURE
No Scale

LAYOUT

CONTINUOUS COMPOSITE PRESTRESSED CONCRETE I-BEAM
BRIDGE: 3 SPANS 49'-3, 50'-0, 49'-3
SKEW=15° RIGHT
44'-0 CL. ROADWAY
S.R.44 OVER MUD CREEK

INDIANA STATE HIGHWAY COMMISSION

RUSH COUNTY

SCALE:-AS NOTED

5-6-1980

SUBMITTED FOR APPROVAL: Raman D. Patel

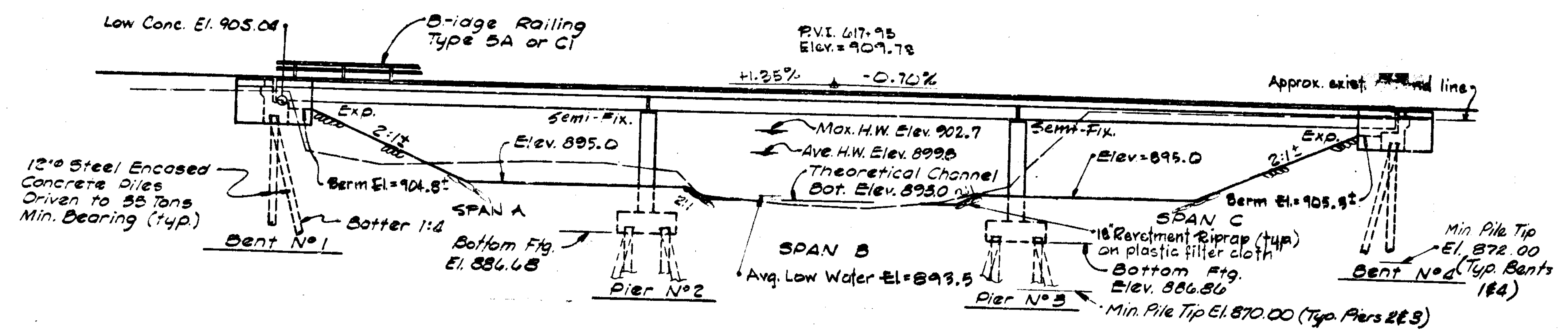
DRAWING:- C1 OF C8
SHEET 7 OF 73
PROJECT:- F-068-6(1)
STATION:-
BRIDGE CONTRACT NO. B-13517
BRIDGE FILE:- 44-70-6634

RAMAN D. PATEL
No 13050

NOTE: FIELD NOTES. BOOK BR 2224

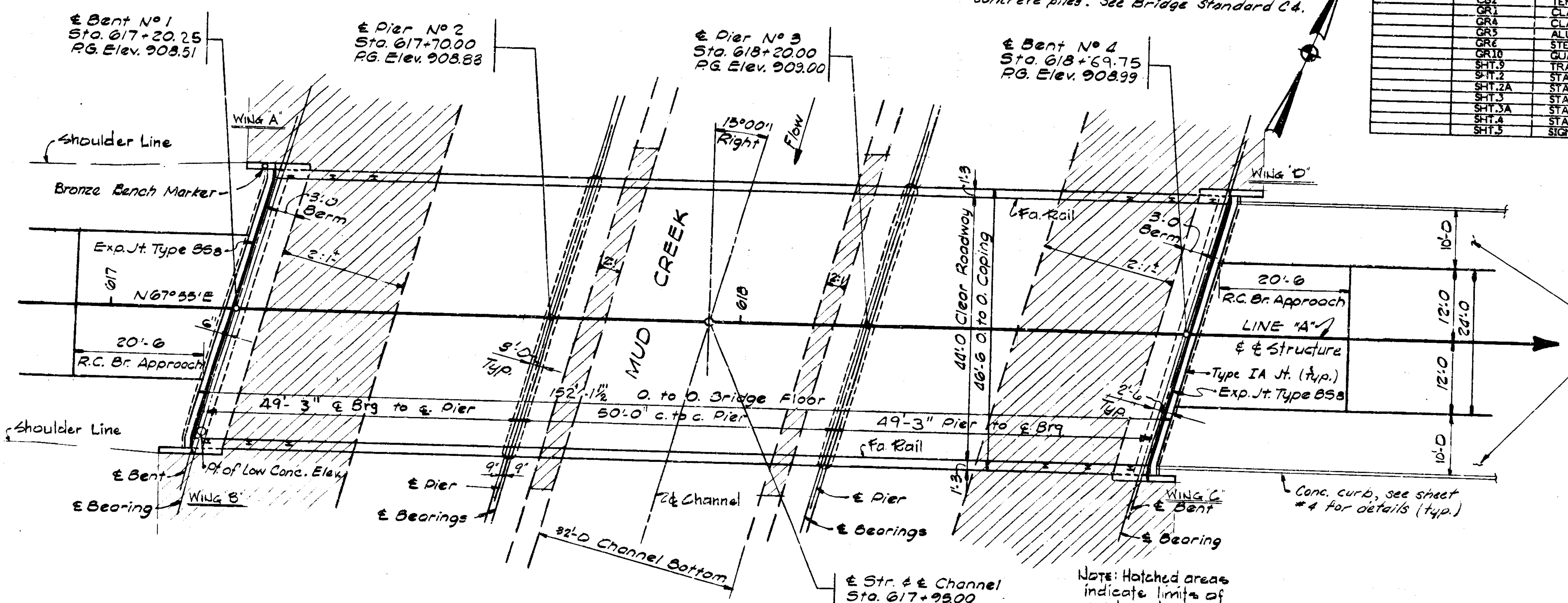
Rev.: 05-06-81; ¼ Section Line, Property Line, Temp. R/W
Rev: 8-18-81; Class II Dr. 619+74 Rt; Constr. Limits

STRUCTURE TO BE BUILT TO A 810 FT. VERTICAL CURVE


Low Conc. El. 905.04
Bridge Railing Type 5A or C1
P.V.I. 617+95
Elev. = 909.78
+1.35%
-0.70%
Approx. exist. ground line
Exp.
2:1±
Semi-Fix.
Elev. 895.0
Max. H.W. Elev. 902.7
Ave. H.W. Elev. 899.8
Theoretical Channel Bot. Elev. 893.0
Semi-Fix.
Elev. = 895.0
2:1±
Exp.
12"⌀ Steel Encased Concrete Piles Driven to 35 Tons Min. Bearing (typ.)
Berm El. = 904.8±
Batter 1:4
SPAN A
Bent N° 1
Bottom Ftg. El. 886.68
SPAN B
Pier N° 2
Avg. Low Water El = 893.5
18" Revetment Riprap (typ.) on plastic filter cloth
Bottom Ftg. Elev. 886.86
Pier N° 3
Min. Pile Tip El. 870.00 (Typ. Piers 2 & 3)
SPAN C
Berm El. = 905.8±
Min. Pile Tip El. 872.00 (Typ. Bents 1 & 4)
Bent N° 4

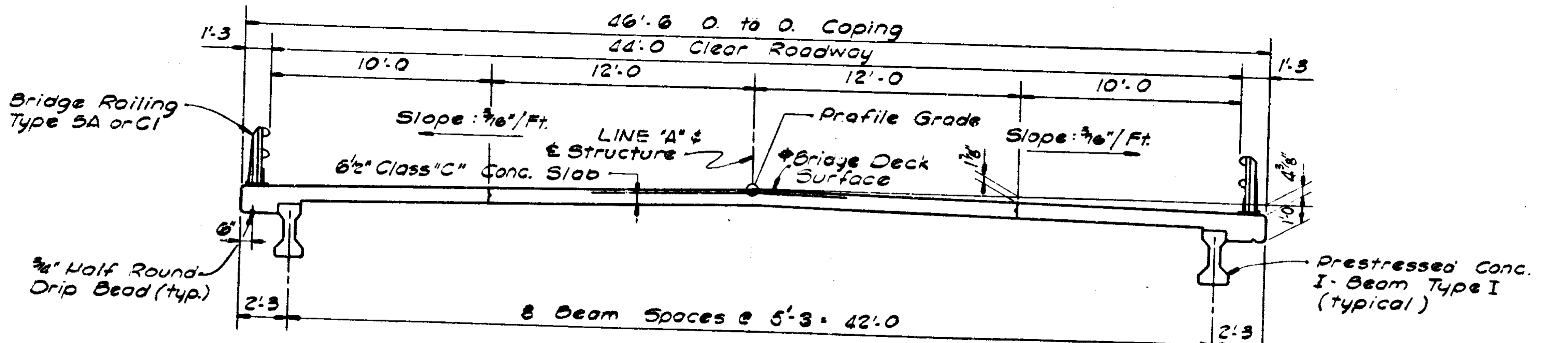
ELEVATION

Note: As an alternate, precast, prestressed concrete piles may be used in place of steel encased concrete piles. See Bridge Standard C4.


℄ Bent N° 1
Sta. 617+20.25
P.G. Elev. 908.51
℄ Pier N° 2
Sta. 617+70.00
P.G. Elev. 908.88
℄ Pier N° 3
Sta. 618+20.00
P.G. Elev. 909.00
℄ Bent N° 4
Sta. 618+69.75
P.G. Elev. 908.99
15°00' Right
Flow
WING 'A'
Shoulder Line
Bronze Bench Marker
3'-0 Berm
2:1
Exp. Jt. Type 85a
N67°55'E
617
20'-6
R.C. Br. Approach
MUD CREEK
618
44'-0 Clear Roadway
46'-6 O. to O. Coping
1'-3
Fa. Rail
WING 'D'
3'-0 Berm
2:1
20'-6
R.C. Br. Approach
LINE "A"
℄ ℄ Structure
Type IA Jt. (typ.)
Exp. Jt. Type 85a
10'-0
12'-0
12'-0
10'-0
44'-0
Curbed shoulders, see sheet 2 for "Typical Section"
8'-0 Typ.
52'-1½ O. to O. Bridge Floor
2'-6 Typ.
49'-3" ℄ Brg to ℄ Pier
50'-0 c. to c. Pier
49'-3" Pier to ℄ Brg
Fa. Rail
Shoulder Line
℄ Bent
℄ Bearing
P.I. of Low Conc. Elev.
WING 'B'
9'
9'
℄ Pier
℄ Bearings
℄ Channel
32'-0 Channel Bottom
℄ Pier
℄ Bearings
℄ Str. & ℄ Channel
Sta. 617+95.00
WING 'C'
℄ Bent
℄ Bearing
Conc. curb, see sheet #4 for details (typ.)

Note: Hatched areas indicate limits of revetment riprap (for limits of riprap, see sheet #7)

PLAN


46'-6 O. to O. Coping
44'-0 Clear Roadway
1'-3
1'-3
10'-0
12'-0
12'-0
10'-0
Bridge Railing Type 5A or C1
Slope: 3/16"/Ft.
LINE "A" ℄
℄ Structure
Profile Grade
Slope: 3/16"/Ft.
6½" Class "C" Conc. Slab
✦ Bridge Deck Surface
6"
3/4" Half Round Drip Bead (typ.)
Prestressed Conc. I-Beam Type I (typical)
2'-3
8 Beam Spaces @ 5'-3 = 42'-0
2'-3

TYPICAL SECTION

Scale: 1/4" = 1'-0

✦ Bridge Deck Surface 1½" Mod. P.C.C. or 2¼" Dense Concrete

STANDARD DRAWINGS

BRIDGE	ROAD	PURPOSE
BR1		ALUMINUM RAILING - TYPE 5A
BR2		ALUMINUM RAILING DETAILS
BR3		STEEL RAILING - TYPE C1
BR4		STEEL RAILING DETAILS
C1		PILES, REINF. BAR NOTES
C3		TYPA IA JOINT, TYPE "A" CONSTR. JT.
C4		PRESTRESSED PRECAST CONC. PILES
M2		TEMPORARY RUNAROUND
PB1		TYPE I CONC. I BEAM
PB10		I-BEAM DETAILS
PB11		ELASTOMERIC BEARING PAD TYPE I
S1		B-BORROW FOR STR. BACKFILL
T, SHT.A		STANDARD TEMPORARY BRIDGE
T, SHT.B		STANDARD TEMPORARY BRIDGE
	A	STANDARD PAVEMENT JOINTS
	MA	R.C. APPROACH DETAILS
	MC	CASTING DETAILS
	MC1	CASTING DETAILS
	MD	INLETS
	MD1	MANHOLES
	ME	CURB
	ME2	PIPE E[illegible]SECTIONS
	M11	TEMPORARY RUNAROUND
	MN	STRUCTURE BACKFILL, UNDERDRAIN
	MP	MISC. PIPE DETAILS
	CB2	TEMPORARY CONC. BARRIER
	GR1	CLASS "A" GUARD RAIL
	GR4	CLASS "G" GUARD RAIL
	GR5	ALUMINUM GUARD RAIL DETAILS
	GR6	STEEL TUBE GUARD RAIL DETAILS
	GR10	GUARD RAIL BURIED ENDS
	SHT.9	TRAFFIC SIGN DETAILS
	SHT.2	STANDARD DETOUR SIGNS
	SHT.2A	STANDARD DETOUR SIGNS
	SHT.3	STANDARD DETOUR SIGNS
	SHT.3A	STANDARD DETOUR SIGNS
	SHT.4	STANDARD DETOUR SIGNS
	SHT.5	SIGN DESIGN DETAILS

DESIGN DATA

DESIGNED FOR HS20-44 LOADING IN ACCORDANCE WITH 1977 A.A.S.H.T.O. SPECIFICATIONS AND 1978 INTERIMS.

DEAD LOAD: ACTUAL LOAD PLUS 35 LBS. PER SQ. FT. FOR FUTURE WEARING SURFACE.

GENERAL NOTES

EXISTING STRUCTURE IS A TWO SPAN STEEL BEAM BRIDGE TO BE REMOVED.

DEPTH OF FOOTINGS TO BE EXTENDED IF FOUND NECESSARY. SEE ARTICLE 206.11 (C) OF SPECIFICATIONS.

PILES SHALL HAVE MINIMUM BEARING VALUE SHOWN ON DETAIL DRAWINGS. DETERMINE PILE LENGTHS BY ARTICLE 701.02 (A) OF SPECIFICATIONS.

FOR DETAILS OF STEEL ENCASED CONCRETE PILES, OR PRESTRESSED CONCRETE PILES AS AN ALTERNATE, SEE BRIDGE STANDARD C1, AND C4.

PILES SHALL BE DRIVEN TO ELEVATION SHOWN ON PLANS OR BELOW IF NECESSARY TO OBTAIN REQUIRED BEARINGS.

REINFORCING STEEL COVERING SHALL BE 2-1/2" IN TOP AND 1" MIN. IN BOTTOM OF FLOOR SLABS, 3" IN FOOTINGS EXCEPT BOTTOM STEEL WHICH SHALL BE 4", AND 2" IN ALL OTHER PARTS, UNLESS NOTED.

CONCRETE IN FOOTINGS TO BE CLASS "B".

CONCRETE IN SUPERSTRUCTURE TO BE CLASS "C."

CONCRETE IN END BENTS, BENT CAPS, WINGS PIER STEM IN PRECAST CONCRETE PILES, AND IN STEEL ENCASED CONCRETE PILES TO BE CLASS "A".

CONCRETE IN STRUCTURE NOT NOTED ABOVE TO BE CLASS "A".

CONTINUOUS CONCRETE POURS SHALL BE REQUIRED BETWEEN CONSTRUCTION JOINTS AS SHOWN IN DETAIL PLANS.

WATERPROOF BACKS OF MUDWALLS, AND WINGWALLS IN ACCORDANCE WITH ARTICLE 702.22 OF THE SPECIFICATIONS.

BEVEL FORMS 1/4" UNDER COPINGS; AND CHAMFER EXPOSED EDGES 1" UNLESS NOTED.

RIPRAP AT LOCATIONS SHOWN ON THE LAYOUT.

TOLERANCE IN POSITION OF PILE HEAD MAXIMUM 2".

ONLY THE TOP OF PIER AND END BENT CAPS, FRONT FACE OF MUDWALLS, FACE OF END DIAPHRAGMS, FACE OF DECK COPING, AND UNDERSIDE OF THE BRIDGE FLOOR FROM COPING FACE ON OUTSIDE BEAM, OUTSIDE FACE OF EXTERIOR CONCRETE BEAMS TO BE SEALED.

THE QUANTITY CROWN-VETCH SEEDING, SHOWN ON THE ESTIMATE OF QUANTITIES SHEET IS TO BE USED AT THOSE LOCATIONS WHERE THE SLOPES ARE 3:1 OR STEEPER OR IN AN AREA REQUIRING SAND CUT OR SAND FILLS OR AS DIRECTED BY THE ENGINEER.

FOR SODDING QUANTITIES, SEE TABLE ON SHEET NO. 15.

GENERAL PLAN
CONT. COMPOSITE PRESTR. CONC. I-BEAM BRIDGE
3 SPANS 49'-3, 50'-0, 49'-3 SKEW: 15°00 Rt.
44'-0 CLEAR ROADWAY
S.R. 44 OVER MUD CREEK

INDIANA STATE HIGHWAY COMMISSION
RUSH COUNTY

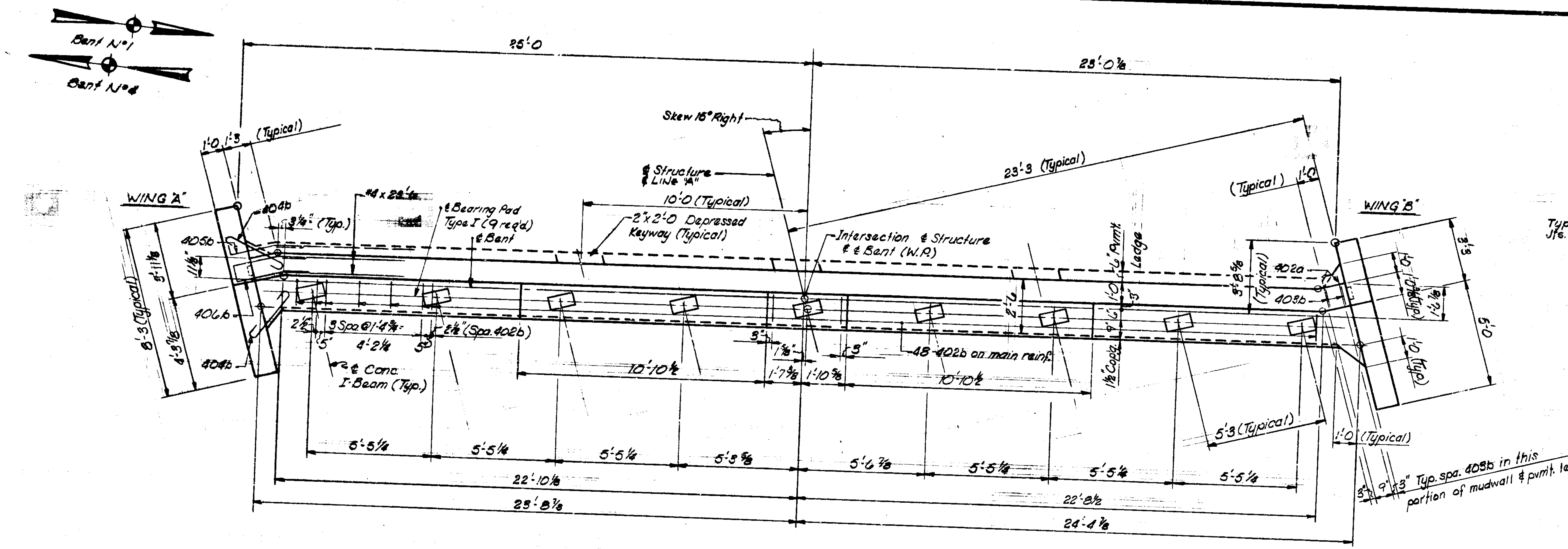
SCALE:- 3/32" = 1'-0 Unless Noted DATE:- 5-16-80
SUBMITTED FOR APPROVAL Raman D. P.
DRAWING:- 02 OF 08 SHEET:- 8 OF 78
PROJECT:- F-068-6(1)
CONTRACT NO. B-13517
BRIDGE FILE:- 44-70-6634

RAMAN D. PATEL
No. 13050
STATE OF INDIANA

DESIGNED RDP 3-78 C'K'D
DRAWN J.E. C'K'D TJN
TRACED C'K'D

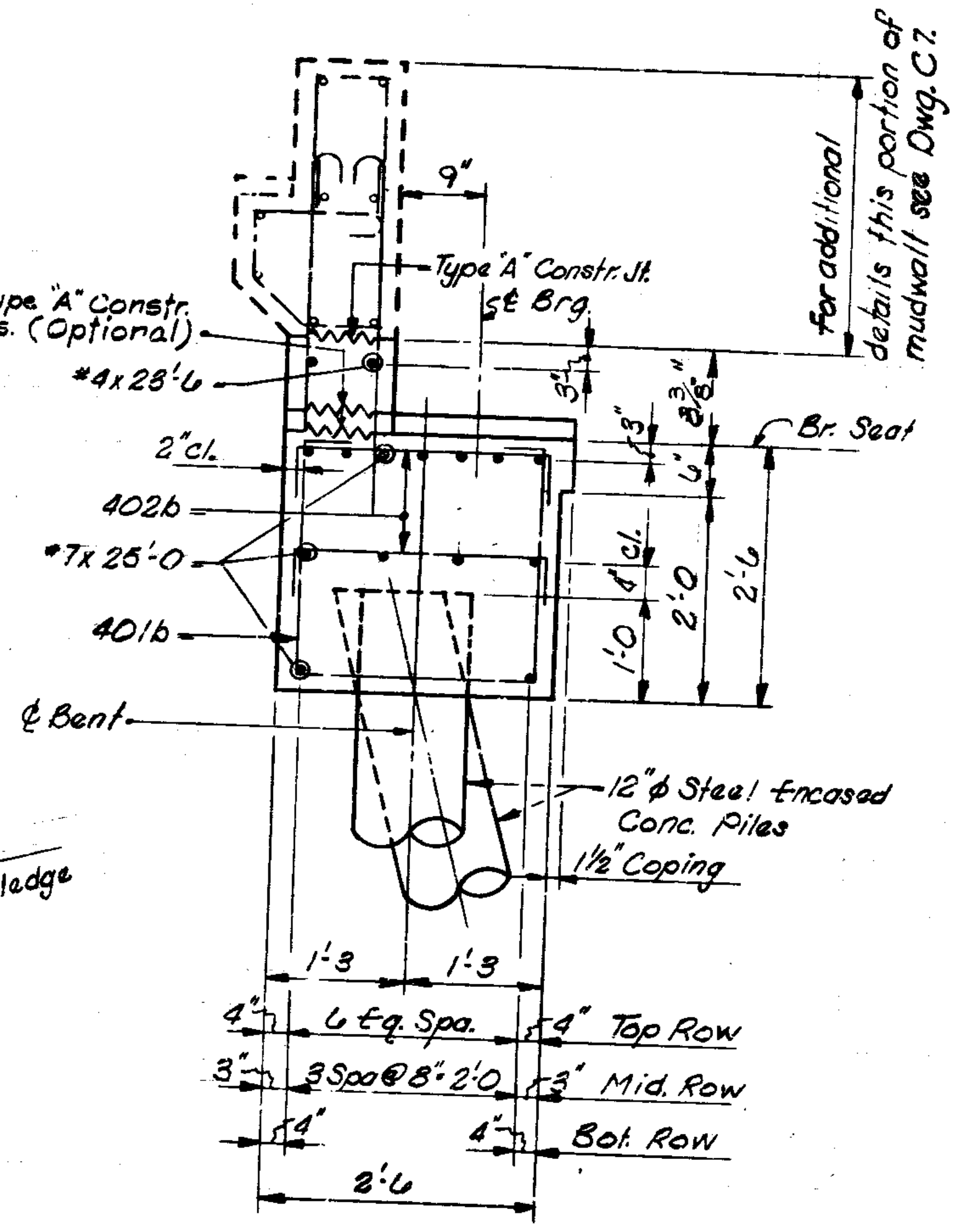
Rev.: 02-22-82 General Notes
Rev.: 01-26-82 General Notes

10-9-79 JLF
10:00-4:30-5½
10-9-79 JLF
7:30-[illegible]
10-10-79 JLF
7:30-4:30-8
10-11-79 JLF
7:30-[illegible]
10-12-79 JLF
7:30-4:00=7½
10-23-79 JLF
10:30-2:00-2½



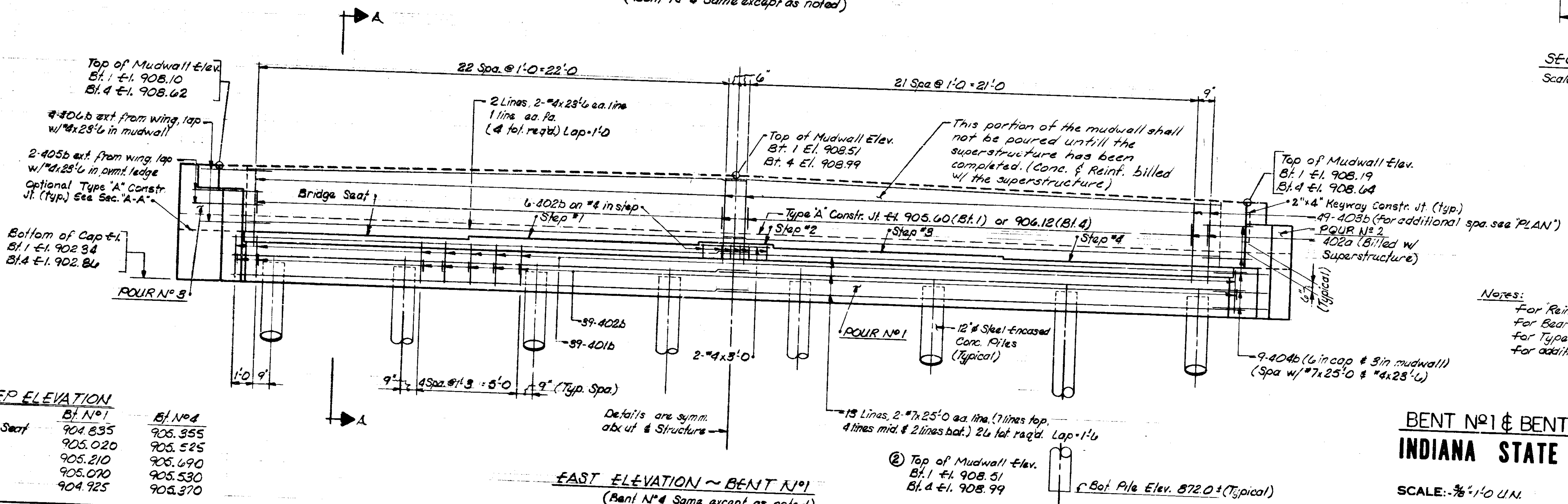
Bent No1
Bent No4
25'-0
23'-0⅛
Skew 16° Right
¢ Structure
¢ Line "A"
10'-0 (Typical)
2"x2'-0 Depressed Keyway (Typical)
Intersection ¢ Structure & ¢ Bent (W.P.)
23'-3 (Typical)
(Typical) 1'-0
1'-0 1'-3 (Typical)
WING "A"
WING "B"
404b
405b
406b
404b
3¼" (Typ.)
#4x23'-6
¢ Bearing Pad Type I (9 req'd)
¢ Bent
8'-3 (Typical)
5'-11⅞
4'-3⅛
1'-1⅛"
2½
3 Spa @ 1'-4¾
5"
4'-2¼
5"
2½" (Spa. 402b)
¢ Conc. I-Beam (Typ.)
10'-10½
3"
1⅝"
1'-7⅝
1'-10⅝
3"
48-402b on main reinf.
10'-10½
2'-6
1'-0
9"-6
1½" Copg.
1'-6" Pvmt. Ledge
3"
3'-8⅝ (Typical)
402a
403b
3'-3
1'-0 (Typ.)
1'-2⅛
1'-0 (Typ.)
5'-0
5'-3 (Typical)
1'-0 (Typical)
3" 9" 3" Typ. spa. 403b in this portion of mudwall & pvmt. ledge
5'-5¼
5'-5¼
5'-5¼
5'-3⅝
5'-6⅞
5'-5¼
5'-5¼
5'-5¼
22'-10⅛
23'-8⅜
22'-8½
24'-4⅞

PLAN ~ BENT No1
(Bent No4 Same except as noted)



For additional details this portion of mudwall see Dwg. C7.
9"
Type "A" Constr. Jt. & ¢ Brg.
Type "A" Constr. Jts. (Optional)
#4x23'-6
2"cl.
402b
#7x25'-0
401b
¢ Bent
3"
8¾"
3"
6"
Br. Seat
4"cl.
1'-0
2'-0
2'-6
12"ϕ Steel Encased Conc. Piles
1½" Coping
1'-3
1'-3
4"
6 Eq. Spa.
4" Top Row
3"
3 Spa @ 8" = 2'-0
3" Mid. Row
4"
4"
Bot. Row
2'-6

SECTION A-A
Scale: ¾"=1'-0



A
A
Top of Mudwall Elev. Bt. 1 El. 908.10 Bt. 4 El. 908.62
22 Spa. @ 1'-0 = 22'-0
6"
21 Spa @ 1'-0 = 21'-0
9"
4-406b ext. from wing, lap w/ #4x23'-6 in mudwall
2-405b ext. from wing, lap w/ #4x23'-6 in pvmt. ledge
Optional Type "A" Constr. Jt. (typ.) See Sec. "A-A"
Bottom of Cap El. Bt. 1 El. 902.34 Bt. 4 El. 902.86
POUR No 3
2 Lines, 2-#4x23'-6 ea. line 1 line ea. fa. (4 tot. req'd) Lap = 1'-0
Bridge Seat
6-402b on #4 in step
Step #1
Top of Mudwall Elev. Bt. 1 El. 908.51 Bt. 4 El. 908.99
This portion of the mudwall shall not be poured untill the superstructure has been completed. (Conc. & Reinf. billed w/ the superstructure)
Type "A" Constr. Jt. El. 905.60 (Bt. 1) or 906.12 (Bt. 4)
Step #2
Step #3
Step #4
Top of Mudwall Elev. Bt. 1 El. 908.19 Bt. 4 El. 908.64
2"x4" Keyway Constr. Jt. (typ.)
49-403b (for additional spa. see "PLAN")
POUR No 2
402a (Billed w/ Superstructure)
6" (Typical)
39-402b
39-401b
2-#4x3'-0
POUR No 1
12"ϕ Steel Encased Conc. Piles (Typical)
9-404b (6 in cap & 3 in mudwall) (Spa w/ #7x25'-0 & #4x23'-6)
1'-0
9"
9"
4 Spa. @ 1'-3 = 5'-0
9" (Typ. Spa.)
Details are symm. abt. ¢ Structure
13 Lines, 2-#7x25'-0 ea. line. (7 lines top, 4 lines mid. & 2 lines bot.) 26 tot. req'd. Lap = 1'-6
② Top of Mudwall Elev. Bt. 1 El. 908.51 Bt. 4 El. 908.99
Bot. Pile Elev. 872.0± (Typical)

EAST ELEVATION ~ BENT No1
(Bent No4 Same except as noted)

STEP ELEVATION

	Bt. No1	Bt. No4
Bridge Seat	904.835	905.355
Step #1	905.020	905.525
Step #2	905.210	905.690
Step #3	905.070	905.530
Step #4	904.925	905.370

Notes:
For "Reinforcing Bar Notes," see Br. Std. C1.
For Bearing Pad Type I, see Br. Std. PB11.
For Type "A" Construction Joint, see Br. Std. C3.
For additional details & Bill of Materials, see Dwg. C-4

DESIGNED		C'K'D	
DRAWN	JLF 10-79	C'K'D	R.P.
TRACED		C'K'D	

BENT No1 & BENT No4 DETAILS

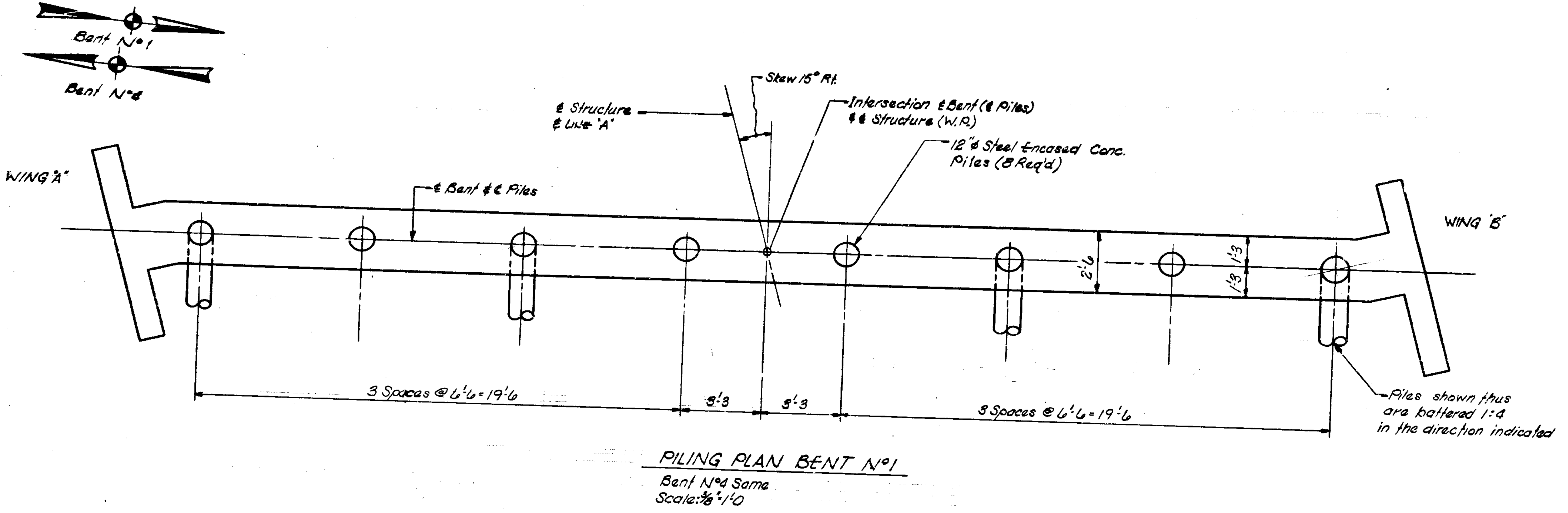
INDIANA STATE HIGHWAY COMMISSION

SCALE:- ⅜"=1'-0 U.N.
DATE:- 5-16-80
SUBMITTED FOR APPROVAL Romand Pela
DRAWING:- C3 OF C8 SHEET:- 9 OF 78
PROJECT:- F-068-6(1)
CONTRACT NO. B-13517
BRIDGE FILE:- 44-70-6634

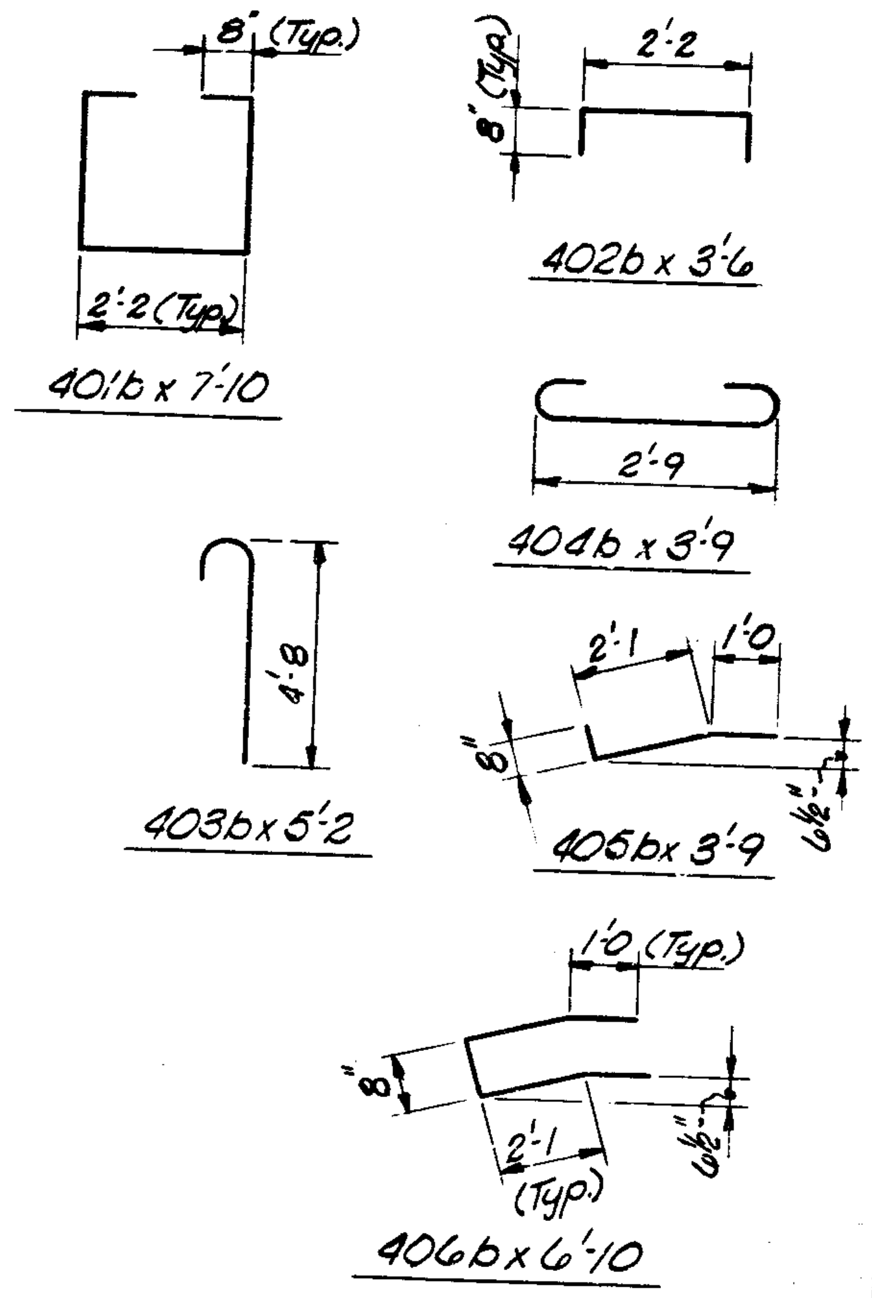
No. 13050

TP118

10-12-79 JLF
4:00-[illegible]
10-15-79 JLF
7:30-10:00
[illegible] JLF
[illegible] 12:30
10-[illegible] JLF
[illegible]
10-23-79 JLF
2:00-4:30-2½
10-26-79 JLF
7:30-11:30-4


Bent Nº1
Bent Nº4
Skew 15° Rt.
℄ Structure
℄ Line "A"
Intersection ℄ Bent (℄ Piles) & ℄ Structure (W.P.)
12"⌀ Steel Encased Conc. Piles (8 Req'd)
WING "A"
℄ Bent & ℄ Piles
WING "B"
2'-6
1'-3
1'-3
3 Spaces @ 6'-6 = 19'-6
3'-3
3'-3
3 Spaces @ 6'-6 = 19'-6
Piles shown thus are battered 1:4 in the direction indicated

PILING PLAN BENT Nº1
Bent Nº4 Same
Scale: ⅜" = 1'-0

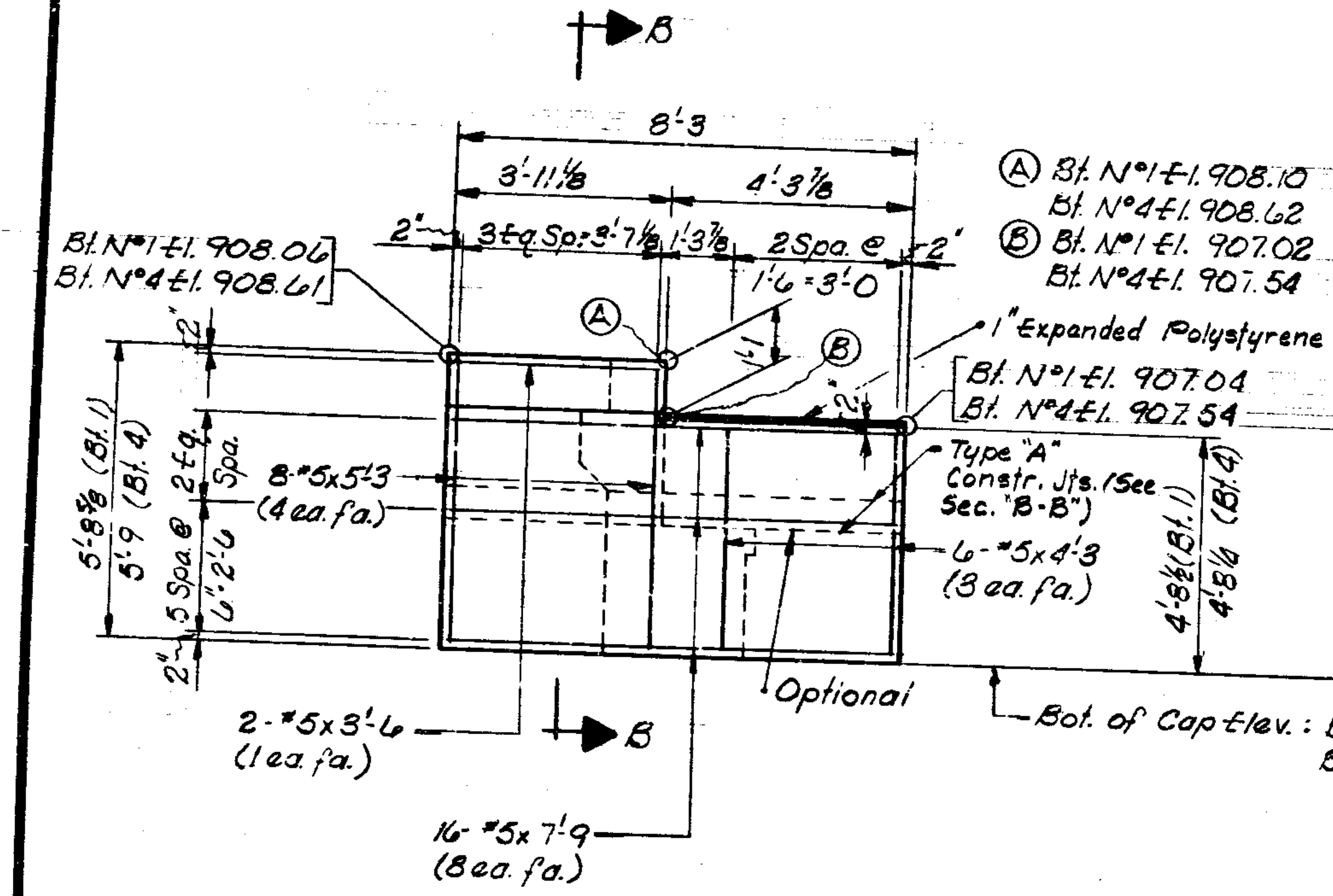

8" (Typ.)
2'-2 (Typ.)
401b x 7'-10
8" (Typ.)
2'-2
402b x 3'-6
2'-9
404b x 3'-9
4'-8
403b x 5'-2
2'-1
1'-0
8"
6½"
405b x 3'-9
1'-0 (Typ.)
8"
2'-1 (Typ.)
6½"
406b x 6'-10

BILL OF MATERIALS ~ BENT Nº1
(Bent Nº4 Same)

REINFORCING STEEL			
Size & Mark	Nº of Bars	Length	Weight (lbs)
#7	26	25'-0	
	TOTAL #7		1,329
#5	32	7'-9	
#5	14	5'-3	
#5	14	4'-3	
#5	2	3'-6	
#5	2	2'-9	
	TOTAL #5		410
401b	39	7'-10	
402b	93	3'-6	
403b	49	5'-2	
404b	18	3'-9	
405b	4	3'-9	
406b	8	6'-10	
#4	4	23'-6	
#4	2	3'-0	
	TOTAL #4		749
	TOTAL REINF.		2,488

CONCRETE CLASS "A"	CYS.
POUR Nº1 ~ BENT Nº1	15.2
POUR Nº1 ~ BENT Nº4	15.1
POUR Nº2	.6
POUR Nº3	.6
TOTAL CLASS "A" CONC. ~ BENT 1	16.4
TOTAL CLASS "A" CONC. ~ BENT 4	16.3
MISCELLANEOUS	**Quan.**
12"⌀ Steel Encased Conc Piles 8 @ 32 Lft. long ea. (approx.)	256 Lft.

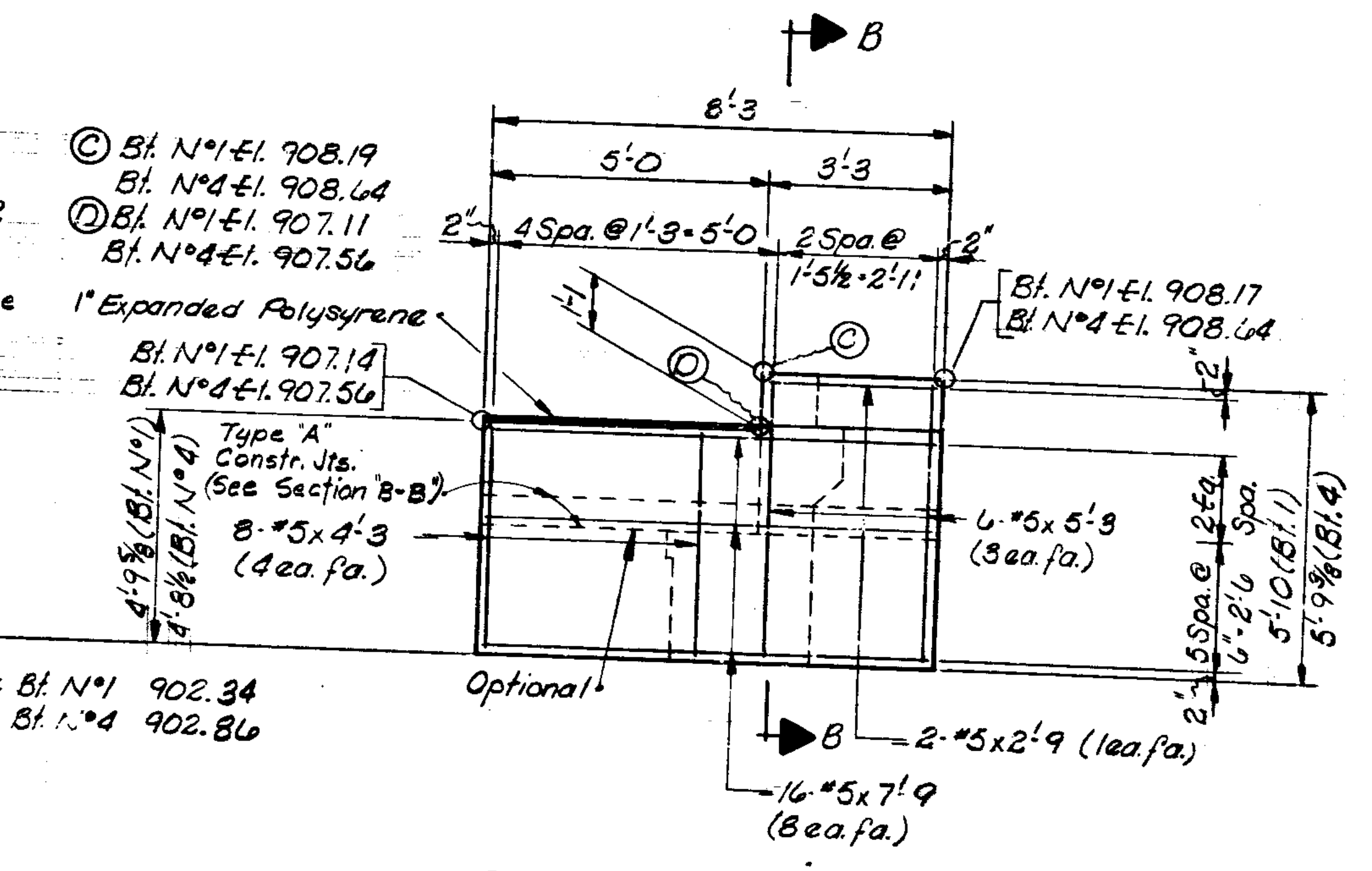
1-7-80 JLF
2:00-4:30-2½
1-8-80 JLF
10:00-11:30-1½


B
8'-3
3'-11⅛
4'-3⅞
2"
3 Eq. Sp. = 3'-7⅛
1'-3⅞
2 Spa. @ 1'-6 = 3'-0
2"
Bl. Nº1 El. 908.06
Bl. Nº4 El. 908.61
A
B
1
1
2"
1" Expanded Polystyrene
Bl. Nº1 El. 907.04
Bl. Nº4 El. 907.54
5'-8⅝ (Bl. 1)
5'-9 (Bl. 4)
2 Eq. Spa.
5 Spa. @ 6" = 2'-6
2"
8-#5x5'-3 (4 ea. fa.)
Type "A" Constr. Jts. (See Sec. "B-B")
6-#5x4'-3 (3 ea. fa.)
4'-8½ (Bl. 1)
4'-8¼ (Bl. 4)
2-#5x3'-6 (1 ea. fa.)
B
Optional
16-#5x7'-9 (8 ea. fa.)

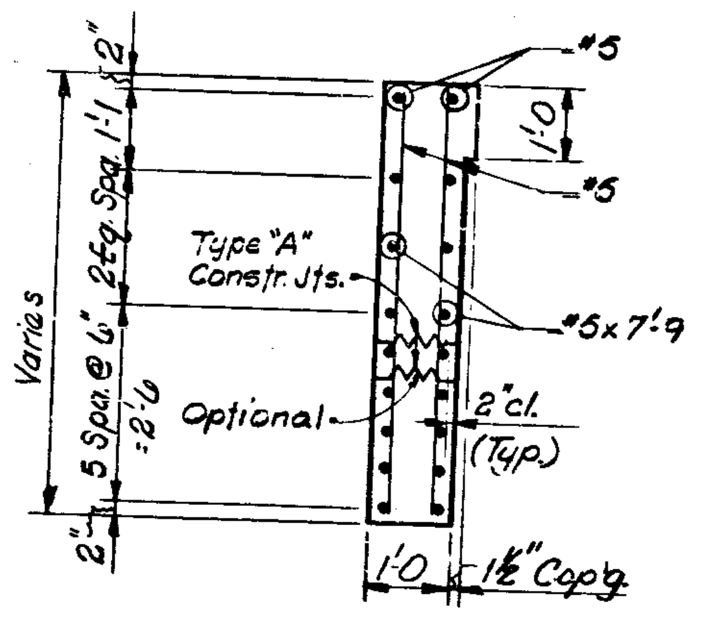
ELEVATION ~ WING "A"
Scale ⅜" = 1'-0

Ⓐ Bl. Nº1 El. 908.10
Bl. Nº4 El. 908.62
Ⓑ Bl. Nº1 El. 907.02
Bl. Nº4 El. 907.54
Ⓒ Bl. Nº1 El. 908.19
Bl. Nº4 El. 908.64
Ⓓ Bl. Nº1 El. 907.11
Bl. Nº4 El. 907.56

Bot. of Cap Elev.: Bl. Nº1 902.34
Bl. Nº4 902.86


B
8'-3
5'-0
3'-3
2"
4 Spa. @ 1'-3 = 5'-0
2 Spa. @ 1'-5½ = 2'-11
2"
1" Expanded Polysyrene
Bl. Nº1 El. 907.14
Bl. Nº4 El. 907.56
1
1
D
C
Bl. Nº1 El. 908.17
Bl. Nº4 El. 908.64
2"
4'-9⅝ (Bl. Nº1)
4'-8¼ (Bl. Nº4)
Type "A" Constr. Jts. (See Section "B-B")
8-#5x4'-3 (4 ea. fa.)
6-#5x5'-3 (3 ea. fa.)
2 Eq. Spa.
5 Spa. @ 6" = 2'-6
5'-10 (Bl. 1)
5'-9⅜ (Bl. 4)
2"
Optional
B
2-#5x2'-9 (1 ea. fa.)
16-#5x7'-9 (8 ea. fa.)

ELEVATION ~ WING "B"
Scale: ⅜" = 1'-0


2"
2 Eq. Spa. 1'-1
5 Spa. @ 6" = 2'-6
2"
Varies
#5
1'-0
#5
Type "A" Constr. Jts.
#5x7'-9
Optional
2" cl. (Typ.)
1'-0
1½" Cap'g.

SECTION B-B
Scale: ½" = 1'-0

NOTES:
For "Reinforcing Bar Notes," see Br. Std. C1.
For additional notes and details, see Dwg. C-3

BENT Nº1 & BENT Nº4 DETAILS &
BILL OF MATERIALS
INDIANA STATE HIGHWAY COMMISSION

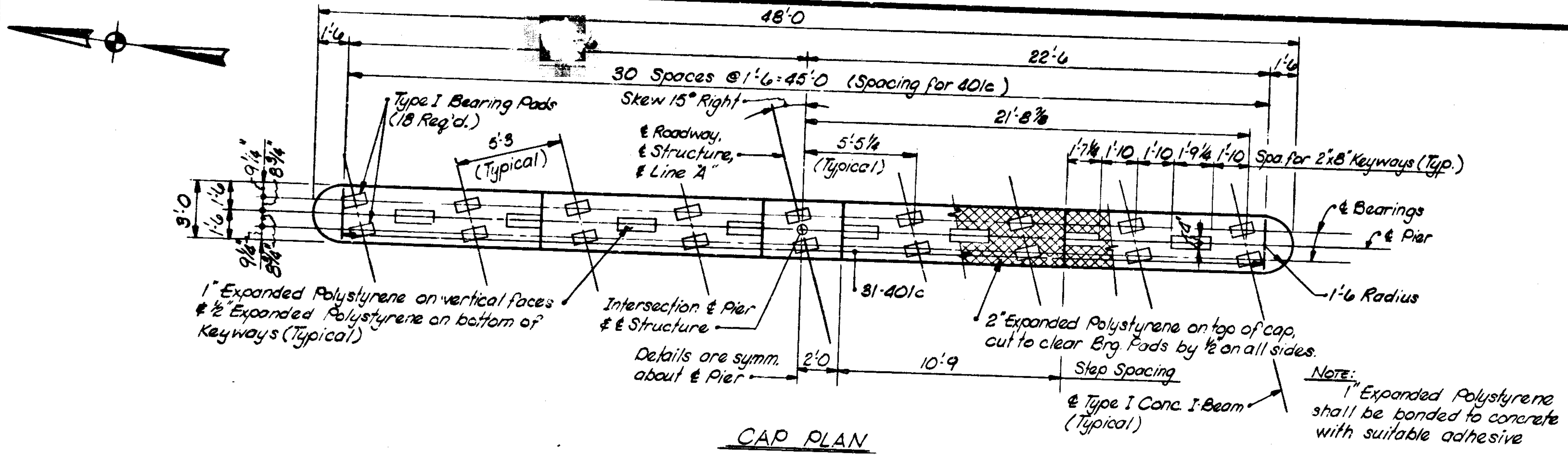
SCALE:- As Noted
DATE:- 5-16-80
Submitted for Approval Raman D Patel
DRAWING:- C4 of C8 SHEET:- 10 of 78
PROJECT:- F-068-6(1)
CONTRACT NO. B-13517
BRIDGE FILE:- 44-70-6634

RAMAN D. PATEL
No. 13050

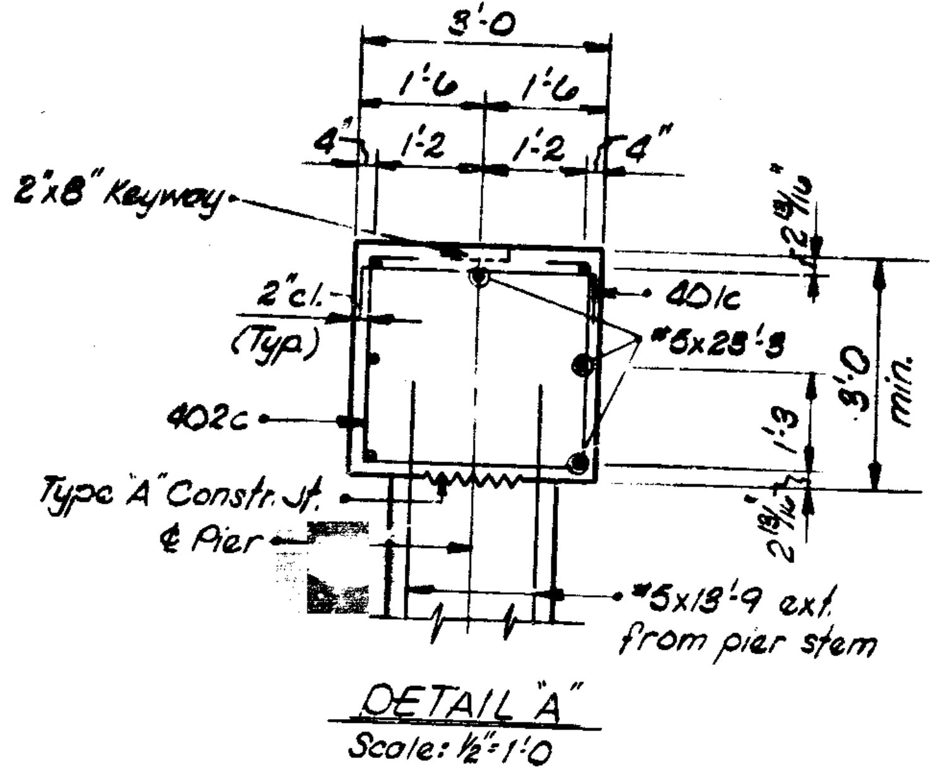
DESIGNED ____ C'K'D ____
DRAWN JLF 10-79 C'K'D R.P.
TRACED ____ C'K'D ____

TP118

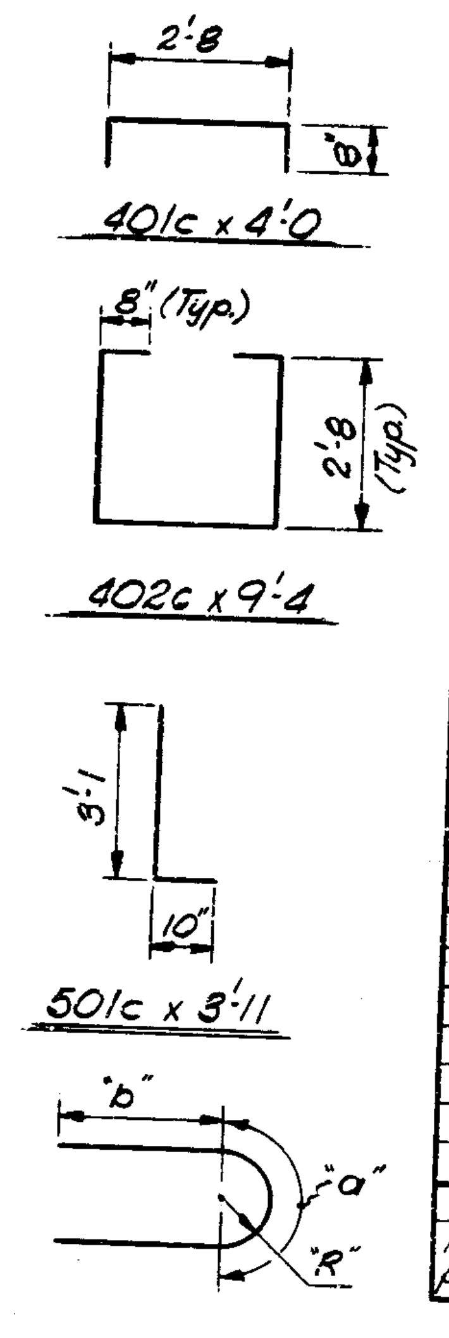
1-8-80 JLF
11:30-4:30-4
11-9-80 JLF
7:30-4:30-8
11-10-80 JLF
7:30-4:30-4
1-18-80 JLF
2:30-4:30-2
1-21-80 JLF
7:30-11:00 3½


48'-0
1'-6
22'-6
30 Spaces @ 1'-6 = 45'-0 (Spacing for 401c)
Skew 15° Right
21'-8⅞
Type I Bearing Pads (18 Req'd.)
6'-3 (Typical)
℄ Roadway, ℄ Structure, ℄ Line "A"
5'-5⅛ (Typical)
Spa. for 2"x8" Keyways (Typ.)
℄ Bearings
℄ Pier
1'-6 Radius
1" Expanded Polystyrene on vertical faces & ½" Expanded Polystyrene on bottom of Keyways (Typical)
Intersection ℄ Pier & ℄ Structure
81-401c
2" Expanded Polystyrene on top of cap, cut to clear Brg. Pads by ½" on all sides.
Details are symm. about ℄ Pier
2'-0
10'-9
Step Spacing
℄ Type I Conc. I-Beam (Typical)
Note: 1" Expanded Polystyrene shall be bonded to concrete with suitable adhesive

CAP PLAN

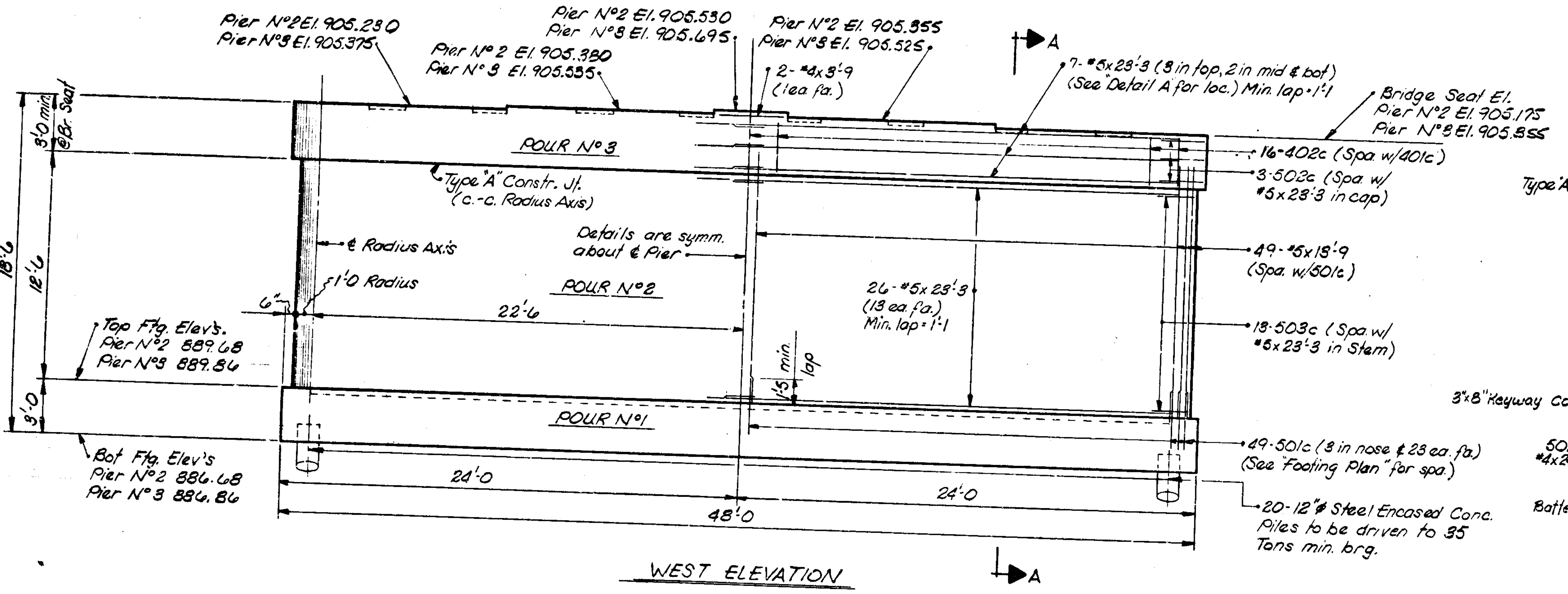

3'-0
1'-6
1'-2
4"
2"x8" Keyway
2" cl. (Typ.)
401c
#5x23'-3
402c
Type "A" Constr. Jt.
℄ Pier
#5x13'-9 ext. from pier stem
3'-0 min.

DETAIL "A"
Scale: ½"=1'-0

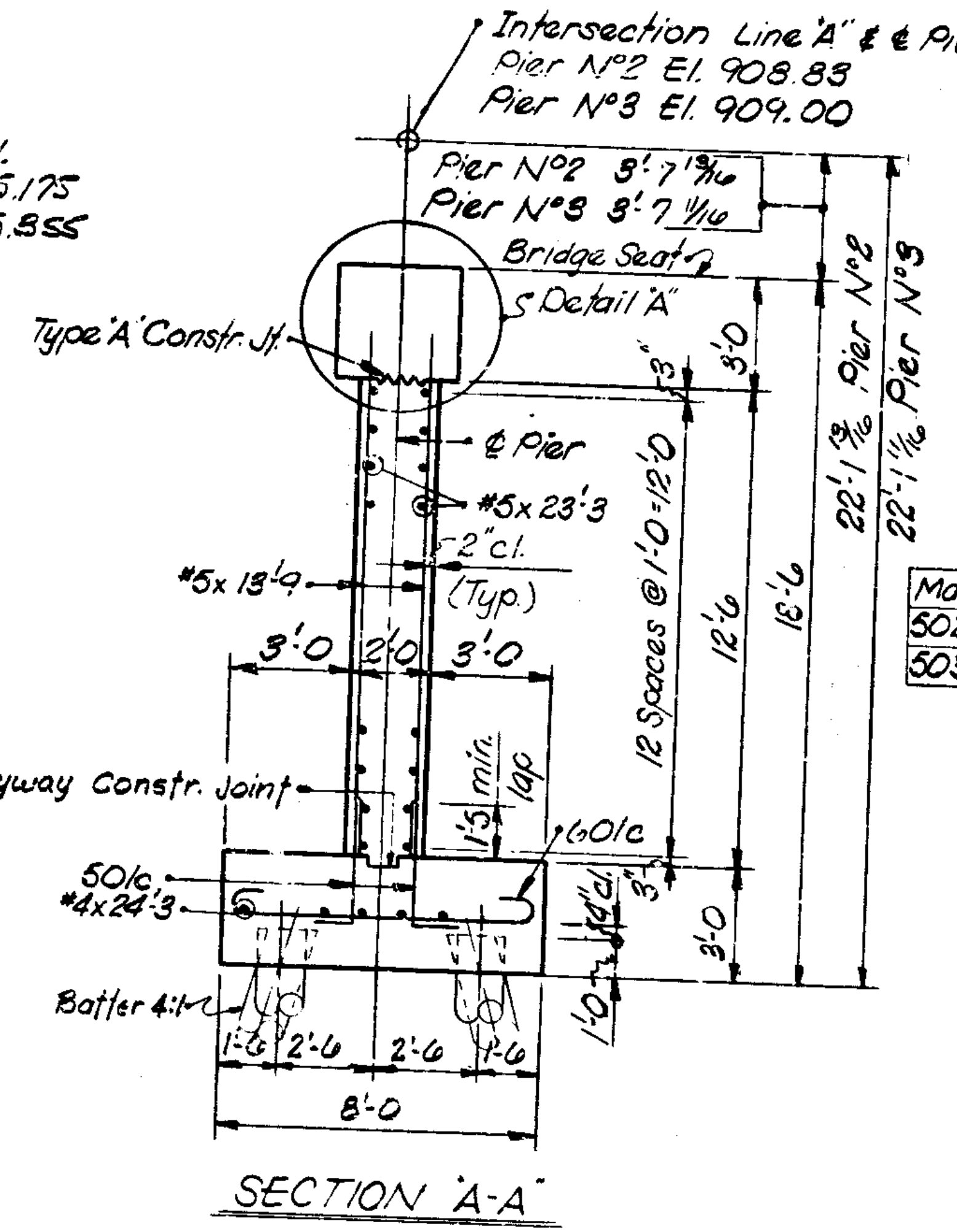

2'-8
8"
401c x 4'-0
8" (Typ.)
2'-8 (Typ.)
402c x 9'-4
3'-1
10"
501c x 3'-11
"b"
"a"
"R"

BILL OF MATERIALS ~ PIER Nº2
(Pier Nº3 Same)

REINFORCING STEEL			
Size & Mark	Nº of Bars	Length	Weight (lbs.)
601c	77	8'-10	
	Total #6		1,022
501c	98	3'-11	
502c	6	6'-3	
503c	26	4'-8	
#5	66	23'-3	
#5	98	13'-9	
#5	64	7'-6	
	Total #5		4,073
401c	31	4'-0	
402c	31	9'-4	
#4	16	24'-3	
#4	2	3'-9	
	Total #4		540
	Total Reinf.		5,635
CONCRETE			CYS.
Class "B" Concrete in Footing (Pour Nº1)			28.4
Class "A" Concrete Pour Nº2			43.1
Pour Nº3			16.4
Total Class "A" Concrete			59.5
MISCELLANEOUS			QUAN.
12"⌀ Steel Encased Conc. Piles 20@20'-0 long ea.			400 Lft.


Pier Nº2 El. 905.230
Pier Nº3 El. 905.375
Pier Nº2 El. 905.380
Pier Nº3 El. 905.535
Pier Nº2 El. 905.530
Pier Nº3 El. 905.695
Pier Nº2 El. 905.355
Pier Nº3 El. 905.525
2-#4x3'-9 (1 ea. fa.)
A
7-#5x23'-3 (3 in top, 2 in mid & bot.) (See Detail A for loc.) Min. lap 1'-1
Bridge Seal El. Pier Nº2 El. 905.175 Pier Nº3 El. 905.355
3'-0 min. @ Br. Seat
POUR Nº3
16-402c (Spa. w/401c)
Type "A" Constr. Jt. (c.-c. Radius Axis)
3-502c (Spa. w/ #5x23'-3 in cap)
℄ Radius Axis
Details are symm. about ℄ Pier
1'-0 Radius
POUR Nº2
49-#5x13'-9 (Spa. w/501c)
18'-6
12'-6
22'-6
26-#5x23'-3 (13 ea. fa.) Min. lap 1'-1
13-503c (Spa. w/ #5x23'-3 in Stem)
6"
Top Ftg. Elev's. Pier Nº2 889.68 Pier Nº3 889.86
1'-5 min. lap
3'-0
POUR Nº1
49-501c (3 in nose & 23 ea. fa.) (See "Footing Plan" for spa.)
Bot. Ftg. Elev's Pier Nº2 886.68 Pier Nº3 886.86
24'-0
24'-0
48'-0
20-12"⌀ Steel Encased Conc. Piles to be driven to 35 Tons min. brg.
A

WEST ELEVATION

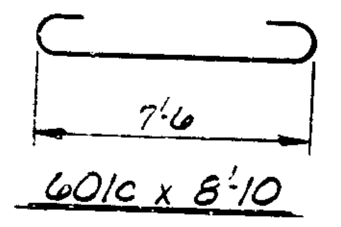

Intersection Line "A" & ℄ Pier
Pier Nº2 El. 908.83
Pier Nº3 El. 909.00
Pier Nº2 3'-7 13/16
Pier Nº3 3'-7 11/16
Bridge Seat ℄
Detail "A"
Type "A" Constr. Jt.
℄ Pier
#5x23'-3
2" cl. (Typ.)
#5x13'-9
12 Spaces @ 1'-0 = 12'-0
3'-0
12'-6
18'-6
22'-1 13/16 Pier Nº2
22'-1 11/16 Pier Nº3
3'-0 2'-0 3'-0
3"x8" Keyway Constr. Joint
1'-5 min. lap
501c
#4x24'-3
601c
Batter 4:1
1'-6 2'-6 2'-6 1'-6
8'-0
3'-0
10"

SECTION "A-A"

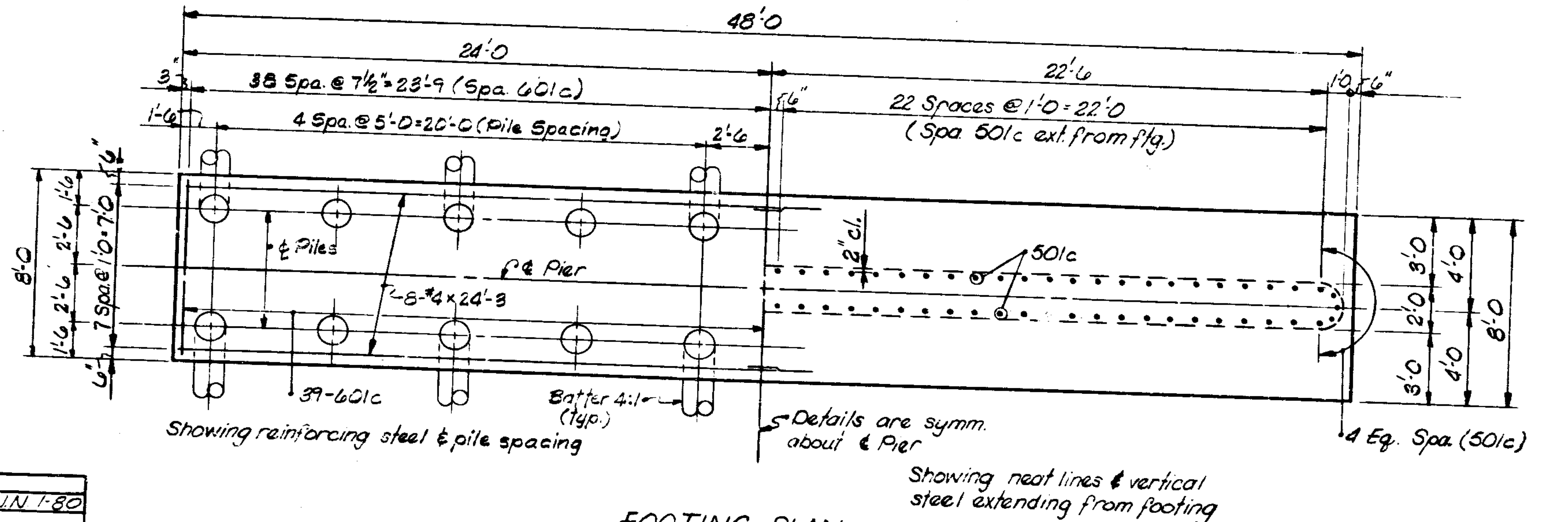
Maximum Soil Pressure Without Piles = 1.7 Ton/Sq. Ft.

Mark	"R"	"a"	"b"	Length
502c	1'-3½	4'-1	1'-1	6'-3
503c	9⅜"	2'-5½	1'-1⅛	4'-8

502c x 6'-3
503c x 4'-8



601c x 8'-10


48'-0
24'-0
22'-6
3"
38 Spa. @ 7½" = 23'-9 (Spa. 601c)
1'-6
4 Spa. @ 5'-0 = 20'-0 (Pile Spacing)
2'-6
1'-6
22 Spaces @ 1'-0 = 22'-0 (Spa. 501c ext. from ftg.)
1'-0
1'-6
8'-0
1'-6 2'-6 2'-6 1'-6
7 Spa. @ 1'-0 = 7'-0
6"
℄ Piles
℄ Pier
8-#4x24'-3
12" cl.
501c
39-601c
Batter 4:1 (Typ.)
3'-0 2'-0 3'-0
4'-0 4'-0
8'-0
Showing reinforcing steel & pile spacing
Details are symm. about ℄ Pier
Showing neat lines & vertical steel extending from footing
4 Eq. Spa. (501c)

FOOTING PLAN

Notes:
For "Reinforcing Bar Notes," see Br. Std. C1.
For Type "A" Construction Joint, see Br. Std. C3.
For Bearing Pad Type I, see Br. Std. PB11.

DESIGNED R.D.P. 1-80 C'K'D
DRAWN J.L.F. 1-80 C'K'D T.J.N. 1-80
TRACED C'K'D

PIERS Nº2 & 3 DETAILS & BILL OF MATERIALS

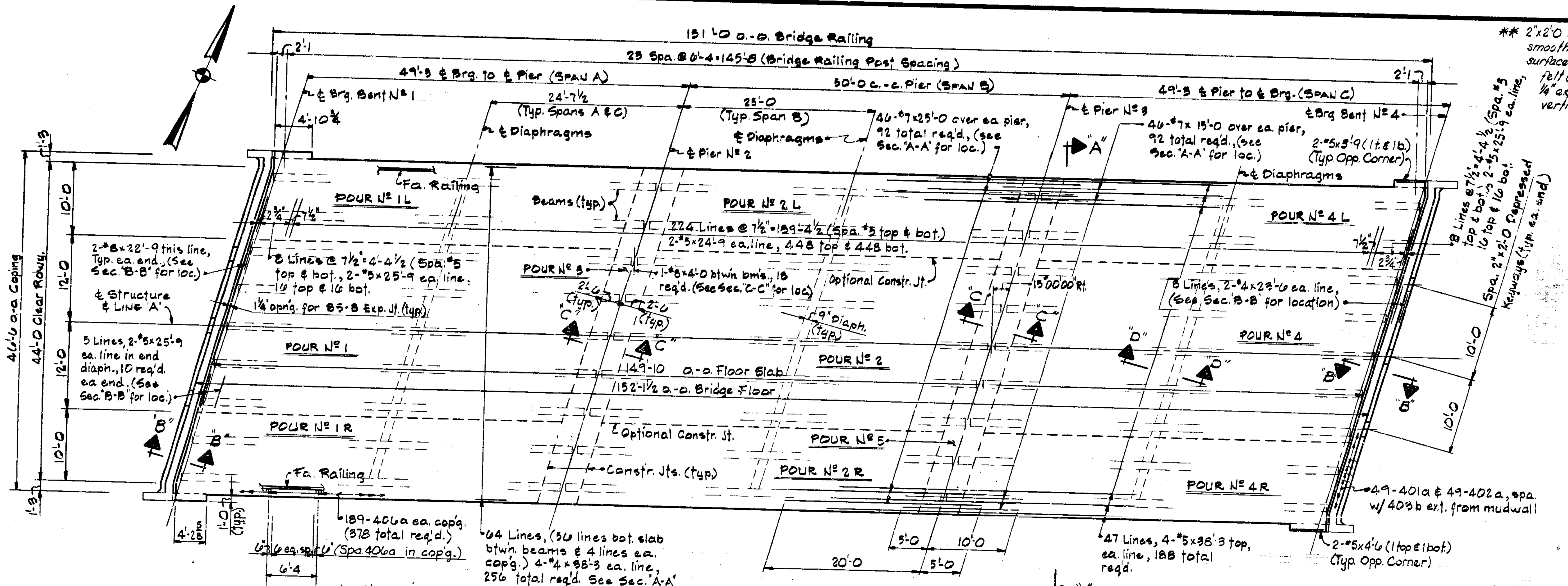
INDIANA STATE HIGHWAY COMMISSION

SCALE:- ¼"=1'-0 U.N.
DATE:- 5-16-80
SUBMITTED FOR APPROVAL: Raman

DRAWING:- C5 OF C7 SHEET:- 11 OF 78
PROJECT:- F-068-6(1)
CONTRACT NO. B-13517
BRIDGE FILE:- 44-70-6634

No. 13050

TP118


131'-0 o.-o. Bridge Railing
23 Spa. @ 6'-4=145'-8 (Bridge Railing Post Spacing)
49'-3 ℄ Brg. to ℄ Pier (SPAN A)
50'-0 c.-c. Pier (SPAN B)
49'-3 ℄ Pier to ℄ Brg. (SPAN C)
2'-1
℄ Brg. Bent Nº 1
4'-10¾
24'-7½ (Typ. Spans A & C)
25'-0 (Typ. Span B)
℄ Diaphragms
℄ Pier Nº 2
℄ Pier Nº 3
℄ Brg. Bent Nº 4
46-#7x25'-0 over ea. pier, 92 total req'd., (see Sec. "A-A" for loc.)
46-#7x 15'-0 over ea. pier, 92 total req'd., (see Sec. "A-A" for loc.)
2-#5x3'-9 (1t. & 1b.) (Typ Opp. Corner)
Fa. Railing
Beams (typ.)
POUR Nº 1L
POUR Nº 2L
POUR Nº 4L
POUR Nº 1
POUR Nº 2
POUR Nº 3
POUR Nº 4
POUR Nº 5
POUR Nº 1R
POUR Nº 2R
POUR Nº 4R
224 Lines @ 7½"=139'-4½ (Spa. #5 top & bot.) 2-#5x24'-9 ea. line, 448 top & 448 bot.
1-#8x4'-0 btwn bm's., 18 req'd. (See Sec. "C-C" for loc.)
Optional Constr. Jt.
15°00'00" Rt.
8 Lines @ 7½"=4'-4½ (Spa. #5 top & bot., 2-#5x25'-9 ea. line, 16 top & 16 bot.
8 Lines, 2-#4x23'-6 ea. line, (See Sec. "B-B" for location)
8 Lines @ 7½"=4'-4½ (Spa. #5 top & bot., 2-#5x25'-9 ea. line, 16 top & 16 bot.
Spa. 2"x2'-0 Depressed Keyways (typ. ea. end)
2-#8x22'-9 this line, Typ. ea. end., (See Sec. "B-B" for loc.)
℄ Structure & LINE 'A'
1¼" opn'g. for BS-8 Exp. Jt. (typ.)
2¾"
7½"
2'-6 (typ.)
2'-6 (typ.)
9" Diaph. (typ.)
5 Lines, 2-#5x25'-9 ea. line in end diaph., 10 req'd. ea end. (See Sec. "B-B" for loc.)
149'-10 o.-o. Floor Slab
152'-1½ o.-o. Bridge Floor
Optional Constr. Jt.
Constr. Jts. (typ.)
46'-6 o-o Coping
44'-0 Clear Rdwy.
10'-0
12'-0
12'-0
10'-0
1'-3
1'-3
"A"
"B"
"C"
"D"
4'-2⅝
1'-0 (typ.)
6" 6 eq. sp. 6" (Spa. 406a in cop'g.)
6'-4
℄ Bridge Railing Posts
189-406a ea. cop'g. (378 total req'd.)
64 Lines, (56 lines bot. slab btwn. beams & 4 lines ea. cop'g.) 4-#4x38'-3 ea. line, 256 total req'd. See Sec. "A-A" for spa.
20'-0
5'-0
10'-0
5'-0
47 Lines, 4-#5x38'-3 top, ea. line, 188 total req'd.
2-#5x4'-6 (1 top & 1 bot.) (Typ. Opp. Corner)
49-401a & 49-402a, spa. w/ 403b ext. from mudwall
10'-0
10'-0

PLAN

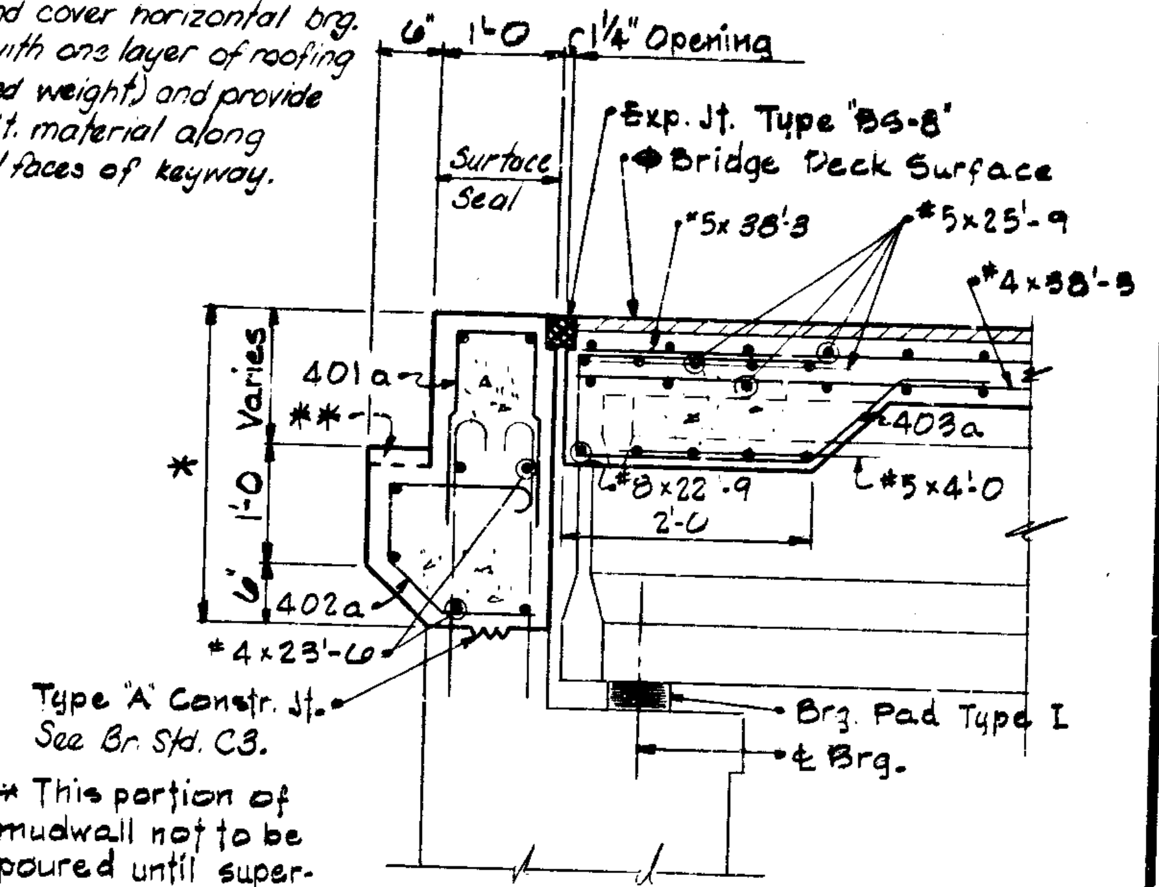
Scale: ⅛"=1'-0

LAP TABLE

#4 Longitudinal = 1'-0
#5 Longitudinal = 1'-1
#5 Transverse = 1'-5
#8 Transverse = 1'-8

Note: Laps shown in the table are minimum lap lengths.

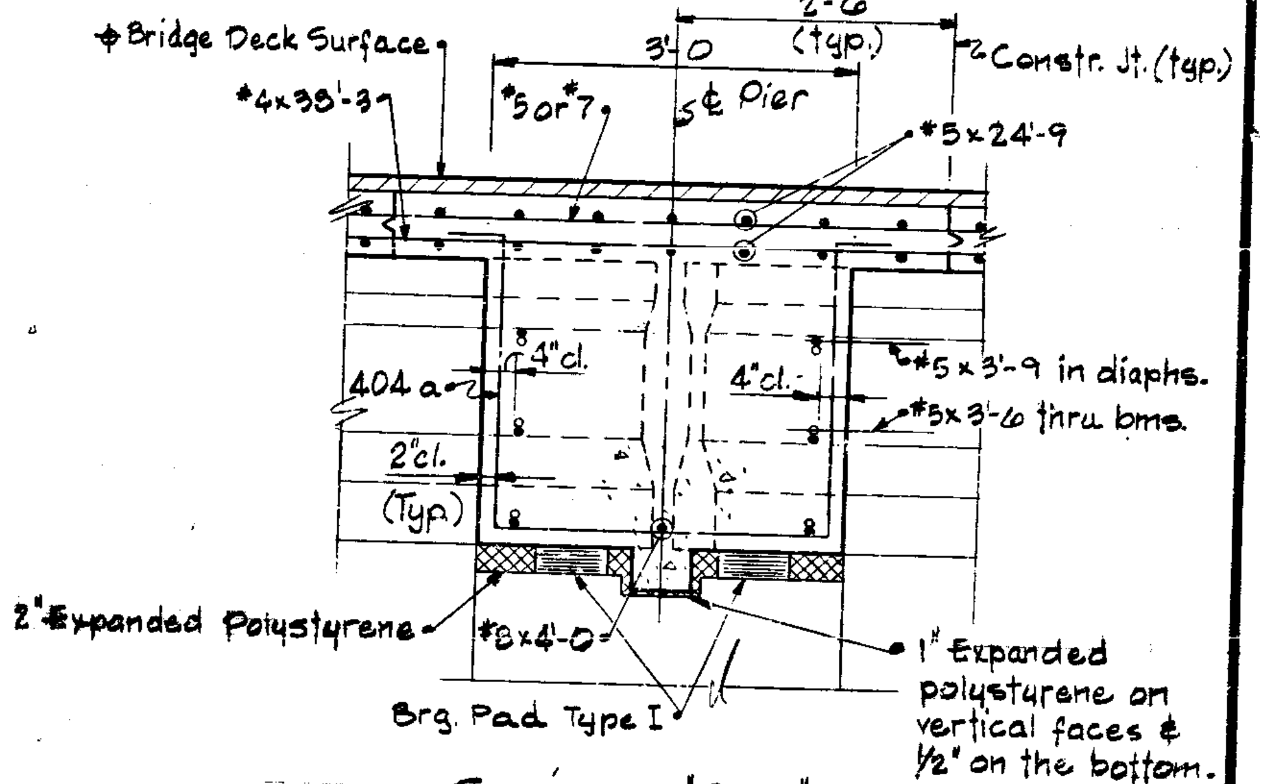
** 2"x2'-0 Depressed Keyway: Trowel smooth and cover horizontal brg. surfaces with one layer of roofing felt (med weight) and provide ¼" exp. jt. material along vertical faces of keyway.


6"
1'-0
1¼" Opening
Surface Seal
Exp. Jt. Type "BS-8"
⌖ Bridge Deck Surface
#5x38'-3
#5x25'-9
#4x38'-3
Varies
401a
**
*
1'-0
6"
402a
403a
#8x22'-9
2'-0
#5x4'-0
#4x23'-6
Type 'A' Constr. Jt. See Br. Std. C3.
Brg. Pad Type I
℄ Brg.

* This portion of mudwall not to be poured until superstructure has been completed.

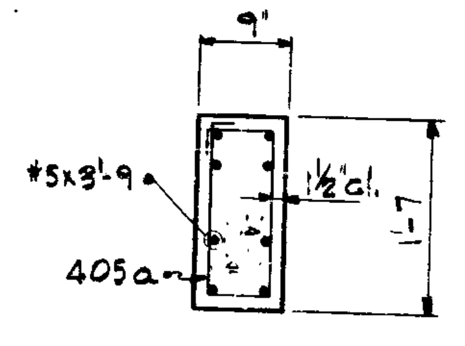
SECTION "B-B"

Scale: ¾"=1'-0


⌖ Bridge Deck Surface
#4x38'-3
#5 or #7
3'-0
2'-6 (typ.)
℄ Pier
Constr. Jt. (typ.)
#5x24'-9
404a
4" cl.
4" cl.
2" cl. (Typ.)
#5x3'-9 in diaphs.
#5x3'-6 thru bms.
2" Expanded Polystyrene
#8x4'-0
Brg. Pad Type I
1" Expanded polystyrene on vertical faces & ½" on the bottom.

SECTION "C-C"

Scale: ¾"=1'-0


9"
#5x3'-9
1½" cl.
1'-7
405a

SECTION "D-D" (TYP. INT. DIAPH.)

Scale: ¾"=1'-0

NOTES:

AFTER BEAMS HAVE BEEN ERECTED, CONCRETE FORMS SHALL NOT BE BLOCKED AGAINST THE EXPANSION END OF THE BEAM IN MAKING ANY POURS ADJACENT TO THE BEAM SPANS.

FOR REINFORCING BAR NOTES SEE BRIDGE STANDARD C1.

POUR NUMBERS INDICATE SEQUENCE OF POURS. POURS OVER INTERIOR SUPPORTS TO BE MADE LAST TO REDUCE THE EFFECT OF THE SLAB DEAD LOAD IN THE NEGATIVE MOMENT AREA. POUR NO. 3 & NO. 5 WILL INCLUDE THE DIAPHRAGM AT SUPPORT, AND WILL BE HELD TO A 5' LENGTH. INTERIOR DIAPHRAGMS TO BE POURED BEFORE SLAB IS POURED. THE LONGITUDINAL CONSTRUCTION JOINTS MAY BE ELIMINATED SUBJECT TO THE APPROVAL OF THE ENGINEER.

BRIDGE SEAT ELEVATIONS WERE CALCULATED USING A DENSE P.C.C. SURFACE (2 1/4"). IF THE CONTRACTOR ELECTS TO USE A MODIFIED P.C.C. SURFACE (1 1/2"), BEAM FILLETS ARE TO BE ADJUSTED ACCORDINGLY. (SEE DWG C7)

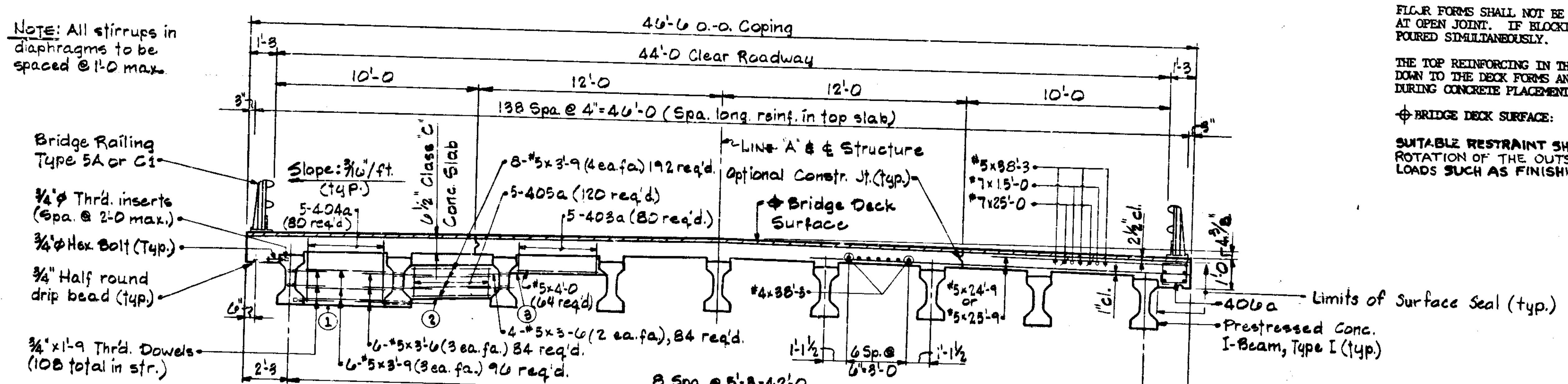
FLOOR FORMS SHALL NOT BE BLOCKED BETWEEN ADJACENT BEAMS AT OPEN JOINT. IF BLOCKING IS USED, BOTH SIDES SHALL BE POURED SIMULTANEOUSLY.

THE TOP REINFORCING IN THE DECK SHALL BE SECURELY TIED DOWN TO THE DECK FORMS AND/OR THE BEAMS TO PREVENT LIFTING DURING CONCRETE PLACEMENT.

⌖ BRIDGE DECK SURFACE: 1 1/2" MODIFIED PORTLAND CEMENT CONCRETE OR
2 1/4" DENSE PORTLAND CEMENT CONCRETE

SUITABLE RESTRAINT SHALL BE PROVIDED TO PREVENT THE ROTATION OF THE OUTSIDE BEAMS FROM CONSTRUCTION LOADS SUCH AS FINISHING MACHINE, FORMS ETC.;

Note: All stirrups in diaphragms to be spaced @ 1'-0 max.


46'-6 o.-o. Coping
44'-0 Clear Roadway
1'-3
10'-0
12'-0
12'-0
10'-0
1'-3
3"
3"
138 Spa. @ 4"=46'-0 (Spa. long. reinf. in top slab)
Bridge Railing Type 5A or C1
Slope: 3/16"/ft. (typ.)
6½" Class "C" Conc. Slab
¾"⌀ Thr'd. inserts (Spa. @ 2'-0 max.)
¾"⌀ Hex. Bolt (Typ.)
¾" Half round drip bead (typ.)
¾"x1'-9 Thr'd. Dowels (108 total in str.)
6"
2'-3
5-404a (80 req'd)
8-#5x3'-9 (4 ea. fa.) 192 req'd.
5-405a (120 req'd.)
5-403a (80 req'd.)
#5x4'-0 (64 req'd)
4-#5x3'-6 (2 ea. fa.), 84 req'd.
6-#5x3'-6 (3 ea. fa.) 84 req'd.
6-#5x3'-9 (3 ea. fa.) 96 req'd.
①
②
③
Line "A" & ℄ Structure
Optional Constr. Jt. (typ.)
⌖ Bridge Deck Surface
#5x38'-3
#7x15'-0
#7x25'-0
#4x38'-3
#5x24'-9 or #5x25'-9
2½" cl.
1" cl.
1'-0
4⅜"
1'-1½
6 Sp. @ 6"=3'-0
1'-1½
8 Spa. @ 5'-3=42'-0
2'-3
406a
Limits of Surface Seal (typ.)
Prestressed Conc. I-Beam, Type I (typ.)

① Typ. Diaphragms @ Piers Nº 2 & 3 (Sec. "C-C")
② Typ Int. Diaphragms @ Mid-Spans (Sec. "D-D")
③ Typ. Diaphragms @ Ends of Slab (Sec. "B-B")

SECTION "A-A"

Scale: ¼"=1'-0

DESIGNED R.D.P. 1/80 CK'D
DRAWN TJN 1/80 CK'D JLF
TRACED CK'D

Rev.; 01-26-82 NOTE ADDED

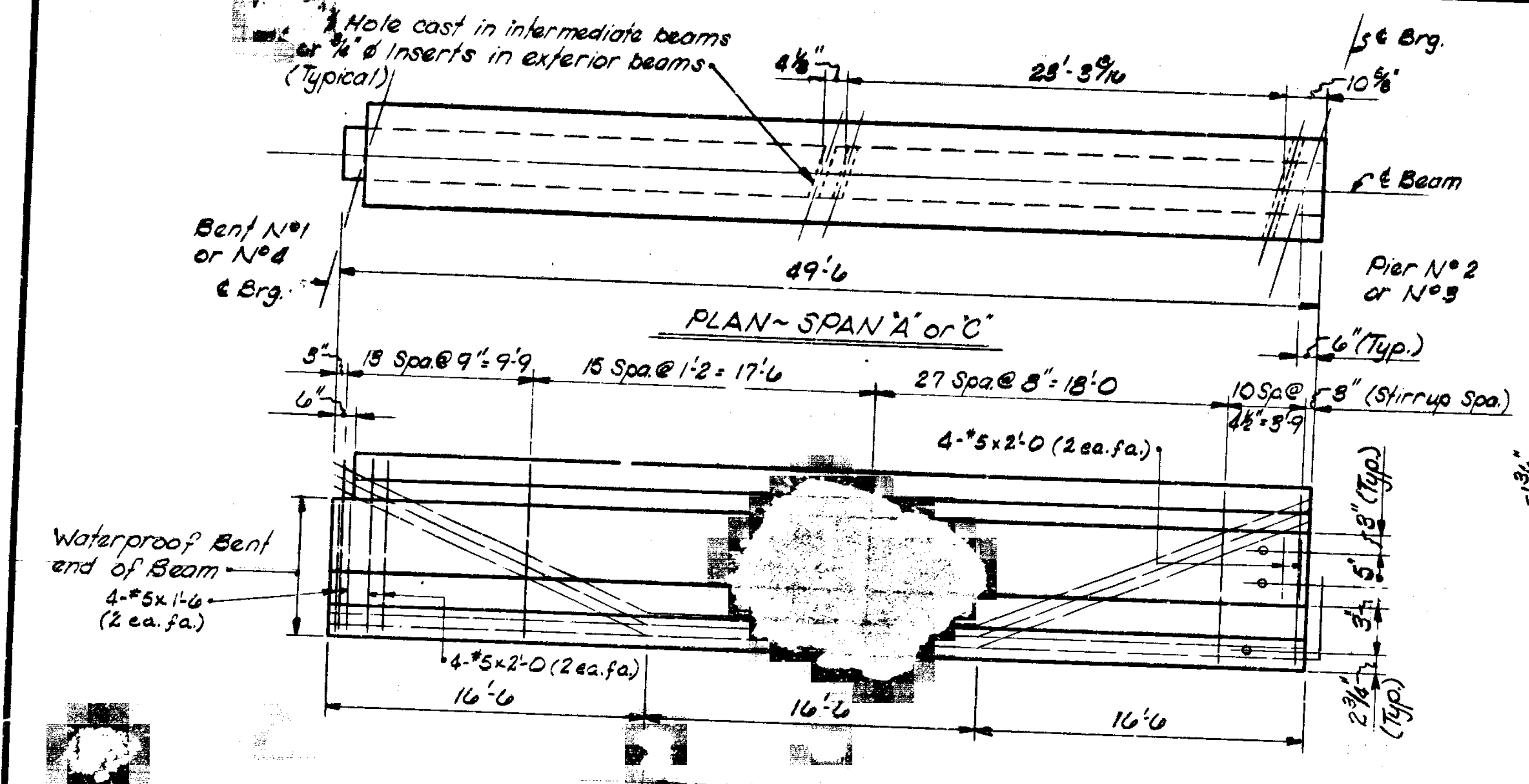
SUPERSTRUCTURE DETAILS

INDIANA STATE HIGHWAY COMMISSION

SCALE:- As Noted
DATE:- 5-16-80
Submitted For Approval: Raman D Patel
DRAWING:- C6 OF C7
SHEET:- 12 OF 78
PROJECT:- F-068-6(1)
CONTRACT NO. B-13517
BRIDGE FILE:- 44-70-6634

RAMAN D. PATEL
No. 13050

TP118

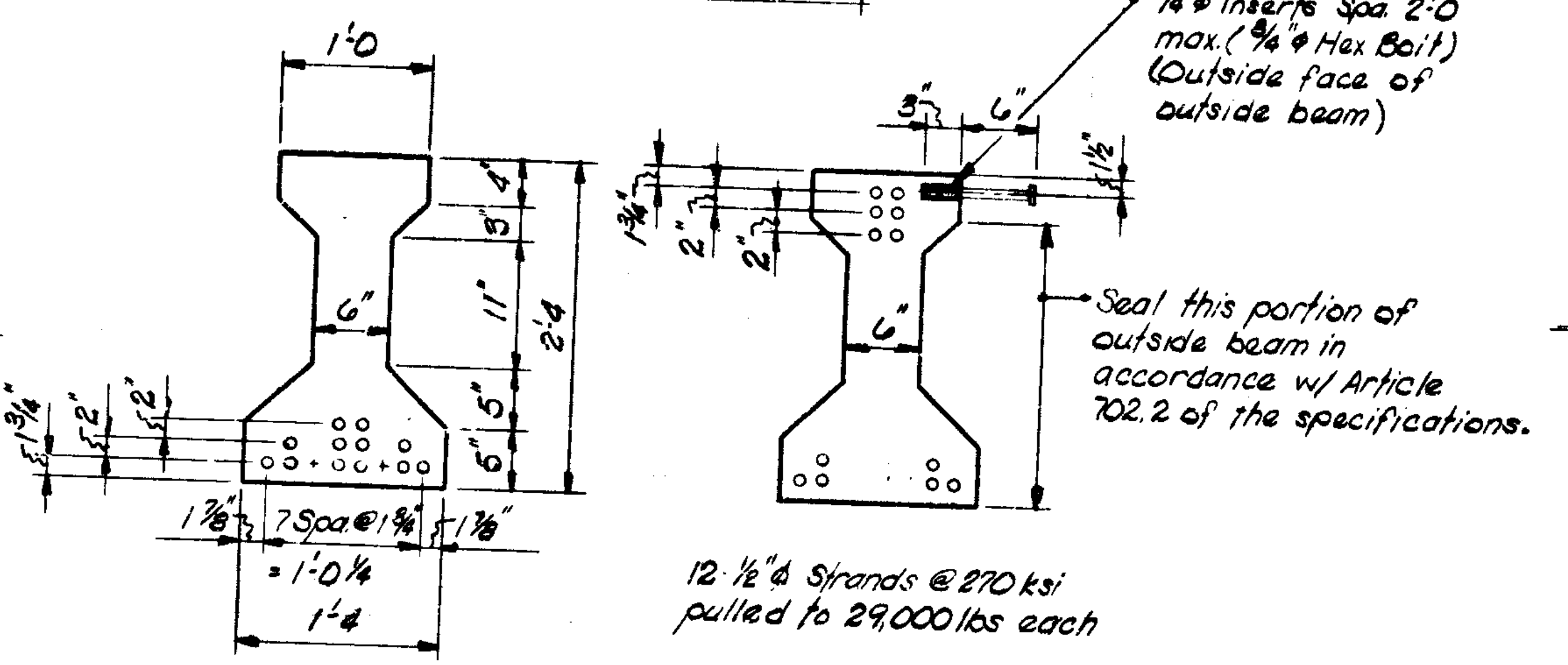


¾" Hole cast in intermediate beams or ¾" ø Inserts in exterior beams (Typical)
4½"
28'-3 9/16
℄ Brg.
10⅝"
℄ Beam
Bent N°1 or N°4 ℄ Brg.
49'-6
Pier N°2 or N°3
6" (Typ.)
3"
13 Spa.@ 9" = 9'-9
15 Spa.@ 1'-2 = 17'-6
27 Spa.@ 8" = 18'-0
10 Spa.@ 4½" = 3'-9
3" (Stirrup Spa.)
6"
4-#5 x 2'-0 (2 ea. fa.)
3" (Typ.)
5"
3½"
2¾" (Typ.)
Waterproof Bent end of Beam
4-#5 x 1'-6 (2 ea. fa.)
4-#5 x 2'-0 (2 ea. fa.)
16'-6
16'-6
16'-6

PLAN~ SPAN "A" or "C"

ELEVATION~ SPAN "A" or "C"

The beams shall be cast a minimum of 15 days before the slab is poured.

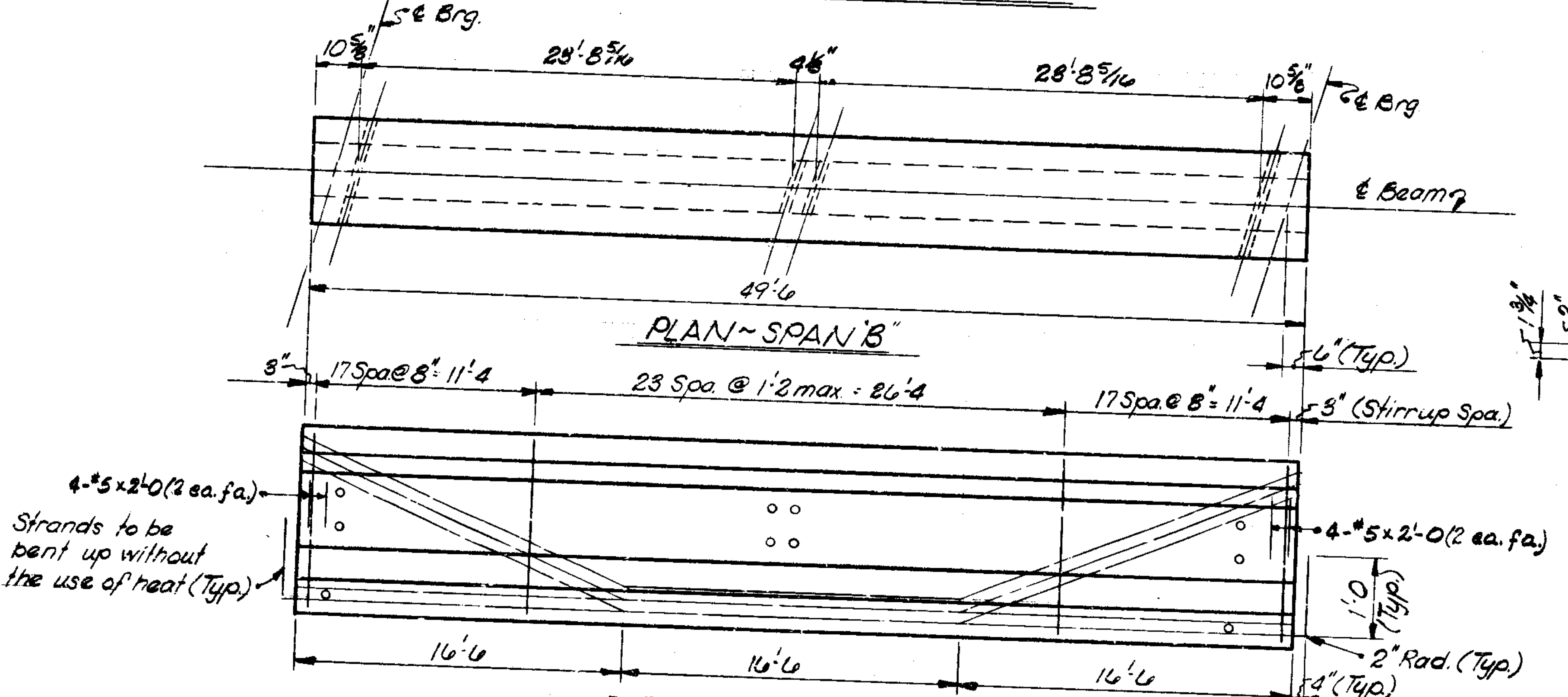


1'-0
6"
4"
3"
11"
2'-4
5"
5"
1¾"
2"
2"
1⅞"
7 Spa.@ 1¾"
1⅞"
= 1'-0¼
1'-4
¾" ø Inserts Spa. 2'-0 max. (¾" ø Hex Bolt) (Outside face of outside beam)
3"
6"
1½"
1¾"
2"
2"
6"
Seal this portion of outside beam in accordance w/ Article 702.2 of the specifications.
12-½" ø Strands @ 270 ksi pulled to 29,000 lbs each

@ MID SPAN

@ END OF SPAN

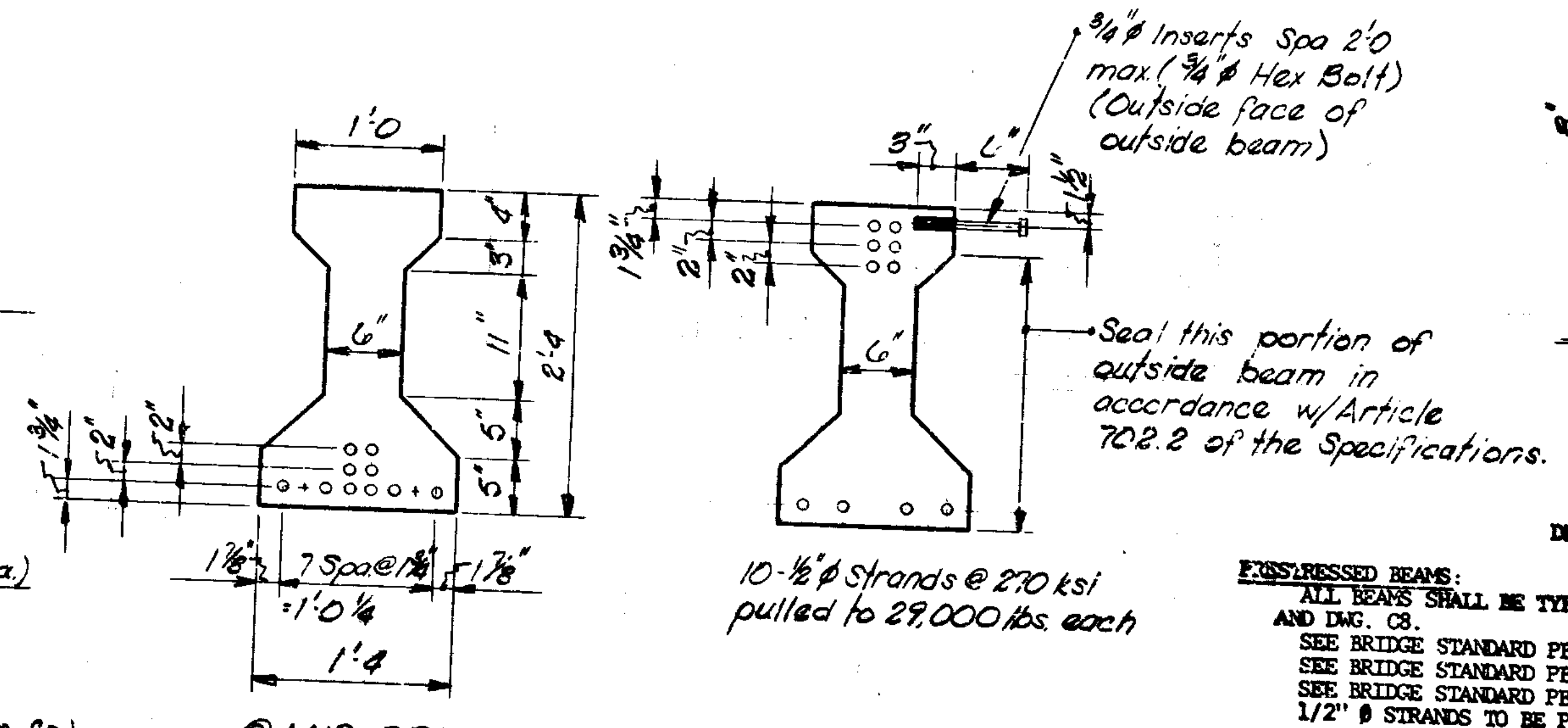
SPANS "A" or "C"



℄ Brg.
10⅝"
28'-8 5/16
4½"
28'-8 5/16
10⅝"
℄ Brg
℄ Beam
49'-6
6" (Typ.)
3"
17 Spa.@ 8" = 11'-4
23 Spa. @ 1'-2 max. = 26'-4
17 Spa.@ 8" = 11'-4
3" (Stirrup Spa.)
4-#5 x 2'-0 (2 ea. fa.)
Strands to be bent up without the use of heat (Typ.)
4-#5 x 2'-0 (2 ea. fa.)
1'-0 (Typ.)
2" Rad. (Typ.)
4" (Typ.)
16'-6
16'-6
16'-6

PLAN~ SPAN "B"

ELEVATION~ SPAN "B"

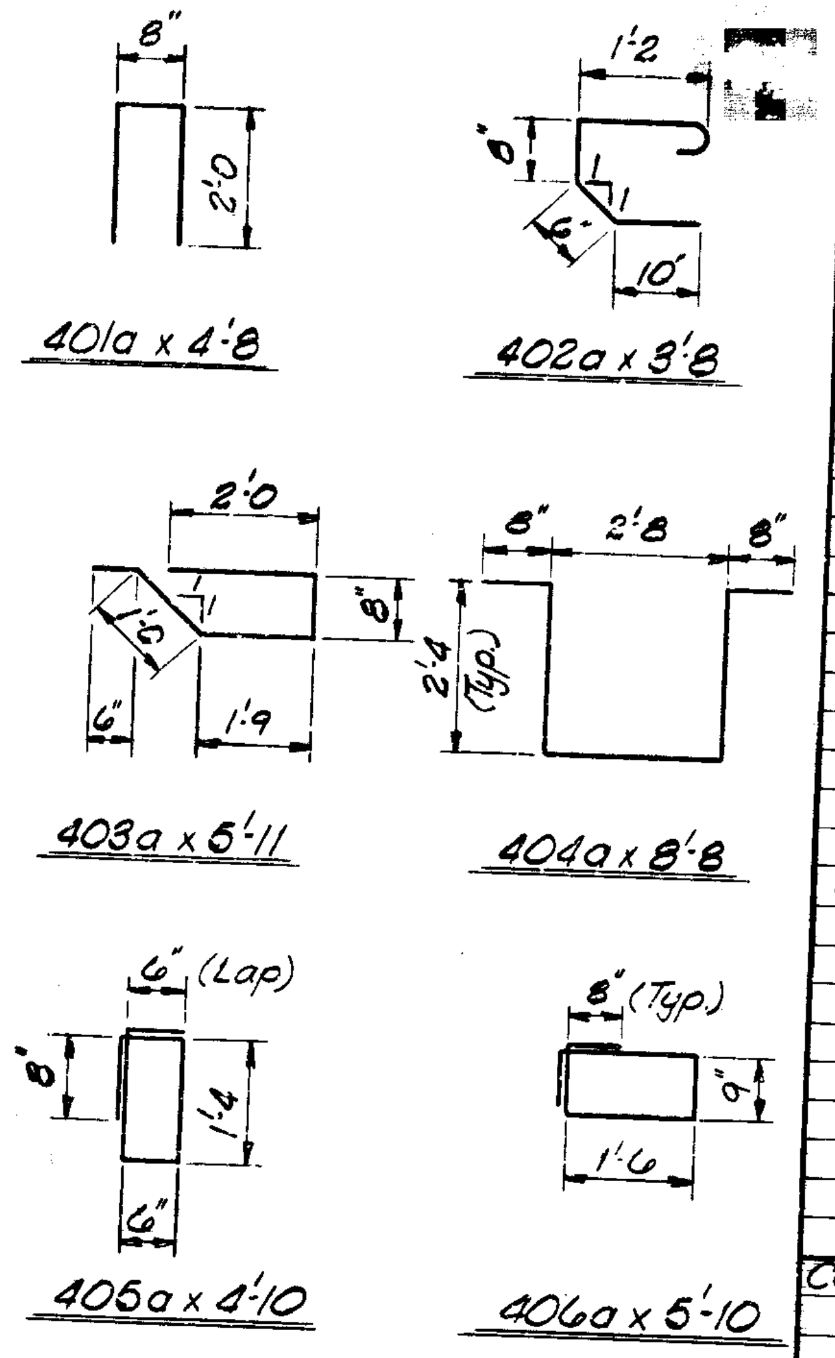


1'-0
6"
4"
3"
11"
2'-4
5"
5"
1¾"
2"
2"
1⅞"
7 Spa.@ 1⅞"
1⅞"
= 1'-0¼
1'-4
¾" ø Inserts Spa 2'-0 max. (¾" ø Hex Bolt) (Outside face of outside beam)
3"
6"
1½"
1¾"
2"
2"
6"
Seal this portion of outside beam in accordance w/ Article 702.2 of the Specifications.
10-½" ø Strands @ 270 ksi pulled to 29,000 lbs. each

@ MID SPAN

@ END OF SPAN

SPAN "B"



8"
2'-0
401a x 4'-8
1'-2
8"
6"
10"
402a x 3'-8
2'-0
1'-5
8"
6"
1'-9
403a x 5'-11
8"
2'-8
8"
2'-4 (Typ.)
404a x 8'-8
6" (Lap)
8"
1'-4
6"
405a x 4'-10
8" (Typ.)
9"
1'-6
406a x 5'-10

BILL OF MATERIALS

REINFORCING STEEL			
Size & Mark	N° of Bars	Length	Weight (lbs)
#8	4	22'-9	
#8	18	4'-0	
	Total #8		435
#7	92	25'-0	
#7	92	15'-0	
	Total #7		7,522
#5	188	38'-3	
#5	84	25'-9	
#5	896	24'-9	
#5	4	4'-6	
#5	64	4'-0	
#5	292	3'-9	
#5	168	3'-6	
	Total #5		34,927
401a	98	4'-8	
402a	98	3'-8	
403a	80	5'-11	
404a	80	8'-8	
405a	120	4'-10	
406a	378	5'-10	
#4	256	38'-3	
#4	32	23'-6	
	Total #4		10,229
Total Reinforcing			53,113

CONCRETE (CLASS "C")	CYS.
Pour N°1	24.2
Pour N°1L	12.3
Pour N°1R	12.3
Pour N°2	21.7
Pour N°2L	11.2
Pour N°2R	11.2
Pour N°3	17.3
Pour N°4	24.2
Pour N°4L	12.3
Pour N°4R	12.3
Pour N°5	17.3
Mudwall Portion @ Bent N°1	5.9
Mudwall Portion @ Bent N°4	5.9
Interior Diaphragms	5.2
Total Class "C" Concrete	193.3

MISCELLANEOUS	QUAN.
Bridge Railing Type 5A or C1	302 Lft.
Expansion Joint Type BS-8	101 Lft.
Structural Members	1 L. Sum
Bridge Deck Surface	775 Sys.
Blasting & Cleaning	775 Sys.
Finishing & Curing	775 Sys.
Surface Seal	733 Sft.

DESIGN DATA

PRESTRESSED BEAMS:
ALL BEAMS SHALL BE TYPE I AS SHOWN ON BRIDGE STANDARD PB1 AND DWG. C8.
SEE BRIDGE STANDARD PB1 AND PB10 FOR ADDITIONAL DESIGN DATA.
SEE BRIDGE STANDARD PB1 FOR ADDITIONAL DETAILS.
SEE BRIDGE STANDARD PB11 FOR BEARING PAD DETAILS.
1/2" ø STRANDS TO BE PULLED TO 29,000 LBS. EACH.

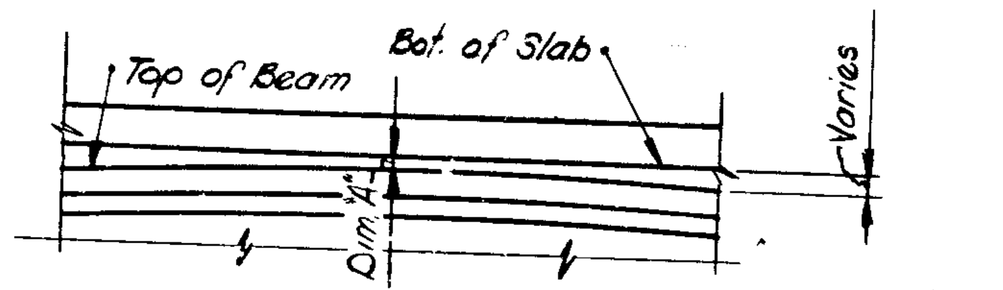
SPAN "A" OR "C"	SPAN "B"
f'c = 5000 PSI	f'c = 5000 PSI
f'ci = 4300 PSI	f'ci = 4000 PSI

APPROXIMATE CAMBER (INCHES):	SPAN "A" OR "C"	SPAN "B"
CAMBER AS ERECTED	= .999	.735
DEFLECTION UNDER SLAB D.L.	= -.616	-.617
RESIDUAL BEAM CAMBER	= .383	.118

LIVE LOAD:
HS 20-44 WITH IMPACT AND DISTRIBUTION OF LOADS IN ACCORDANCE WITH 1977 AASHTO SPECIFICATIONS AND INTERIM SPECIFICATIONS

DEAD LOAD:
ACTUAL WEIGHT PLUS 54#/SQ. FT. ROADWAY TO PROVIDE FOR THE WEIGHT OF THE BRIDGE DECK SURFACE PLUS THE FUTURE WEARING SURFACE.

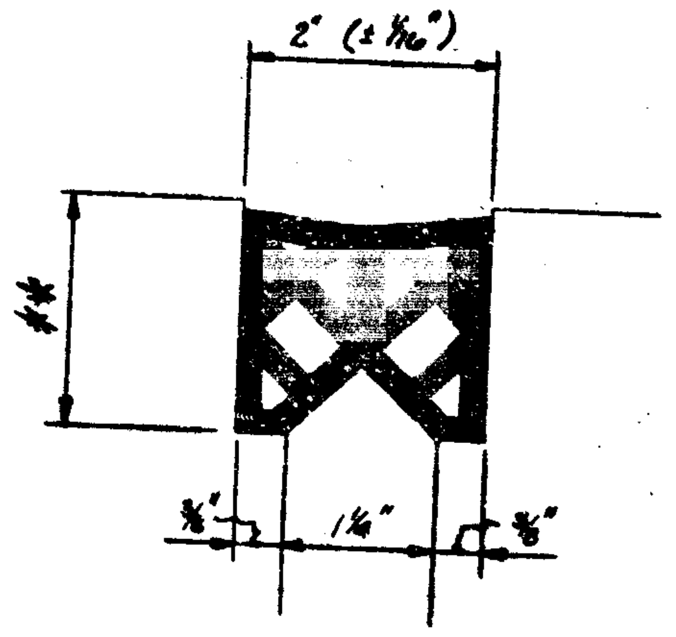
NOTES:
THE COST OF PRESTRESSED CONC. I BEAMS TYPE I, ELASTOMERIC BEARING PADS, 3/4" ø x 2'-6 THREADED DOWELS, 3/4" ø THREADED INSERTS, 3/4" ø HEX BOLTS, SEALING EXTERIOR FACE OF EXTERIOR BEAMS, AND EXPANDED POLYSTYRENE TO BE INCLUDED IN LUMP SUM BID FOR STRUCTURAL MEMBERS.
FOR REINF. BAR NOTES, SEE BRIDGE STANDARD C1.



Top of Beam
Bot. of Slab
Dim. "A"
Varies

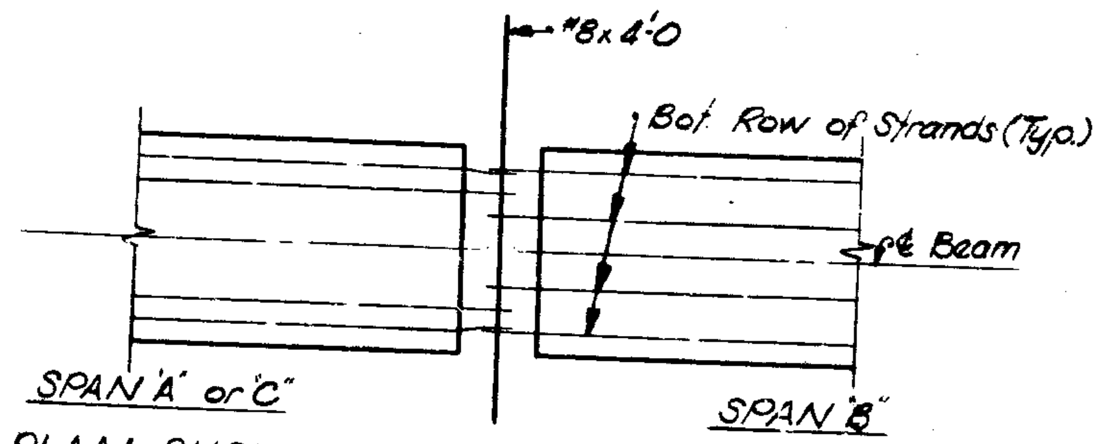
BEAM ELEVATION

Dimension "A"
¾" w/ 1½" Surface
0" w/ 2¼" Surface



EXP. JOINT ~ TYPE "BS-8"

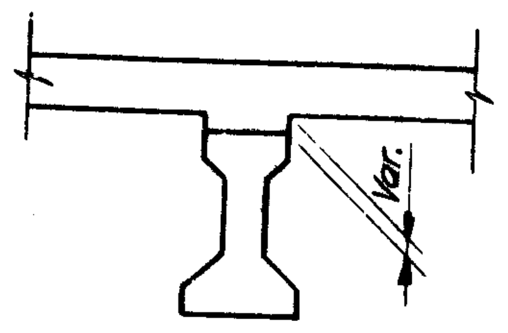
** To be determined in the field. See the Special Provision.



#8 x 4'-0
Bot. Row of Strands (Typ.)
℄ Beam
SPAN "A" or "C"
SPAN "B"

PLAN SHOWING EXTENDED STRANDS

Piers N°2 & N°3



TYPICAL SECTION

NOTE: BRIDGE SEAT ELEVATIONS WERE CALCULATED USING DESIGN CAMBER & DEADLOAD DEFLECTION OF SLAB (RESIDUAL BEAM CAMBER) WITH TOP OF BEAM AT DIM. "A" FROM BOTTOM OF SLAB ELEVATION AT CENTERLINE OF SPAN. ACTUAL CAMBERS WHICH ARE GREATER THAN DESIGN CAMBERS WILL BE TAKEN CARE OF BY PERMITTING THE TOP OF THE BEAM TO EXTEND INTO THE SLAB (LIMIT OF 1 INCH).

DESIGNED RP	C'K'D
DRAWN JLF 1-80	C'K'D R.P
TRACED	C'K'D

SUPERSTRUCTURE DETAILS &
BILL OF MATERIALS

INDIANA STATE HIGHWAY COMMISSION

SCALE:- No Scale
DATE:- 5-16-80
SUBMITTED FOR APPROVAL: Raman D. Patel
DRAWING:- C7 OF C7
SHEET:- 13 OF 73
PROJECT:- F-068-6(1)
CONTRACT NO. B-13517
BRIDGE FILE:- 44-70-6634

RAMAN D. PATEL
No. 13050

TP118

FEDERAL ROAD REGION NO.	STATE	PROJECT NO.	FISCAL YEAR	SHEET NO.	TOTAL SHEETS
5	IND	F-068-6(1)	1979	15	73

STRUCTURE DATA

* IF CONTRACTOR ELECTS TO USE METAL PIPE THICKNESS AS SHOWN BELOW ARE TO BE USED.

STRUCTURE NUMBER	LOCATION	LEFT	RIGHT	CROSS	SIZE INCHES	DESCRIPTION GROUP	DESCRIPTION SEE STD. SHEET "MP" FOR ACCEPTABLE TYPE OF PIPE WITHIN EACH GROUP.	LENGTH FT.	SKEW	COVER FT.	FLOW LINE UP STREAM ELEV.	FLOW LINE DOWN STREAM ELEV.	CONCRETE CLASS "A" CU. YDS.	"B" BORROW CU. YDS.	METHOD OF BACKFILL CLASS	THICKNESS STEEL *	THICKNESS ALUM. *	VELOCITY	RIPRAP SYS.	PIPE END SECTION EA.	REINF. STEEL LBS.	REMARKS	PLANS ON SHEET NO.
1	616+50'1'	✓																				18" C.M.P. To Be Removed	4
2	619+45'A'		✓		12	L	Inlet Type J-10 &	40		2	904.66	904.66 903.87			A							Connect To STR. № 6	4
3	620+20'A'		✓		12	P	Pipe Catch Basin 12"	6		1	904.75	904.75 904.63			A							Connect To STR. № 4	4
4	620+20'A'		✓		12	L	Inlet Type M-10 &	40		2	904.63	904.63 904.23			A							Connect To STR. № 3 Connect To STR. № 5	4
5	620+20'A'	✓			12	L	Inlet Type M-10 &	72		2	904.23	904.23 903.87			A							Connect To STR. № 4 Connect To STR. № 6	4
6	619+45'A'				12	L	Inlet Type M-10 &	96		4	903.87	903.87 895.00			B				16	1		Connect To STR. № 5 & 2	4
7				✓	6		F.B.C.P.C.S.	70							A							Drains End Bent	8
8				✓	6		F.B.C.P.C.S.	70							A							Drains End Bent	8

LEGEND FOR ABBREVIATION

F.B.C.C.S./P.I.	FULLY BITUMINOUS COATED CORRUGATED STEEL WITH PAVED INVERT.
F.B.C.C.A.A./P.I.	FULLY BITUMINOUS COATED CORRUGATED ALUMINUM ALLOY WITH PAVED INVERT.
F.B.C.C.S.	FULLY BITUMINOUS COATED CORRUGATED STEEL.
C.S.	CORRUGATED STEEL.
C.A.A.	CORRUGATED ALUMINUM ALLOY.
S.P.S.	STRUCTURAL PLATE STEEL.
F.B.C.P.C.S.	FULLY BITUMINOUS COATED PERFORATED CORRUGATED STEEL.
F.B.C.S.A./P.I.	FULLY BITUMINOUS COATED CORRUGATED STEEL ARCH WITH PAVED INVERT.
F.B.C.C.A.A.A./P.I.	FULLY BITUMINOUS COATED CORRUGATED ALUMINUM ARCH WITH PAVED INVERT.
F.B.C.C.A.	FULLY BITUMINOUS COATED CORRUGATED STEEL ARCH.
F.B.C.C.A.A.A.	FULLY BITUMINOUS COATED CORRUGATED ALUMINUM ALLOY ARCH.
C.S.A.	CORRUGATED STEEL ARCH.
C.A.A.A.	CORRUGATED ALUMINUM ALLOY ARCH.
S.P.S.A.	STRUCTURAL PLATE STEEL ARCH.

PAVED SIDE DITCH AND SODDING SUMMARY TABLE

STATION TO STATION	PAVED SIDE DITCH (Lineal Feet) TYPE	PAY LENGTH	NUMBER OF LUGS	PAY LENGTH	CUT OFF WALLS	PAY LENGTH	TOTAL PAY LENGTH	SODDING (Square Yards) FOR PAVED SIDE DITCH	FOR DITCHES	OTHER	TOTAL SQUARE YARDS SOD	SHEET NUMBER
615+00 To 617+00 [illegible]									332		332	4
619+45 To 620+30 [illegible]										114	114	4
619+78 To 620+45 Lt.										110	110	4

TOTALS:
SODDING = 556 sys.

RIGHT-OF-WAY MARKERS

STATION	OFFSET	LEFT	RIGHT
615+20'A'	℄	✓	
616+89'A'	108±	✓	
617+85'A'	98±	✓	
619+48'A'	54±	✓	
619+75'A'	49±	✓	
619+75'A'	45±	✓	
620+50'A'	45±	✓	
51+00'S-1-A'	℄		✓
50+80'S-1-A'	50		✓
616+75'A'	℄		✓
619+10'A'	55±		✓
619+15'A'	40±		✓
620+20'A'	40±		✓
620+34'A'	70±		✓

APPROACH TABLE

LOCATION	DESCRIPTION	RADIUS FT.	WIDTH FT.	LENGTH FT.	BITUMINOUS BASE-660#/SYS. TON	BITUMINOUS SURFACE-110#/SYS. TON	BITUMINOUS MIXTURE FOR APPROACHES-6" TON	3" COMPACTED AGGREGATE BASE	BITUMINOUS MATERIAL FOR TACK COAT SYS
STA. 616+25.30'A' Rt.	Spl. Public Road App.		20	100	172	27.7			1,023
STA. 619+67'A' Lt.	Cl. IV Comm. Drive		20	20			26.9		82
Sta 619+74 Rt.	Cl. II Drive	25+15	12	18			4.4	4.5	
TOTALS					172	27.7	31.3	4.5	1,105

UNDERDRAIN TABLE

RUN NO.	LOCATION	6" GROUP "K" PIPE	6" NON-PREF. FBCCS PIPE	OUTLET	WYES	BENDS	SODDING SYS.	DELIN. POSTS
1	STA. 615+00 To STA. 617+20'A' Lt.	220	14	Ditch Lt. of Sta. 615+00		1-90°	2	1
2	STA. 615+00 To STA. 617+14'A' Rt.	214	14	Ditch Rt. of Sta. 615+00		1-90°	2	1
3	STA. 618+75 To STA. 619+45'A' Lt.	64	8	Connect To STR. № 6		1-45°		
4	STA. 618+68 To STA. 619+45'A' Rt.	70	8	Connect To STR. № 2		1-45°		
5	STA. 619+45 To STA. 620+20'A' Lt.	66	8	Connect To STR. № 5		1-45°		
6	STA. 619+45 To STA. 620+20'A' Rt.	66	8	Connect To STR. № 3		1-45°		
	6 Bends @ 2 Lft./Bend		12					
	TOTAL =	700	72				4	2

AGGREGATE FOR UNDERDRAINS
772 Lft. × 0.09 cys/Lft. = 69.48 cys.

RIPRAP

SHT. NO.	LOCATION	TYPE	SYS.
9	STA. 617+35'A' Lt. & Rt.	Revetment	331
9	STA. 617+77'A' Lt. & Rt.	Revetment	42
9	STA. 618+13'A' Lt. & Rt.	Revetment	42
9	STA. 618+54'A' Lt. & Rt.	Revetment	24
9	STRUCTURE № 6	Revetment	16
	TOTAL		680

Rev: 8-18-81; Class II Dr. 619+74 Rt.

MISCELLANEOUS TABLES

F-068-6(1)
B-13517
44-70-6634

D. PATEL
No. 13050

BRIDGE FILE 44-70-6634

STRUCTURE QUANTITIES

ITEM	CONCRETE CLASS C SUPERSTR CU. YDS.	CONCRETE CLASS A SUBSTR CU. YDS.	CONCRETE CLASS B ABOVE FTG. CU. YDS.	CONCRETE CLASS B IN FTG. CU. YDS.	CONCRETE RAILING CLASS C CU. YDS.	CONCRETE RAILING CLASS C LIN. FT.	REINF. STEEL TOTAL LBS.	STRUCT. STEEL *** LBS.	ANCHOR RODS MK-AR EACH	ANCHOR PLATES MK-AP EACH	PILES UNTREATED TIMBER NO.	PILES UNTREATED TIMBER LIN. FT.	PILES TREATED TIMBER NO.	PILES TREATED TIMBER LIN. FT.	PILES 12" STEEL ENCASED CONC. NO.	PILES 12" STEEL ENCASED CONC. LIN. FT.	PILES STEEL H BEARING NO.	PILES STEEL H BEARING LIN. FT.
BENT N°1		16.4					2,488								8	250		
PIER N°2		59.5		42.7			5,635								20	400		
PIER N°3		59.5		42.7			5,635								20	400		
BENT N°4		16.3					2,488								8	256		
SUPERSTRUCTURE	193.3																	
							53,113											
Reinf. Steel for Approach Structures																		
Reinf. Steel for R.C. Bridge Approaches																		
Reinf. Steel for Lip Gutter, Pvmt., Tapers, etc.							2,611											
TOTALS	193.3	151.7		85.4			71,970								56	1,312		

ITEM	CAST IRON DRAIN PIPE LBS.	CAST IRON, GRATES, BASINS, & FITTINGS LBS.	RAILING TYPE 5A OR CI LIN. FT.	EXP. JOINT TYPE BSB LIN. FT.	EXP. JOINT CLASS LIN. FT.	CONC. STR. MEMBERS BOX BEAMS TYPE SQ. FT.	CONC. STR. MEMBERS CONC. I BEAM TYPE I LUMP SUM	APPLIED SHEET MEMBRANE SQ. FT.	BITUM MIXTURE FOR APPROACHES TONS	MOD. P.C. CONCRETE SURFACE SQ. YDS	DECK DRAIN EACH	SURFACE SEAL SQ. FT.	BLASTING & CLEANING SQ. YDS.	FINISHING & CURING SQ. YDS.
			302	101			1							
										775		733	775	775
TOTALS			302	101			1			775		* 733	775	775

* ESTIMATED QUANTITY
SEE SPECIAL PROVISIONS

APPROACH STRUCTURES

BRIDGE FILE

STRUCT. NO.	LOCATION	DESCRIPTION SIZE	DESCRIPTION KIND	LENGTH LIN. FT.	CONCR. CL. A IN STRS. CU. YDS.	REINF. STEEL LBS.	PIPE END SEC. EACH		REMARKS
			TOTALS						Total of Reinforcing Steel Carried to "Structure Quantities"

SEE SHEET Nº 14 FOR INFORMATION IN THIS TABLE

PAVED SIDE DITCH & SODDING SUMMARY

LT. OR RT.	STATION TO STATION	PAVED SIDE DITCH (LIN. FT.) TYPE	PAY LENGTH	NO. OF LUGS	PAY LENGTH	CUT OFF WALLS	PAY LENGTH	TOTAL PAY LENGTH	SODDING (SQ. YD.) FOR PSD	FOR DITCHES	SHOULDERS	OTHER	TOTAL SOD

SEE SHEET Nº 14 FOR INFORMATION IN THIS TABLE

APPROACH TABLE

LOCATION LT/RT	LOCATION STATION	DESCRIPTION	WIDTH FT.	RADII FT.	GRADE %	LENGTH FT.	DIST. BEYOND R/W FT.	EXCAVATION (CYS) CUT	EXCAVATION (CYS) FILL	BITUM. SURFACE #/SQ.YD.	BITUM. SURFACE TONS	BITUM. BINDER #/SQ.YD.	BITUM. BINDER TONS	BITUM. BASE #/SQ.YD.	BITUM. BASE TONS	COMP. AGG. BASE Depth (In)	COMP. AGG. BASE TONS

SEE SHEET Nº 14 FOR INFORMATION IN THIS TABLE

REVISIONS

DATE	ITEM
01-26-82	SURFACE SEAL NOTE

MARCH 1975

SUMMARIZED JLF C'K'D DEA
TRACED DEA C'K'D JLF

TP103

NOTES:
Weight of Spirals includes weight of 1½ extra turns top and bottom.
Spacers and 1½ turns at laps included in cost of Spiral.
*** The weight of structural steel is approximate only, and it shall be the Contractor's responsibility to determine the weight on which he bases his bid.
For Test Bar Samples See Bridge Standard C1.

BRIDGE SUMMARY
INDIANA STATE HIGHWAY COMMISSION

DATE 5-16-80

SUBMITTED FOR APPROVAL:

SHEET 15 OF 73

PROJECT: F-068-6 (1)
CONTRACT NO: B-13517
BRIDGE FILE: 44-70-6634

RAMAN B. PATEL
No. 13050

ESTIMATE OF QUANTITIES

STRUCTURE PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
	MOBILIZATION AND DEMOBILIZATION	L.SUM			1
51002	CONCRETE CLASS C IN SUPERSTRUCTURE	CYS.			193.3
51001	CONCRETE, CLASS A IN SUPERSTRUCTURE	CYS.			
51005	CONCRETE, CLASS A IN SUBSTRUCTURE	CYS.			151.7
51010	CONCRETE, CLASS B ABOVE FOOTINGS	CYS.			
51015	CONCRETE, CLASS B IN FOOTINGS	CYS.			65.4
31875	SPECIAL CLASS A CONCRETE	SFT			
51045	CONCRETE STRUCTURAL MEMBERS	LSUM			1
51030	REINFORCING STEEL	LBS.			71,970
51032	EPOXY COATED REINFORCING STEEL	LBS			
51035	STRUCTURAL STEEL	LBS.			
51038	STRUCTURAL STEEL	LSUM			
51090	BRONZE PLATES	LBS.			
51070	ANCHOR PLATES (MK-AP 1)	EACH			
51075	ANCHOR PLATES (MK-AP 2)	EACH			
51080	ANCHOR PLATES (MK-AP 3)	EACH			
51085	ANCHOR PLATES (MK-AP 4)	EACH			
51112	ANCHOR BOLTS	EACH			
51068	TIE DOWN ASSEMBLY MK-UA	EACH			
51095	CAST IRON, DRAIN PIPE, 4 INCH	LBS.			
51100	CAST IRON, DRAIN PIPE, 6 INCH	LBS.			
51105	CAST IRON, DRAIN PIPE, 8 INCH	LBS.			
51110	CAST IRON, GRATES, BASINS AND FITTINGS	LBS.			
51134	REMOVAL OF PRESENT RAILING	LFT.			
51132	RAILING RESET	LFT.			
51115	RAILING (TYPE 5 OR C)	LFT.			
51120	RAILING (TYPE 5A OR C1)	LFT.			302
51125	RAILING (TYPE 6 OR D)	LFT.			
51130	RAILING (TYPE 7 OR E)	LFT.			
51020	CLASS C, CONCRETE RAILING	CYS.			
51025	CLASS C, CONCRETE RAILING	LFT.			
51131	BARRIER RAILING TYPE X	LFT.			
51215	CLASS X EXCAVATION	CYS.			
51220	WET EXCAVATION	CYS.			293
51223	WATERWAY EXCAVATION	CYS.			
51224	WATERWAY EXCAVATION	LSUM			
51225	DRY EXCAVATION	CYS.			64
51230	FOUNDATION EXCAVATION (UNCLASSIFIED)	CYS.			50
51231	FOUNDATION EXCAVATION (UNCLASSIFIED)	LSUM			
51813	PNEUMATICALLY PLACED MORTAR	SFT.			
51870	REPOINTING MASONRY IN STR'S.	SFT.			
51814	WELDED STEEL WIRE FABRIC	SFT.			
51859	PAINTING OLD STEEL BRIDGE	LSUM			
51881	EXPANSION JOINT, TYPE BS2	LFT.			
51885	EXPANSION JOINT, TYPE BS8	LFT.			
51887	EXPANSION JOINT, TYPE BS8	LFT.			101
51888	EXPANSION JOINT, TYPE BS9	LFT.			
51890	EXPANSION JOINT, TYPE BS11	LFT.			
51925	EXPANSION JOINT, CLASS S-S	LFT.			
51926	EXPANSION JOINT, CLASS T-S	LFT.			

SUMMARIZED JLF C'K'D DEA & TJN

TRACED DEA C'K'D JLF

NOVEMBER, 1978

STRUCTURE PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
51135	TIMBER PILES FURNISHED, UNTREATED	LFT.			
51140	TIMBER PILES DRIVEN, UNTREATED	LFT.			
51145	TIMBER PILES FURNISHED, TREATED	LFT.			
51150	TIMBER PILES DRIVEN, TREATED	LFT.			
51155	PILE SHELLS FURNISHED AND DRIVEN (12 INCH)	LFT.			1,312
51160	PILE SHELLS FURNISHED AND DRIVEN (14 INCH)	LFT.			
51185	STEEL H PILES FURNISHED AND DRIVEN (8 BP 36)	LFT.			
51190	STEEL H PILES FURNISHED AND DRIVEN (10 BP 42)	LFT.			
51195	STEEL H PILES FURNISHED AND DRIVEN (12 BP 53)	LFT.			
51210	PILE ENCASEMENT (CONCRETE)	LFT.			
51328	REMOVAL OF PRESENT STRUCTURE (PORTIONS)	LSUM			
51330	REMOVAL OF PRESENT STRUCTURE	LSUM			1
51335	TEMPORARY BRIDGE AND APPROACHES	LSUM			1
51366	CONCRETE SLOPEWALL 5 INCH	SYS.			
51365	SLOPEWALL	SYS.			
51370	RIPRAP	SYS.			
51375	REVETMENT RIPRAP	TON.			595
51371	HANDLAID RIPRAP 12 INCH	SYS.			
51372	DUMPED RIPRAP	TON			
51374	PLASTIC FILTER CLOTH	SYS.			680
51106	DECK DRAINS	EACH			
51395	STEEL DRAIN PIPE (6 INCH)	LSUM			
51400	STEEL DRAIN PIPE (8 INCH)	LSUM			
51092	STEEL PIPE CONDUIT (2 INCH)	LFT.			
51566	RIVETS REMOVED	EACH			
51864	FIELD DRILLED HOLES	EACH			
51867	STRUCTURAL STEEL CUTTING	SIN			
51863	FIELD DRILLED HOLES IN CONCRETE	EACH			
*	SURFACE SEAL	LSUM			1
51826	SURFACE SEAL	SFT.			
51827	BRIDGE DECK MEMBRANE	LSUM			
51842	BRIDGE DECK OVERLAY	SYS.			
51845	BRIDGE DECK SURFACE	SYS.			775
		CYS.			
51843	BRIDGE DECK PATCHING	SFT.			
51833	CONCRETE SCARIFYING	SYS.			
51840	ADDITIONAL CONCRETE SCARIFYING	SYS.			
51837	BLASTING AND CLEANING	SYS.			775
51838	FINISHING AND CURING	SYS.			775

APPROACH PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
	EXCAVATION FOR SUBGRADE TREATMENT	CYS			605
02020	UNCLASSIFIED EXCAVATION	CYS.			
52240	COMMON EXCAVATION	CYS.			1900
52245	BORROW	CYS.			
52250	B BORROW	CYS.			
52255	B BORROW FOR STRUCTURE BACKFILL	CYS.			53
52303	REMOVAL OF PAVEMENT	SYS.			982
02235	BREAKING PAVEMENT	SYS.			
52490	TERMINAL JOINT	LFT.			
52495	CONTRACTION JOINT, TYPE D-1	LFT.			
	CONCRETE CURB	LFT.			535
52280	CONCRETE PAVEMENT REINFORCED (7 INCH)	SYS.			
52285	CONCRETE PAVEMENT REINFORCED (8 INCH)	SYS.			
52290	CONCRETE PAVEMENT REINFORCED (9 INCH)	SYS.			
52300	CONCRETE PAVEMENT REINFORCED (10 INCH)	SYS.			127
06070	CONCRETE SIDEWALK	SYS.			
52305	TYPE P COMPACTED AGGREGATE FOR BASE (SIZE NO. 53)	TON			
52600	COVER AGGREGATE	TON			
52806	COVER AGGREGATE (SIZE NO. 12)	TON			
52605	AGGREGATE FOR SHOULDER DRAINS	TON			
52610	AGGREGATE FOR UNDER DRAINS	CYS.			70
52308	TYPE O COMPACTED AGGREGATE FOR BASE (SIZE NO. 53)	TON			4.5
52310	SUBBASE	CYS.			24
52315	BITUMINOUS STABILIZED SUBBASE TYPE I, II, OR III	TON			
52320	BITUMINOUS STABILIZED SUBBASE	TON			
52445	BITUMINOUS BASE	TON			826
5 04260	BITUMINOUS BASE (SIZE NO 5D)	TON			653
	OPEN GRADED BITUM. BASE Nº 5	TON			43.7
4 52450	BITUMINOUS SURFACE	TON			128
	BITUMINOUS MATERIAL FOR TACK COAT	SYS.			4,969
52461	BITUMINOUS MATERIAL FOR PRIME COAT	SYS.			
04348	SEAL COAT TYPE 2	SYS.			770
52470	BITUMINOUS MIXTURE FOR APPROACHES	TON			31.3
52475	BITUMINOUS MIXTURE FOR SHOULDER	TON			
52480	BITUMINOUS MATERIAL, APPLIED	TON			
52500	GUARD RAIL, TYPE A	LFT.			30
52505	GUARD RAIL, TYPE B	LFT.			
52510	GUARD RAIL, TYPE C	LFT.			
52515	GUARD RAIL, TYPE D	LFT.			
52520	GUARD RAIL, TYPE E	LFT.			
52525	GUARD RAIL, TYPE F	LFT.			
52530	GUARD RAIL, TYPE G	LFT.			350
52531	GUARD RAIL, TYPE H	LFT.			
	GUARD RAIL END TREATMENT	EACH			3
52380	SODDING	SYS.			560
	MULCHED SEEDING (LEGUME)	SYS.			2,037
	MULCHED SEEDING "TR"	SYS.			
		SYS.			992
1. 52400	MULCHING MATERIAL	TON			
2. 52405	FERTILIZER	TON			
52410	WATER	M.G.			6
3. 52415	AGRICULTURAL LIMESTONE	TON			
52398	SEED MIXTURE "CV"	LBS.			.2
52401	MULCHING MATERIAL (WOOD CELLULOSE FIBER)	TON			
52640	MAINTAINING TRAFFIC	LSUM			
52370	CLEARING RIGHT-OF-WAY	LSUM			
52434	BREAKAWAY CABLE TERMINAL TYPE "A"	EACH			

1. INCLUDES ——— TONS FOR SEED MIXTURE "R"
2. INCLUDES ——— TONS FOR SEED MIXTURE "R"
3. INCLUDES ——— TONS FOR SEED MIXTURE "R"
4. INCLUDES 46 TONS FOR TEMPORARY RUNAROUND.
5. INCLUDES 409 TONS FOR TEMPORARY RUNAROUND.

* ESTIMATED QUANTITY 733 SFT.
SEE SPECIAL PROVISIONS

APPROACH PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
07025	PIPE: GR. A (0.064" FBCCS) 12"	LFT.			
07075	PIPE: GR. A (0.064" FBCCS) 15"	LFT.			
07125	PIPE: GR. A (0.064" FBCCS) 18"	LFT.			
07175	PIPE: GR. A (0.064" FBCCS) 24"	LFT.			
07225	PIPE: GR. A (0.064" FBCCS) 30"	LFT.			
07275	PIPE: GR. A (0.064" FBCCS) 36"	LFT.			
07325	PIPE: GR. A (0.064" FBCCS) 42"	LFT.			
10000	PIPE: GR. D (0.064" CS) 12"	LFT.			
10025	PIPE: GR. D (0.064" CS) 15"	LFT.			
10050	PIPE: GR. D (0.064" CS) 18"	LFT.			
10075	PIPE: GR. D (0.064" CS) 24"	LFT			
10100	PIPE: GR. D (0.064" CS) 30"	LFT.			
10125	PIPE: GR. D (0.064" CS) 36"	LFT.			
10150	PIPE: GR. D (0.064" CS) 42"	LFT.			
34000	PIPE: 0.052" FBC PERF. CS 6"	LFT.			140
29000	PIPE: 0.064" FBCCS 12"	LFT.			
52375	CONCRETE CLASS A IN STRUCTURE	CYS.			
52376	CONCRETE CLASS C IN STRUCTURE	CYS.			
46000	PIPE END SECTION 12"	EACH			1
46005	PIPE END SECTION 15"	EACH			
46010	PIPE END SECTION 18"	EACH			
46015	PIPE END SECTION 21"	EACH			
46020	PIPE END SECTION 24"	EACH			
46025	PIPE END SECTION 27"	EACH			
46030	PIPE END SECTION 30"	EACH			
46035	PIPE END SECTION 33"	EACH			
46040	PIPE END SECTION 36"	EACH			
45000	INLET, TYPE A-1	EACH			
45025	INLET, TYPE D-6	EACH			
45030	INLET, TYPE E-7	EACH			
45070	INLET, TYPE P-12A	EACH			
06335	PAVED SIDE DITCH TYPE A	LFT.			
06340	PAVED SIDE DITCH TYPE B	LFT.			
06345	PAVED SIDE DITCH TYPE C	LFT.			
06350	PAVED SIDE DITCH TYPE D	LFT.			
06355	PAVED SIDE DITCH TYPE E	LFT.			
06360	PAVED SIDE DITCH TYPE F	LFT.			
06365	PAVED SIDE DITCH TYPE G	LFT.			
52831	DRILLED HOLES FOR MUDJACKING	EACH			
52830	MATERIAL FOR MUDJACKING	CYS.			
	PIPE: GR. K 6"	LFT.			700
	PIPE NON-PERF F.B.C.C.S. 6"	LFT.			72
	INLET, TYPE J-10	EACH			1
	INLET, TYPE M-10	EACH			3
	PIPE: GR. L 12"	LFT.			248
	PIPE: GR. P 12	LFT.			6
	E.E.O. TRAINEE (1)	HRS.			1,000

APPROACH PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
06040	47" FENCE, F. FIELD	LFT.			
06045	48" FENCE, CHAIN LINK	LFT.			
52340	CONSTRUCTION SIGNS (TYPE A)	EACH			26
52345	CONSTRUCTION SIGNS (TYPE B)	EACH			
52350	STANDARD BARRICADES (TYPE III)	EACH			2
	STANDARD BARRICADES (TYPE II)	EACH			8
06650	STOP SIGN, TYPE R-1A	EACH			1
06652	DO NOT PASS SIGN, TYPE R-11-A	EACH			
06655	YIELD SIGN, TYPE R-101	EACH			
06657	PASS WITH CARE SIGN, TYPE R-12A	EACH			
06660	CURVE SIGN, TYPE W-2AR	EACH			
06665	CURVE SIGN, TYPE W-2AL	EACH			
06670	REVERSE CURVE SIGN, TYPE W-4AR	EACH			
06675	REVERSE CURVE SIGN, TYPE W-4AL	EACH			
06680	LARGE ARROW SIGN, TYPE W-11A	EACH			
06685	STOP AHEAD SIGN, TYPE W-13A	EACH			
06725	DELINEATOR WITH POST, TYPE D-1 3 INCH	EACH			22
06740	DELINEATOR WITH POST, TYPE D-2 3 INCH	EACH			
	DO NOT SPRAY SIGN	EACH			6
06770	DELINEATOR POST	EACH			2
06771	FLEXIBLE DELINEATOR POST	EACH			
52366	TEMPORARY PAVEMENT MARKING TAPE	LFT.			6200
52367	TEMPORARY PAVEMENT MARKING PAINT	LFT.			800
52360	RIGHT-OF-WAY MARKERS	EACH			14
06500	MONUMENT, TYPE A	EACH			
06505	MONUMENT, TYPE B	EACH			
06510	MONUMENT, TYPE C	EACH			
06515	MONUMENT, TYPE D	EACH			
52821	FLASHING ARROW SIGN	EACH			
06713	LINE, SOLID, WHITE, 4"	LFT.			1115
06714	LINE, SOLID, YELLOW, 4"	LFT.			70
06702	THERMOPLASTIC LINE, SKIP WHITE, 5"	LFT.			
06716	REMOVAL OF LINE, SOLID, WHITE, 4"	LFT.			
06717	REMOVAL OF LINE, SOLID, YELLOW, 4"	LFT.			
06718	REMOVAL OF LINE, SKIP, WHITE, 4"	LFT.			
52536	TEMPORARY CONCRETE BARRIER	LFT.			
52819	TEMPORARY CROSSOVER TYPE "B"	EACH			
	LINE, SKIP, YELLOW, 4"	LFT.			218
	SPECIAL MARKING, STOP LINE 24"	EACH			1

END STR.

REVISIONS

DATE	ITEM
8-18-81	52495, 52308, 52470, 52530, Mulched Seeding "TR" deleted
01-26-82	52370 AND 51826 DELETED SURFACE SEAL ADDED 51230 ADDED
02-22-82	Revetment Riprap, Open Graded Bitum. Base Nº 5, Inlet, Type J-10, Inlet, Type M-10, E.E.O. Trainee (1) Added. SPECIAL MARKING, STOP LINE 24", 06650 ADDED 52445, 04260, 52310, 52398, SURFACE SEAL

BRIDGE ESTIMATE OF QUANTITIES

INDIANA STATE HIGHWAY COMMISSION

DATE 5-16-80

SUBMITTED FOR APPROVAL: Raman D Patel

RAMAN D. PATEL
REGISTERED
No. 13050
STATE OF INDIANA
PROFESSIONAL ENGINEER

PROJECT: F-068-6(1)
CONTRACT NO: B-13517
BRIDGE FILE: 44-70-6634

SHEET 16 OF 73

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